

DUBAI RAPID LINK CONSORTIUM

DUBAI METRO PROJECT OFFICE

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**Dubai Metro, UAE
Green Line – Al-Qusais Depot**

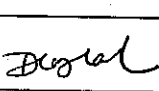
**Diesel Engine Workshop
(Building 6)**

**Structural Design Calculations
(DCP3 Submission)**

July 2007

Client: JT Metro JV	Contract No. (if any):
Project Title: Dubai Metro, UAE	Project No.: JRD4561
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Revision, Review and Approval/Authorization Records

				
A1	DCP3 Submission	D. Shah July 07	P. Scott July 07	J. Hill July 07
Revision	Description	Prepared by / date	Reviewed by / date	App. or Auth. by / date

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Note: App. and Auth. mean "Approved" and "Authorized" respectively.

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1 Introduction

The Dubai Metro Project is under construction by the Dubai Rapid Link Consortium (DURL). The Civil Works are contracted to the Japan Turkey Metro Joint Venture (JTMJV), and the Rail Systems Works are contracted to Mitsubishi Heavy Industries (MHI). The rolling stock will be provided by Kinki Sharyo.

The Diesel Engine Workshop is located in a building 96m long, 20m wide, and 12m high. The facility is equipped with two maintenance tracks, one having a double pit construction at 1.1m and 1.6m below top of rail. Both maintenance tracks will have overhead gantry cranes with a maximum lifting capacity of 6t.

This submission for Diesel Engine Workshop (Building 6) in Al-Qusais Depot is made by the Designers for the JTMJV, and as such concentrates on the scope of the Civil Works Contract, which includes the Architecture, Structure and M&E Design.

2 Building Form and Structural System

The Diesel Engine Workshop is located in a building 96m long, 20m wide, and 12m high. This building is identical as that of Red Line Rashidiya Main Depot, Building 6 Diesel Engine workshop.

For this building the proposed structural system comprises of traditional portal steel frame to take all the gravity loads and integrated cross bracing system to provide stability for lateral forces. Purlins at roof and at side walls will give additional restraints to long span beams and columns in lateral direction. These purlins will be connected to bracing system in vertical and horizontal direction.

For functionally efficient solution, a 20 m span steel portal frame spaced at @ 8 m c/c is proposed. 8 m span I Beam spaced @ 4 m c/c connected on steel portal frame provides support to cold form Z shape steel section purlin spaced at 2.0 m c/c. Structural system of a portal frame in one direction and vertical bracing in other direction provides stability against lateral loads ie. wind and earthquake forces.

The building is to be clad with insulated standing seam aluminium sheeting, fixed to cold formed purlins which span onto structural frame.

A 200 mm to 600 mm thick non-suspended ground level slab monolithic with RC Beams is proposed to resist various forces from railway track, trenches and pits. The RC Pedestal raised from RC pad footing will support structural steel columns with holding down bolts fixed in position at formation level.

3 Peer Review of Red Line building and comparison

Essentially this building is identical as that of Red Line Rashidiya Main Depot, Building 6 Diesel Engine workshop, except that it is an exact mirror placement along its longitudinal axis. So the form and structural system has been replicated from the Rashidiya building.

The structural report DM001-E-MAG-CDP-DR-DCC-371943 (Ref.Sect.5) for the Rashidiya building 6-Diesel Engine Workshop has been peer reviewed for loadings, analysis, strength & deflection checks for all the members and foundation design. The drawings have also been peer reviewed for the design intent translation on to the details, which have then been replicated onto the Green Line Depot building-6-Diesel Engine Workshop as listed in sect.4.

The basis of this report is to utilise Red Line Main Depot Diesel Engine Workshop (Bldg-6) to Green Line Al-Qusais Depot Diesel Engine Workshop (Bldg-6) with sufficient review to allow submission of Green Line DCP3 Phase.

4 Drawings

To complement the purpose of this submission, the following Architectural and Structural drawings are submitted along with this report.

	<u>Drawing Title</u>	<u>Drawing No.</u>
Structural Drawings		
1	Al-Qusais-Depot, Building-6 - Diesel Engine Workshop - Structural Drawing List	DM001-D-GAG-WDE-DD-DCC-551299
2	Al-Qusais-Depot - General Notes And Typical Details Sheet 1 Of 3	DM001-D-GDP-CDP-DD-DCC-551990
3	Al-Qusais-Depot - General Notes And Typical Details Sheet 2 Of 3	DM001-D-GDP-CDP-DD-DCC-551991
4	Al-Qusais-Depot - General Notes And Typical Details Sheet 3 Of 3	DM001-D-GDP-CDP-DD-DCC-551992
5	Al-Qusais-Depot, Building-6 - Diesel Engine Workshop - Foundation Layout Plan	DM001-D-GAG-WDE-DD-DCC-551225

	<u>Drawing Title</u>	<u>Drawing No.</u>
6	Al-Qusais-Depot, Building-6 - Diesel Engine Workshop - Ground Slab Layout Plan Sheet 1	DM001-D-GAG-WDE-DD-DCC-551226
7	Al-Qusais-Depot, Building-6 - Diesel Engine Workshop - Ground Slab Layout Plan Sheet 2	DM001-D-GAG-WDE-DD-DCC-551227
8	Al-Qusais-Depot, Building-6 - Diesel Engine Workshop -Crane Beam Level Layout Plan	DM001-D-GAG-WDE-DD-DCC-551228
9	Al-Qusais-Depot, Building-6 - Diesel Engine Workshop - Roof Layout Plan	DM001-D-GAG-WDE-DD-DCC-551229
10	Al-Qusais-Depot, Building-6 - Diesel Engine Workshop - Section A-A & B-B	DM001-D-GAG-WDE-DD-DCC-551230
11	Al-Qusais-Depot, Building-6 - Diesel Engine Workshop - Section C-C	DM001-D-GAG-WDE-DD-DCC-551231
12	Al-Qusais-Depot, Building-6 - Diesel Engine Workshop – Elevations	DM001-D-GAG-WDE-DD-DCC-551232
13	Al-Qusais-Depot, Building-6 - Diesel Engine Workshop - Bracing Connection Details	DM001-D-GAG-WDE-DD-DCC-551233
14	Al-Qusais-Depot, Building-6 - Diesel Engine Workshop -Footing Details - Sheet 1	DM001-D-GAG-WDE-DD-DCC-551234
15	Al-Qusais-Depot, Building-6 - Diesel Engine Workshop -Footing Details - Sheet 2	DM001-D-GAG-WDE-DD-DCC-551235
16	Al-Qusais-Depot, Building-6 - Diesel Engine Workshop -Ground Slab And Pit Reinforcement Details - Sheet 1	DM001-D-GAG-WDE-DD-DCC-551236
17	Al-Qusais-Depot, Building-6 - Diesel Engine Workshop -Ground Slab And Pit Reinforcement Details - Sheet 2	DM001-D-GAG-WDE-DD-DCC-551237
18	Al-Qusais-Depot, Building-6 - Diesel Engine Workshop -Ground Slab And Pit Reinforcement Details - Sheet 3	DM001-D-GAG-WDE-DD-DCC-551238
19	Al-Qusais-Depot, Building-6 - Diesel Engine Workshop - Beam Reinforcement Details	DM001-D-GAG-WDE-DD-DCC-551239
20	Al-Qusais-Depot, Building-6 - Diesel Engine Workshop - Roof Setting Out-Plan & Sections	DM001-D-GAG-WDE-DD-DCC-551240

Appendix - A
Red Line Rashidiya Depot
Diesel Engine Workshop Building - 6
Design Calculations (Enhanced Design)

**Dubai Metro, UAE
Rashidiya Depot**

**Design Calculation
(Enhanced Design)
Building 6 – Diesel Engine Workshop**

June 2007

Client: JT Metro JV	Contract No. (if any):
Project Title: Dubai Metro, UAE	Project No.: JRD4561
Document No.: DM001-E-MAG-CDP-DR-DCC-371943	Controlled Copy No.:
Document Title: Main Depot- Rashidiya Design Calculation (Enhanced Design) Building 6 – Design Engine Workshop	
Covering Letter / Transmittal Ref. No.: To be advised	Date of Issue: 12 Jun 2007

Revision, Review and Approval/Authorization Records

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A2	Issued for Approval	CK Lam Oct 06	P.Shepherd-Smith Oct 06	B. Maney Oct 06
A1	Internal Review	CK Lam Sep 06	P.Shepherd-Smith Sep 06	B. Maney Sep 06
Revision	Description	Prepared by / date	Reviewed by / date	App. or Auth. by / date

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Document No.: **DM001-E-MAG-CDP-DR-DCC-371943**

Rev A3

Date: 12 June 2007

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1 3D FRAME ANALYSIS AND DESIGN

Document No.: **DM001-E-MAG-CDP-DR-DCC-371943**

Rev A3

Date: 12 June 2007

Page

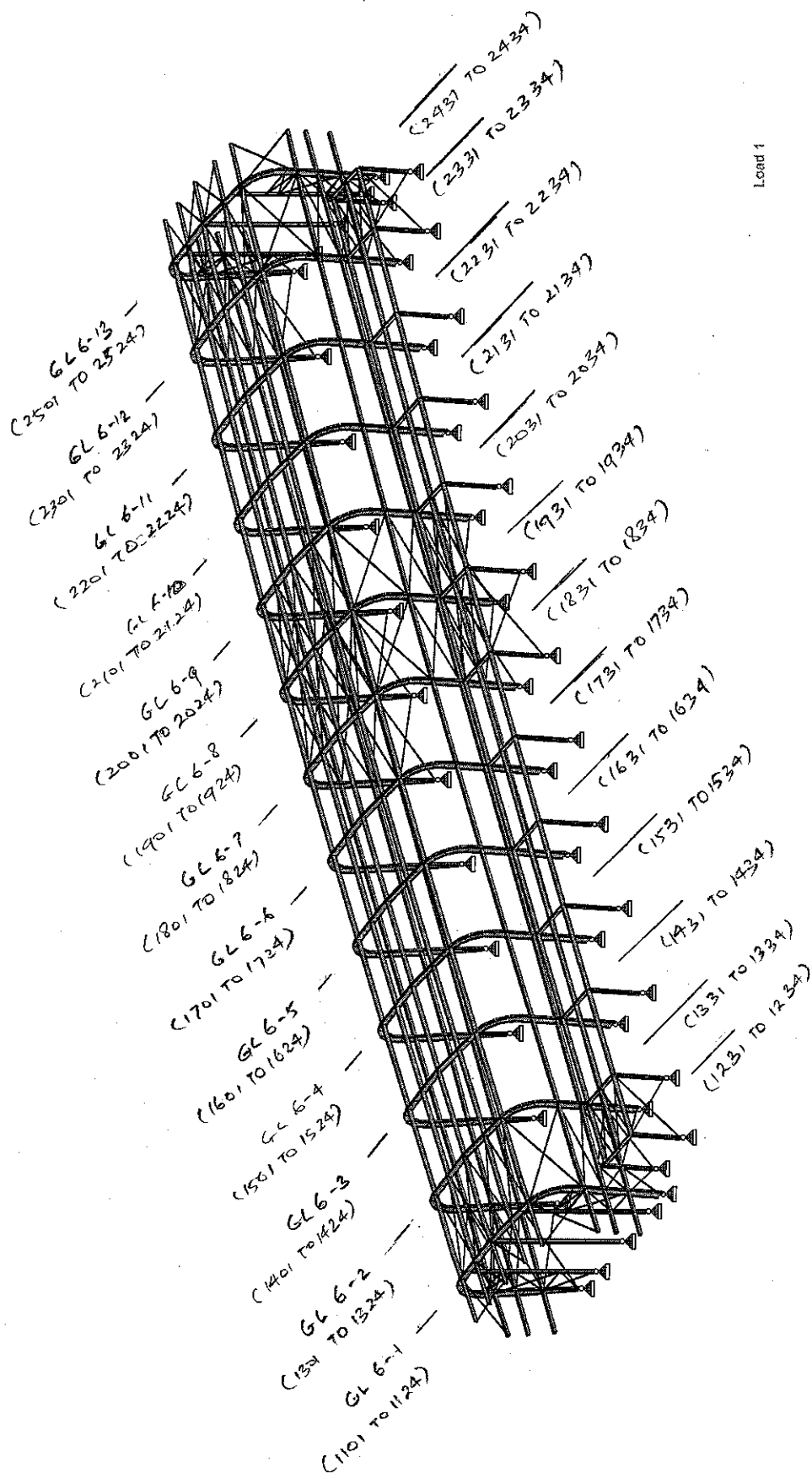


Software licensed to Meinhardt Infrastructure Pte Ltd

Job Title

Client

Job No	Sheet No	Rev
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Part		
Ref		
By	Date	Chd
	April 2007	
File Diesel Engine Workshop.		
Date/Time 08-Jun-2007 12:30		





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Job Title

Part

Ref

By

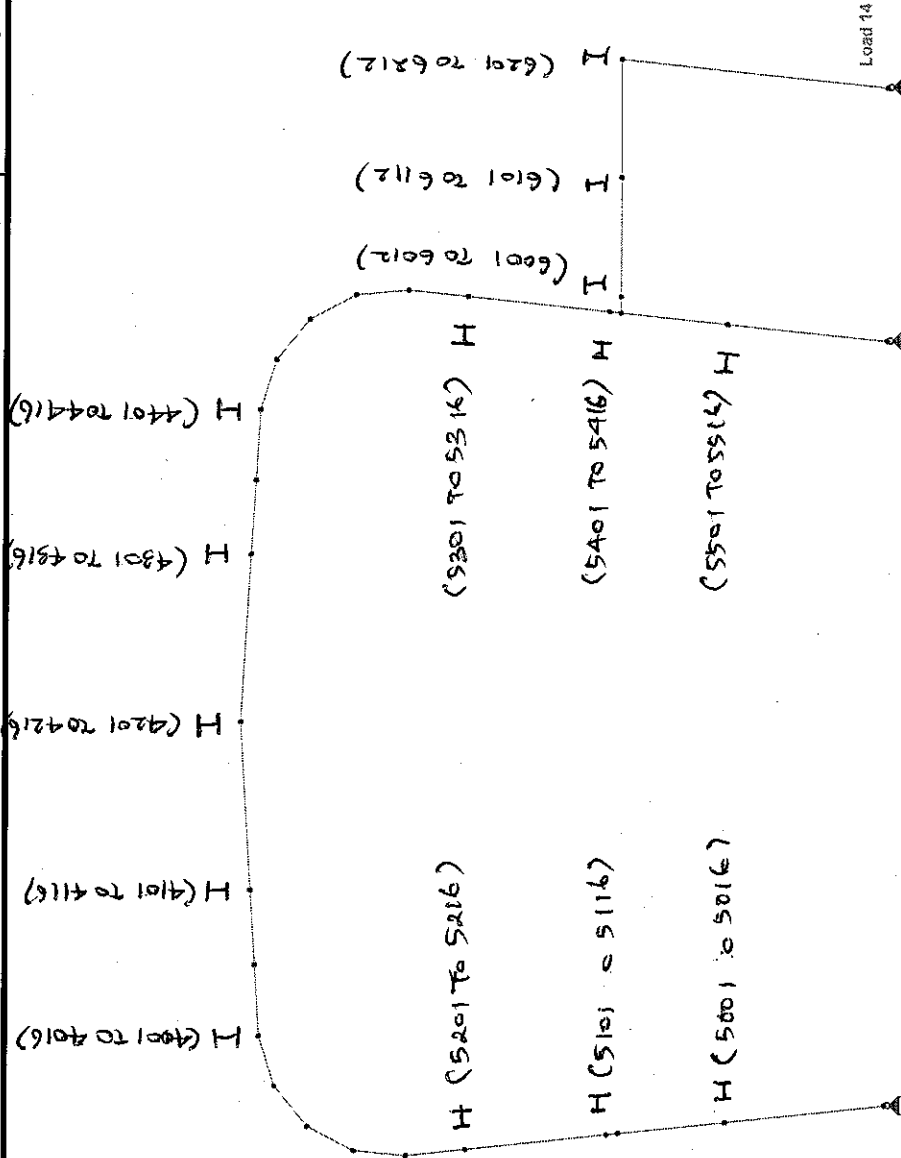
Date April 2007

Chd

File Diesel Engine Workshop

Date/Time 08-Jun-2007 12:30

Client





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Job Title

Client

Job No

Sheet No

1

Rev

Part

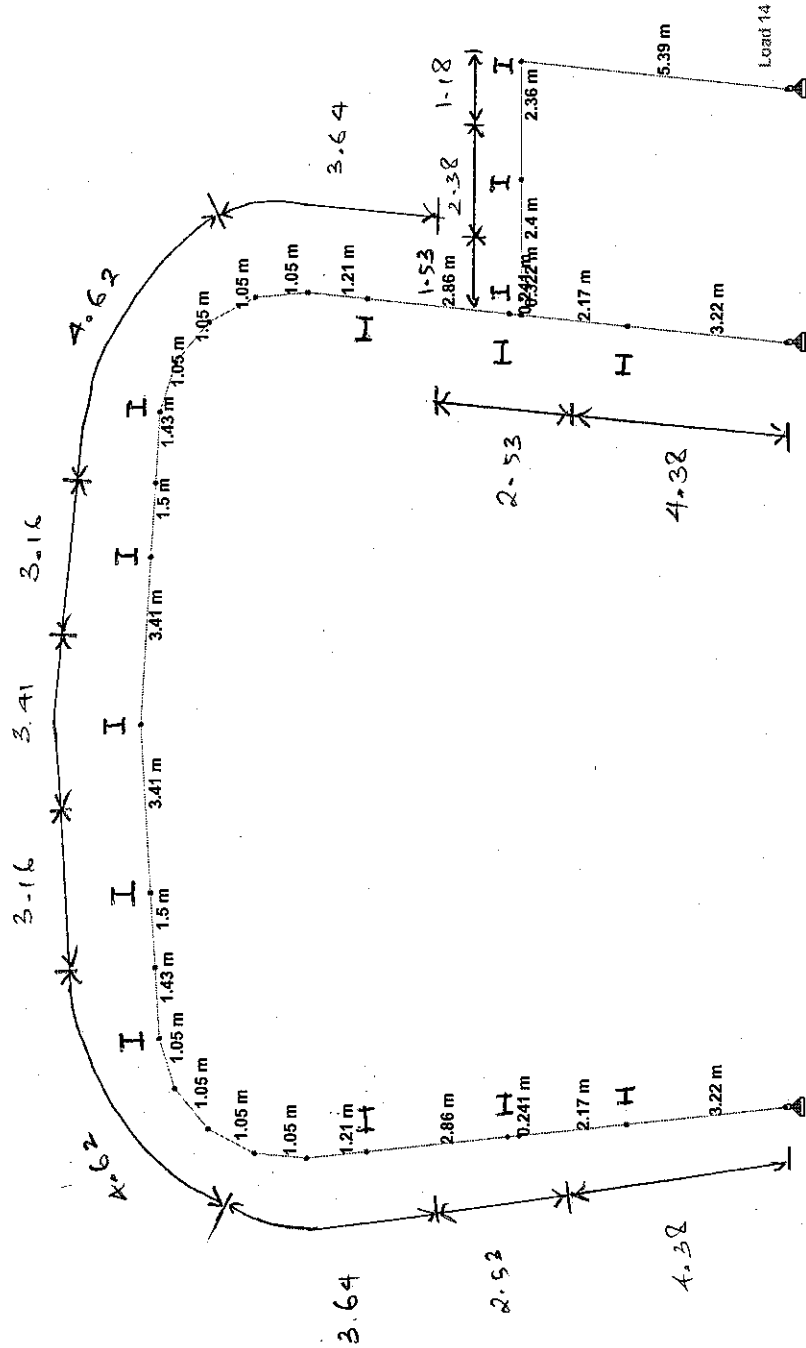
Ref

By

Date April 2007

Chd

File Diesel Engine Workshop. | Date/Time 08-Jun-2007 12:30





Site:	Dubai Metro Project - Rashidiya	Date:	Jun '07	Job No.:	68010.027
	Diesel Engine Workshop - Main Hall				
Loadings		Designed by:	Rudy	Sheet No.:	

Roof Loadings :**DL** W_{DL} :

Sheeting	0.1 kPa		
Purlins	0.1 kPa		
Insulation	0.3 kPa		
M&E Finishes	0.5 kPa		
	<u>1 kPa</u>	x 1 =	1 kN/m ²

P_{LL1} : A/C equipment S/W
 = 5 kN per 8m bays

Total no of AC equipment = 12
 Therefore, total P_{LL2} = 60 kN

Notional horizontal load, Ph

(Load Case of 1.4DL + 1.6LL)

Tributary load area for the roof
 = $(12*8+2*4)*(14.65)$
 = 1523.6 m²

Total Dead Load
 1524 kN
 (Including
 overhang roof)

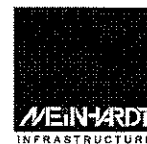
Total Live Load
 974 kN
 (Including A/C equipment &
 overhang roof)

Tributary load area for the wall
 = $13*(12*8+2*14.65)$
 = 1628.9 m²

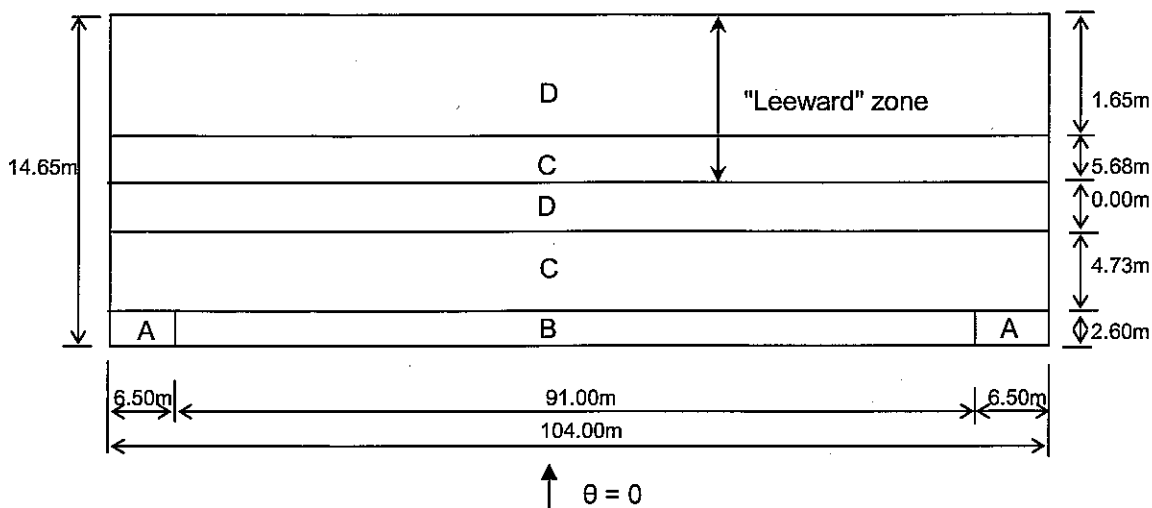
Total Dead Load
 814 kN
 (excluding the M&E Finishes)

Therefore, total ULS load, W_u = 4832 kN

Ph = 1.5 % of W_u
 = 72.5 kN



Site:	Dubai Metro Project - Rashidiya Diesel Engine Workshop - Main Hall Wind Load Computation	Date:	Jun '07	Job No.:	68010.027
		Designed by:	Rudy	Sheet No.:	

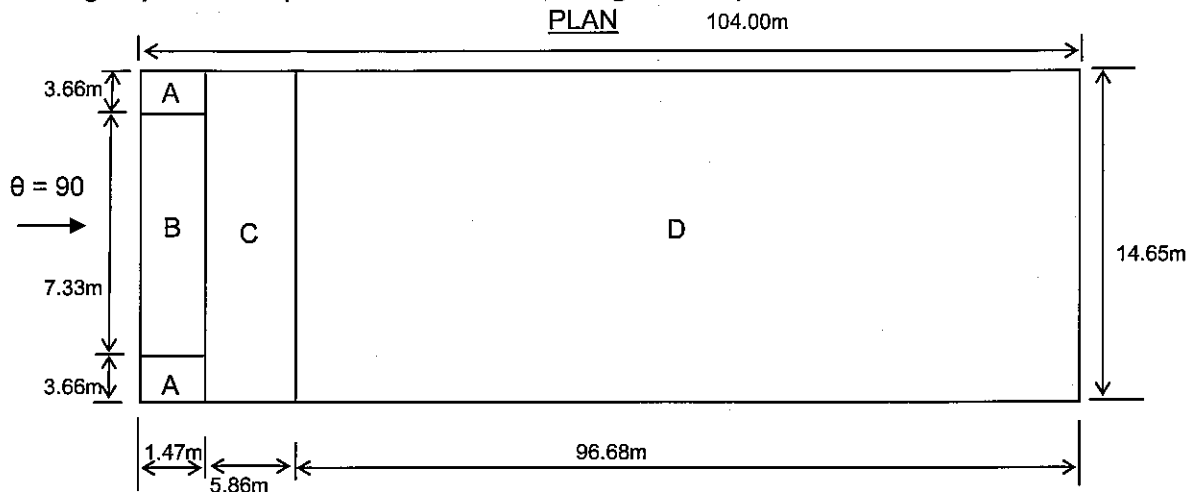
Average Cpe for roof (Wind direction normal to eave)
PLAN


Design as flat roof (with sharp eaves)

$$\begin{aligned}
 B &= 104 \text{ m} \\
 H &= 13 \text{ m} \\
 b &= \text{smaller of } (B, 2H) \\
 &= 2 \times 13 \\
 &= 26 \text{ m}
 \end{aligned}$$

Zone	Windward		Leeward		
	Area (m ²)	Cpe	Area	Cpe1	Cpe2
A	33.8	-2			
B	236.6	-1.4			
C	491.4	-0.7	590.2	-0.7	-0.7
D	0.0	-0.2	171.6	-0.2	0.2
Total	761.8		761.8		
Average		-0.98		-0.59	-0.50

Site: Dubai Metro Project - Rashidiya Diesel Engine Workshop - Main Hall Wind Load Computation	Date: Jun '07	Job No.: 68010.027 Designed by: Rudy Sheet No.:
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Average Cpe for roof (Wind direction normal to gable end)


Design as flat roof (with sharp eaves)

$$\begin{aligned}
 B &= 14.65 \text{ m} \\
 H &= 13 \text{ m} \\
 b &= \text{smaller of } (B, 2H) \\
 &= 1 \times 14.65 \\
 &= 14.65 \text{ m}
 \end{aligned}$$

For maximum uplift, assume zone C coefficient =>

$$C_{pe} = -0.7$$

For maximum downward force, assume zone D coefficient =>

$$C_{pe} = 0.2$$

Wind Load Computation

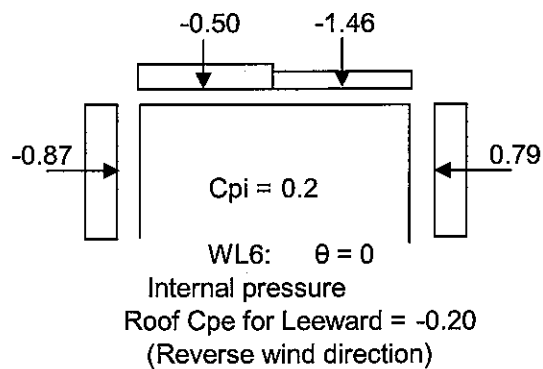
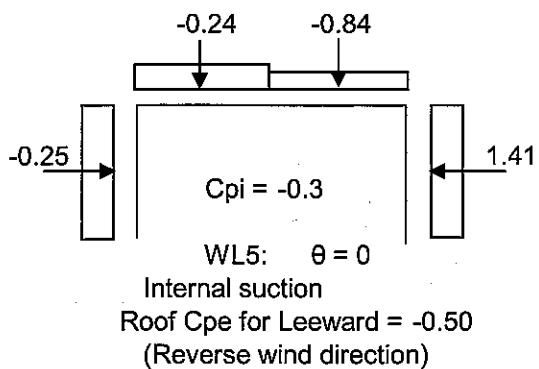
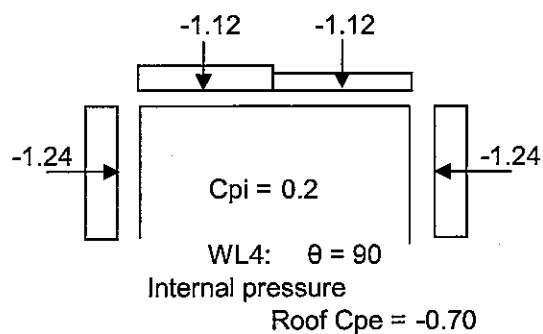
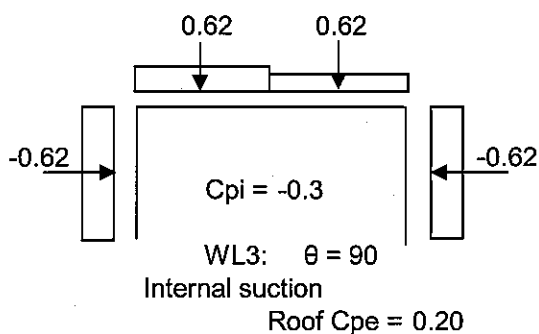
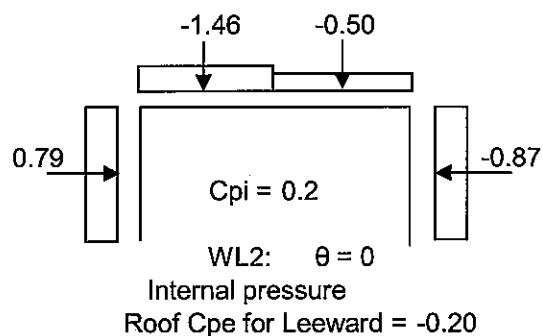
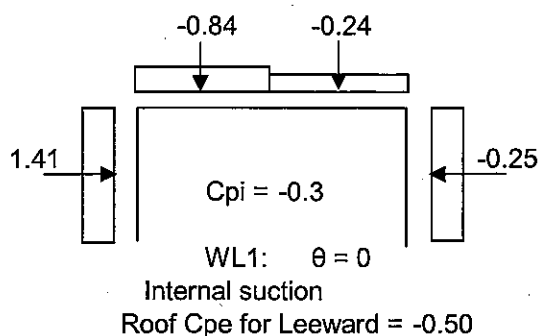
$$\begin{aligned}
 D &= 14.65 \text{ m} \\
 H &= 13 \text{ m} \\
 D/H &= 1.13
 \end{aligned}$$

Elements	Effective Wind Speed V_e m/s	Dynamic Wind Pressure q kN/m ²	External Pressure Coefficient, C_{pe}			
			$\theta = 0$ (normal to eave)		$\theta = 90$ (normal to gable end)	
			Windward	Leeward		
Roof	45	1.24	-0.98	-0.59 -0.50	-0.7 (Zone C)	0.2 (Zone D)
Wall	45	1.24	0.84	-0.5 -0.5	-0.8 (Zone B)	-0.8 (Zone B)



Site:	Dubai Metro Project - Rashidiya Diesel Engine Workshop - Main Hall Wind Load Computation	Date:	Jun '07	Job No.:	68010.027
		Designed by:	Rudy	Sheet No.:	

Wind Load = 1 x q x (Cpe - Cpi) kN/m (For unit width)



Site: Dubai Metro Project - Rashidiya
Diesel Engine Workshop - Main Hall
Earthquake Loading

Date: Jun '07

Job No.: 68010.01

Designed by: Rudy

Sheet No.:

Estimation of Earthquake Loading

Total Dead Load on the Building Column W = Given below (including part of Live Load)
Height of the Column h_n = Given below

Design Information:

- 1) Seismic Zone = 2A (from Design Brief)
2) Occupancy Category = 1 (Tanle- 16-K, Conservatively adopt 1)
3) Lateral Force procedure = **Static Procedure** (Section 1629.8 seismic zone 2A, structures not more than 5 stories)

Earthquake Loads

- 4) **Base Shear** $V_{base-S} = (C_v \times I \times W) / (R \times T)$ (Section 1630.2.1)
Seismic Coefficient $C_v = 0.32$ (Table 16-R assuming soil profile type S_D)
Importance Factor $I = 1.00$ (Table 16-K)
Numeric. Coef: Strength & Ductility $R = 4.50$ (Table 16-N for OMRT steel only)
Elastic Fundamental Period of Vibration $T = C_t (h_n)^{3/4}$ Sec.
Numerical Coefficient $C_t = 0.0853$ (Steel frame t.b.c)

- 5) **Limiting Values of Base shear** $V_{base-S-Max} = (2.5 \times C_a \times I \times W) / R = 345.29$
 $V_{base-S-Min} = 0.11 \times C_a \times I \times W = 68.37$

Design Earthquake Loads

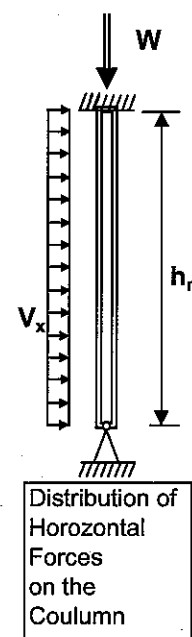
(Section 1630.6)

- $E_x = \rho \times \text{Min. } (E_h \text{ or } V_{base-S-Max}) + E_v$
 $E_{x-max} = \Omega \times \text{Min. } (E_h \text{ or } V_{base-S-Max})$
 $V_{base-S} = E_h$ = Earthquake load in the element due to the base shear, V , as set out above (for non-structural components refer to section 1632)
Vert. component of EQ ground motion $E_v = 0.5 C_a I D$ (Table 16-K)
Seismic Coefficient $C_a = 0.22$ (Table 16-Q, for soil type S_D)
Seismic Force Amplification Factor $\Omega = 2.80$ (Table 16-N)
Dead Load on the element $W = D$ = Given below
 $\rho = 2 - 6.1 / (r_{max} \sqrt{A_B}) = 1.25$ (conservative) ($1.0 \leq \rho \leq 1.5$)
 r_{max} = The maximum element-storey shear ratio
 A_B = Ground floor area of the structure

Distribution of Horizontal Forces:Distribution of Horizontal Force on the Column $V_x = E_x / h_n$

Col	Gr. Load, W , kN	Ht h_n , m	Time, T sec.	V_{base-S} kN	V_{base-S} kN	E_v kN	E_x kN	E_{x-max} kN	V_x kN/m	V_{x-max} kN/m
	2825	13.00	0.58	344.0	345.29	310.764	740.8	963.2	57.0	74.1

Load per column = $13 \times 57 / (13 \times 2 + 6)$
= 23.1 kN





Site:	Dubai Metro Project - Rashidiya	Date:	Jun '07	Job No.:	68010.027
	Diesel Engine Workshop - Office Area				
Loadings		Designed by:	Rudy	Sheet No.:	

Roof Loadings :**DL** W_{DL} :

Sheeting	0.1 kPa		
Purlins	0.1 kPa		
Insulation	0.3 kPa		
M&E Finishes	0.5 kPa		
	<u>1 kPa</u>	x 1 =	1 kN/m ²

LL

0.6 kPa	x 1 =	0.6 kN/m ²
---------	-------	-----------------------

P_{LL1} : A/C equipment S/W
 = 5 kN per 8m bays

Total no of AC equipment = 12
 Therefore, total P_{LL2} = 60 kN

Notional horizontal load, Ph

(Load Case of 1.4DL + 1.6LL)

Tributary load area for the roof

= $(10*8+2*5.6)*(5.4)$ = 492.48 m²

Total Dead Load

492 kN

(Including overhang roof)

Total Live Load

355 kN

(Including A/C equipment & overhang roof)

Tributary load area for the wall

= $5.4*[(10*8+2*5.6)+2*5.4]$ = 550.8 m²

Total Dead Load

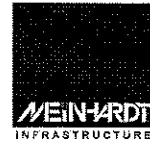
275 kN

(excluding the M&E Finishes)

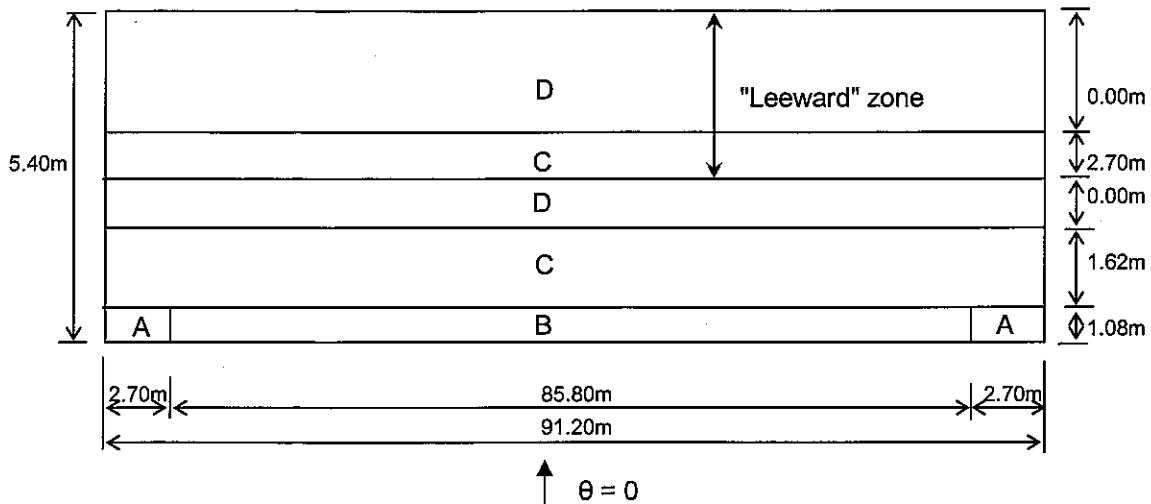
Total Live Load

Therefore, total ULS load, W_u = 1644 kN

Ph = 1.5 % of W_u
 = 24.7 kN



Site:	Dubai Metro Project - Rashidiya	Date:	Jun '07	Job No.:	68010.027
	Diesel Engine Workshop - Office Area				
	Wind Load Computation	Designed by:	Rudy	Sheet No.:	

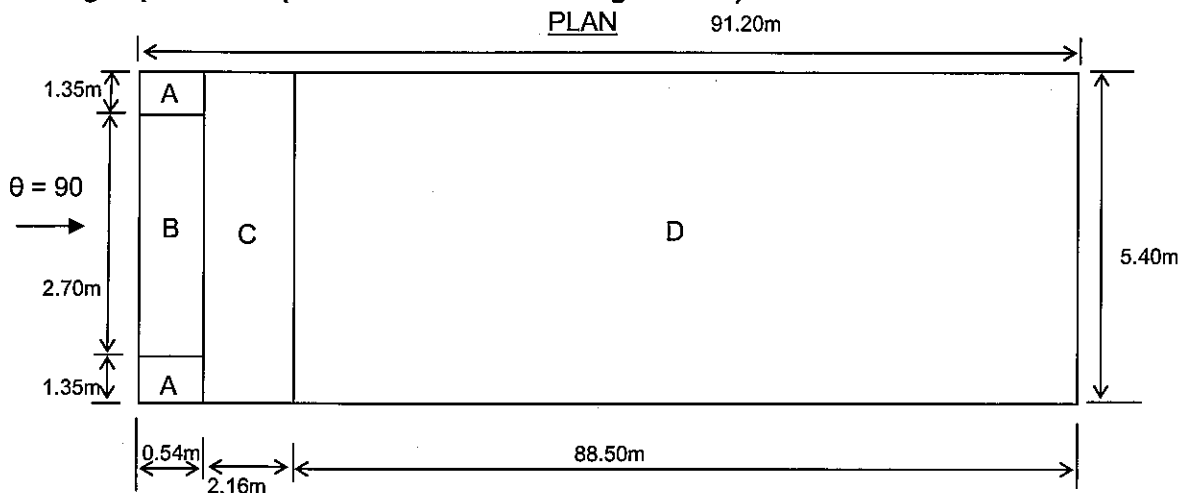
Average Cpe for roof (Wind direction normal to eave)
PLAN


Design as flat roof (with sharp eaves)

$$\begin{aligned}
 B &= 91.2 \text{ m} \\
 H &= 5.4 \text{ m} \\
 b &= \text{smaller of } (B, 2H) \\
 &= 2 \times 5.4 \\
 &= 10.8 \text{ m}
 \end{aligned}$$

Zone	Windward		Leeward		
	Area (m ²)	Cpe	Area	Cpe1	Cpe2
A	5.8	-2			
B	92.7	-1.4			
C	147.7	-0.7	246.2	-0.7	-0.7
D	0.0	-0.2	0.0	-0.2	0.2
Total	246.2		246.2		
Average		-0.99		-0.70	-0.70

Site: Dubai Metro Project - Rashidiya	Date: Jun '07	Job No.: 68010.027
Diesel Engine Workshop - Office Area		
Wind Load Computation	Designed by: Rudy	Sheet No.:

Average Cpe for roof (Wind direction normal to gable end)


Design as flat roof (with sharp eaves)

$$\begin{aligned}
 B &= 5.4 \text{ m} \\
 H &= 5.4 \text{ m} \\
 b &= \text{smaller of } (B, 2H) \\
 &= 1 \times 5.4 \\
 &= 5.4 \text{ m}
 \end{aligned}$$

For maximum uplift, assume zone C coefficient =>

$$C_{pe} = -0.7$$

For maximum downward force, assume zone D coefficient =>

$$C_{pe} = 0.2$$

Wind Load Computation

$$\begin{aligned}
 D &= 5.40 \text{ m} \\
 H &= 5.4 \text{ m} \\
 D/H &= 1.00
 \end{aligned}$$

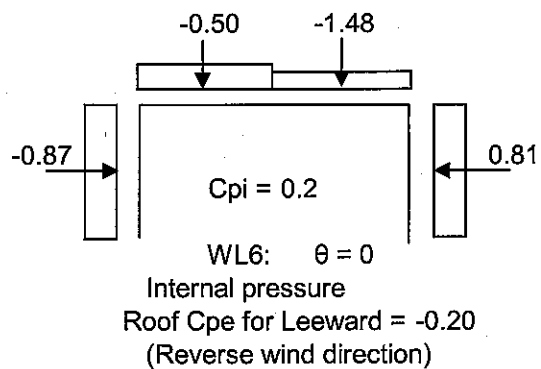
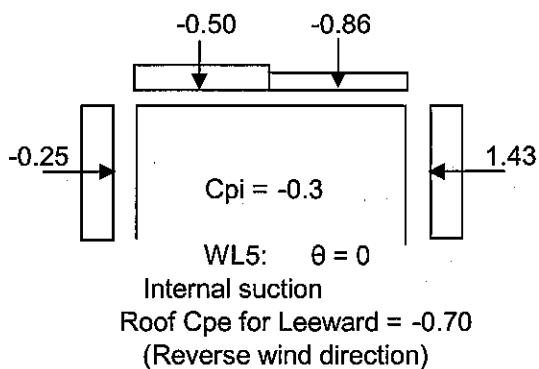
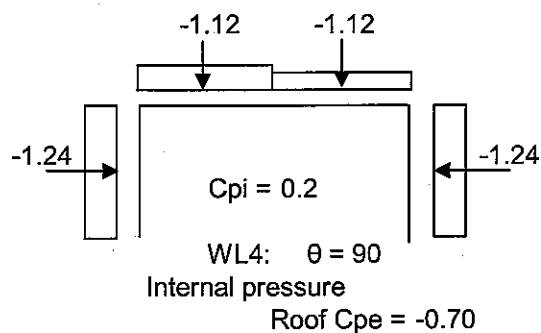
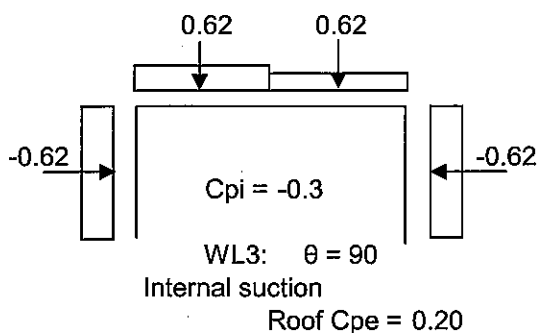
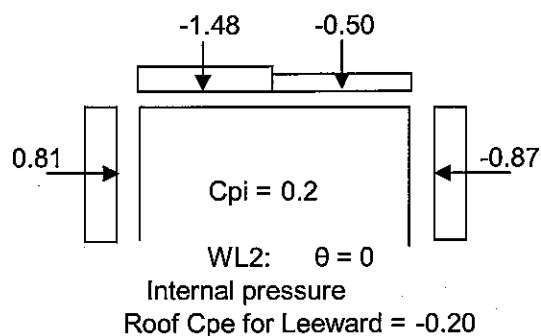
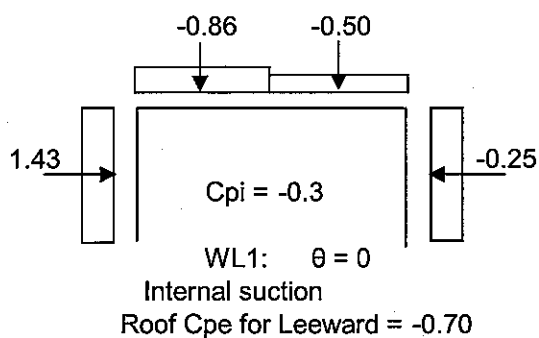
Elements	Effective Wind Speed V_e m/s	Dynamic Wind Pressure q kN/m ²	External Pressure Coefficient, C_{pe}			
			$\theta = 0$ (normal to eave)		$\theta = 90$ (normal to gable end)	
			Windward	Leeward		
Roof	45	1.24	-0.99	-0.70 -0.70	-0.7 (Zone C)	0.2 (Zone D)
Wall	45	1.24	0.85	-0.5 -0.5	-0.8 (Zone B)	-0.8 (Zone B)

Site: Dubai Metro Project - Rashidiya
 Diesel Engine Workshop - Office Area
 Wind Load Computation

Date: Jun '07 Job No.: 68010.027

Designed by: Rudy Sheet No.:

Wind Load = 1 x q x (Cpe - Cpi) kN/m (For unit width)



Site: Dubai Metro Project - Rashidiya
Diesel Engine Workshop - Office Area
Earthquake Loading

Date: Jun '07

Job No.: 68010.01

Designed by: Rudy

Sheet No.:

Estimation of Earthquake Loading

Total Dead Load on the Building Column W = Given below (including part of Live Load)
Height of the Column h_n = Given below

Design Information:

- 1) Seismic Zone = 2A (from Design Brief)
2) Occupancy Category = 1 (Table- 16-K, Conservatively adopt 1)
3) Lateral Force procedure = Static Procedure (Section 1629.8 seismic zone 2A, structures not more than 5 stories)

Earthquake Loads

- 4) **Base Shear** $V_{base-S} = (C_v \times I \times W) / (R \times T)$ (Section 1630.2.1)
Seismic Coefficient $C_v = 0.32$ (Table 16-R assuming soil profile type S_D)
Importance Factor $I = 1.00$ (Table 16-K)
Numeric. Coef: Strength & Ductility $R = 4.50$ (Table 16-N for OMRT steel only)
Elastic Fundamental Period of Vibration $T = C_t (h_n)^{3/4}$ Sec.
Numerical Coefficient $C_t = 0.0853$ (Steel frame t.b.c)

- 5) **Limiting Values of Base shear** $V_{base-S-Max} = (2.5 \times C_a \times I \times W) / R = 115.58$
 $V_{base-S-Min} = 0.11 \times C_a \times I \times W = 22.88$

Design Earthquake Loads

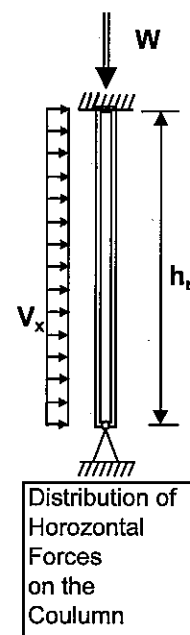
(Section 1630.6)

- $E_x = \rho \times \text{Min. } (E_h \text{ or } V_{base-S-Max}) + E_v$
 $E_{x-max} = \Omega \times \text{Min. } (E_h \text{ or } V_{base-S-Max})$
 $V_{base-S} = E_h = \text{Earthquake load in the element due to the base shear, } V, \text{ as set out above (for non-structural components refer to section 1632)}$
Vert. component of EQ ground motion $E_v = 0.5 C_a I D$ (Table 16-K)
Seismic Coefficient $C_a = 0.22$ (Table 16-Q, for soil type S_D)
Seismic Force Amplification Factor $\Omega = 2.80$ (Table 16-N)
Dead Load on the element $W = D = \text{Given below}$
 $\rho = 2 - 6.1 / (r_{max} \sqrt{A_B}) = 1.25$ (conservative) ($1.0 \leq \rho \leq 1.5$)
 $r_{max} = \text{The maximum element-storey shear ratio}$
 $A_B = \text{Ground floor area of the structure}$

Distribution of Horizontal Forces:Distribution of Horizontal Force on the Column $V_x = E_x / h_n$

Col	Gr. Load, W, kN	Ht h_n , m	Time, T sec.	V_{base-S} kN	V_{base-S} kN	E_v kN	E_x kN	E_{x-max} kN	V_x kN/m	V_{x-max} kN/m
	946	5.40	0.30	222.5	115.58	104.019	248.5	323.6	46.0	59.9

$$\begin{aligned} \text{Load per column} &= 5.4 \times 46 / (13) \\ &= 19.1 \text{ kN} \end{aligned}$$





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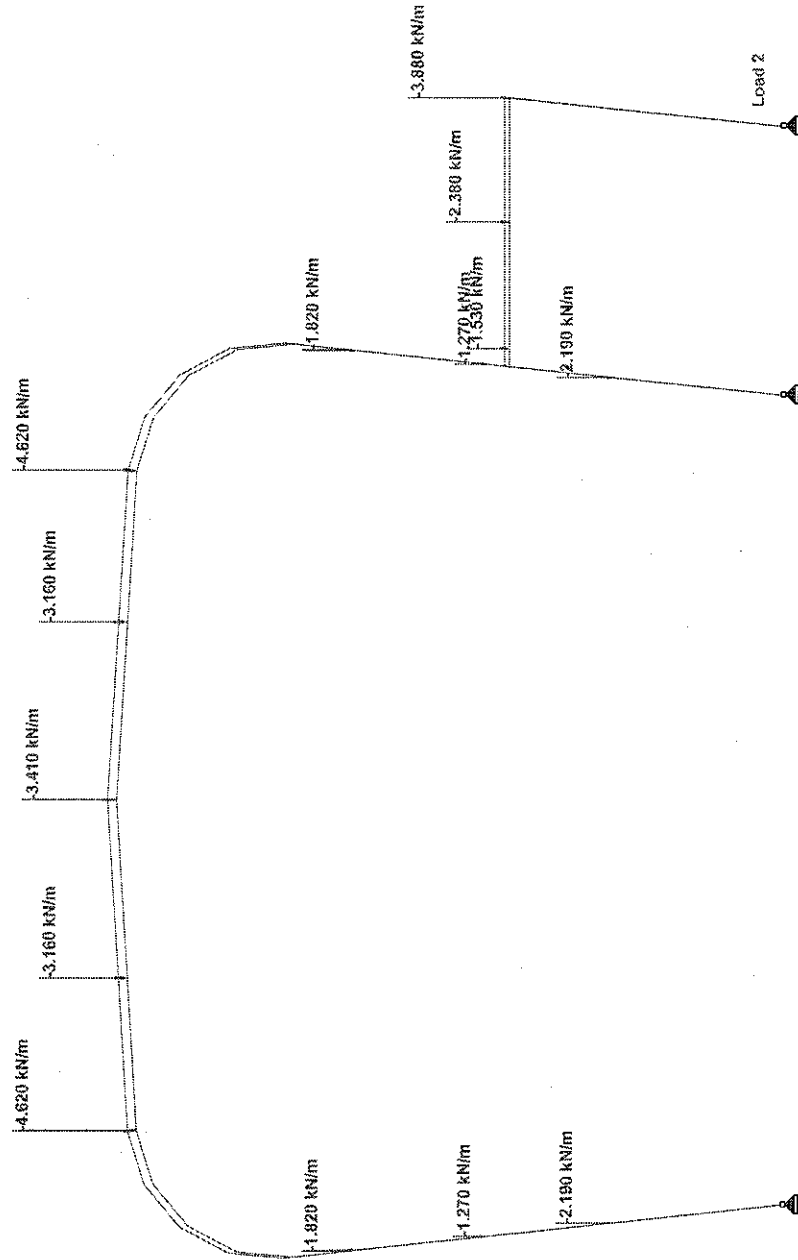
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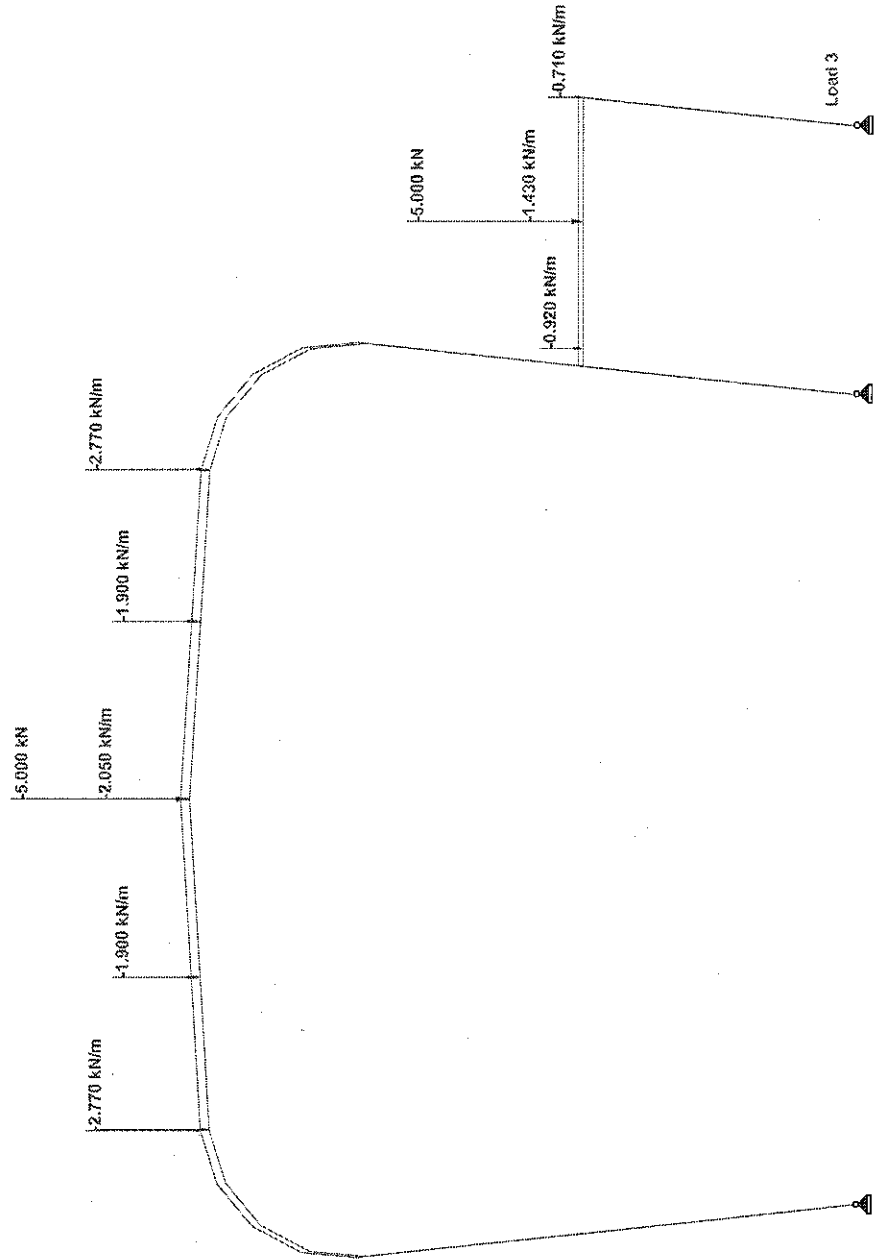
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File Diesel Engine Workshop

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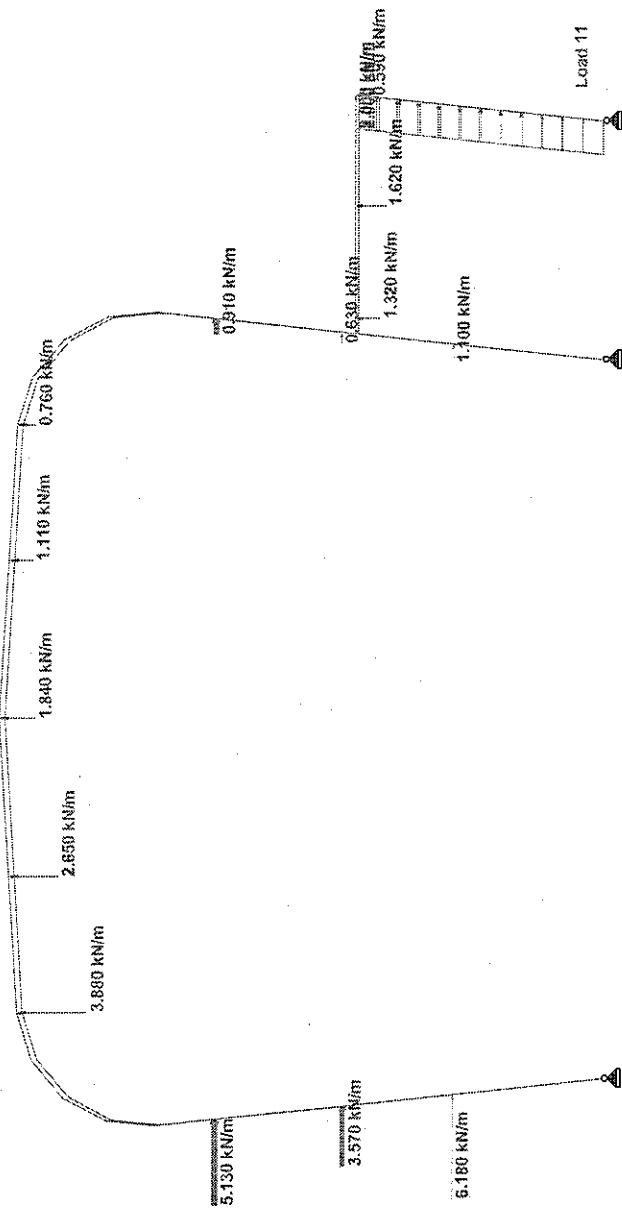
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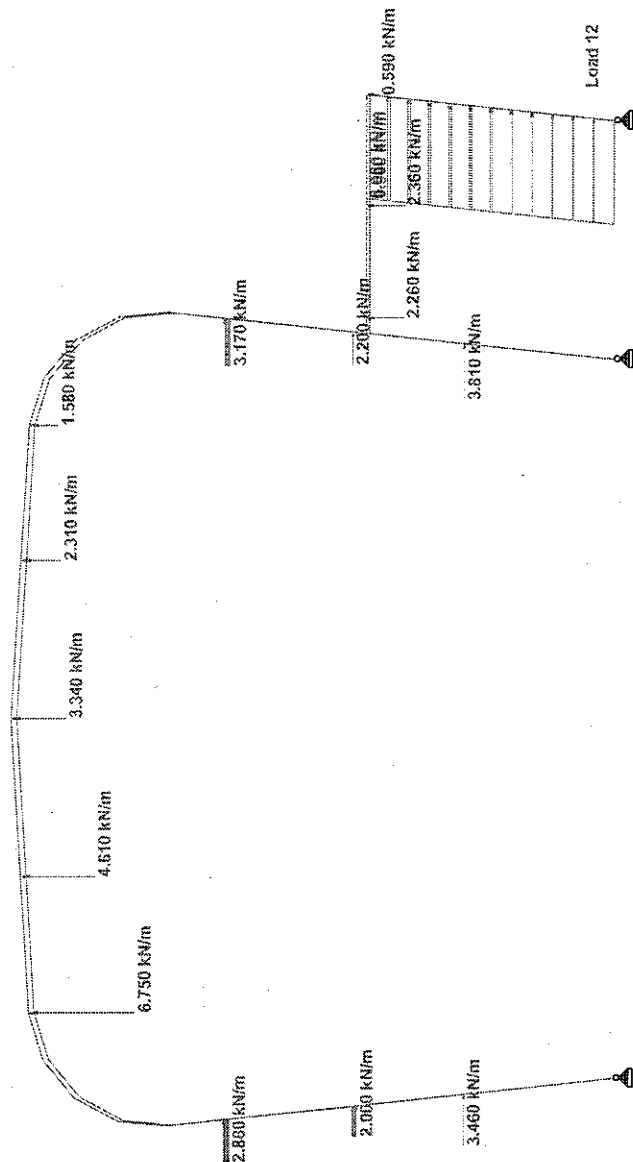
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Load 12



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Diesel Engine Workshop

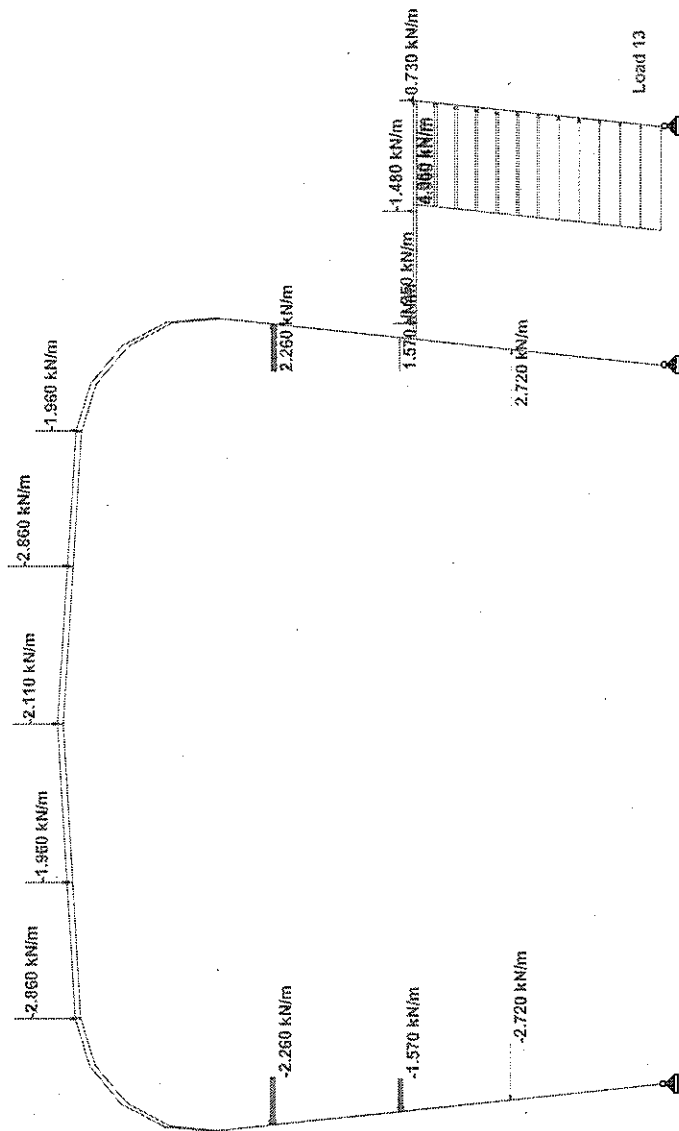
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Job No

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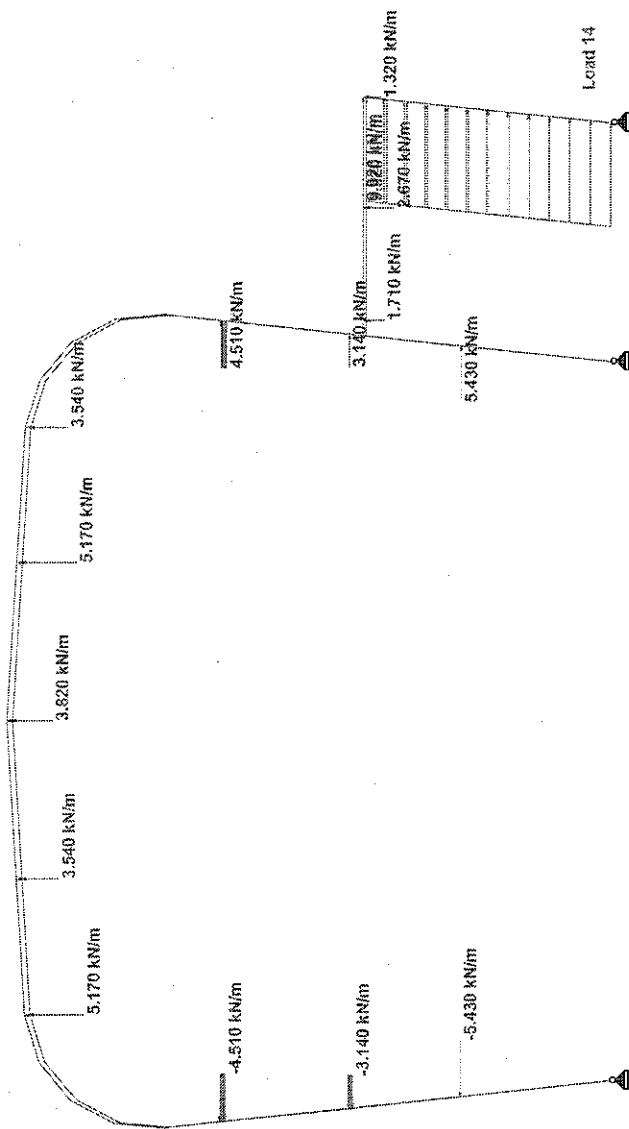
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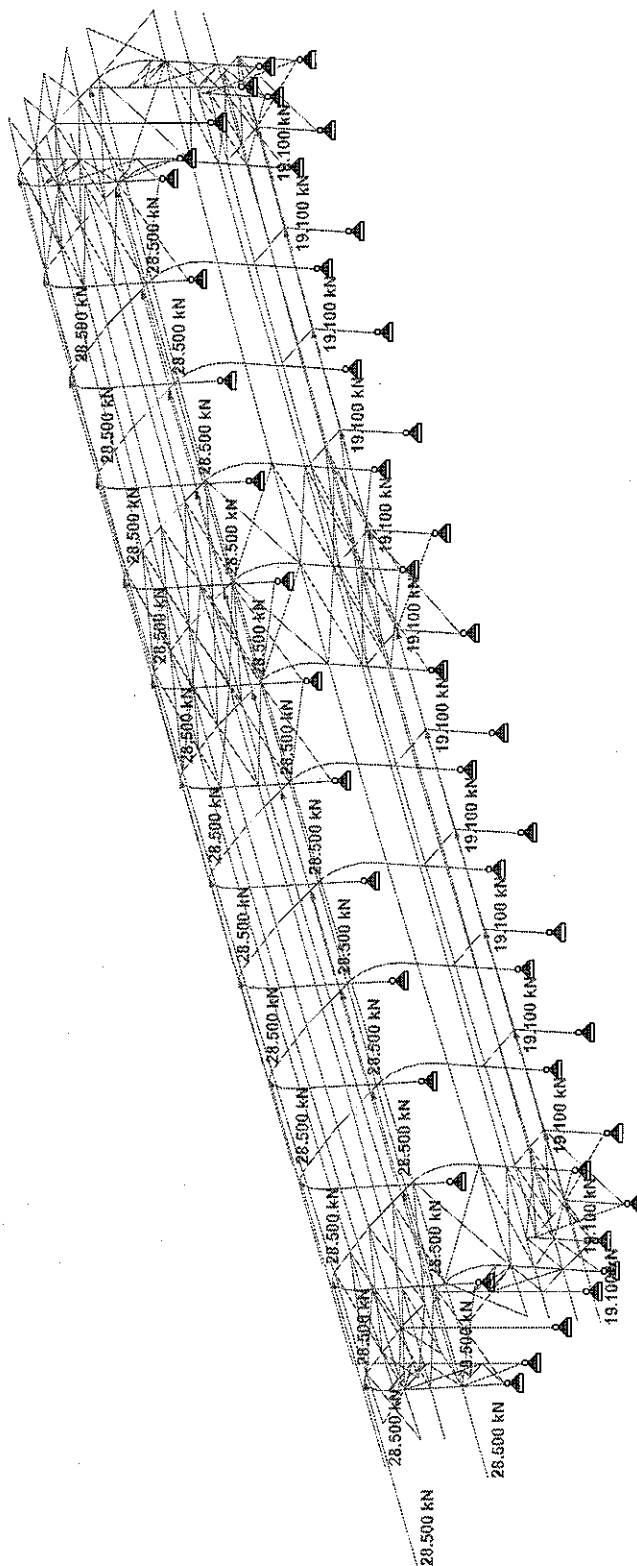
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Load 21



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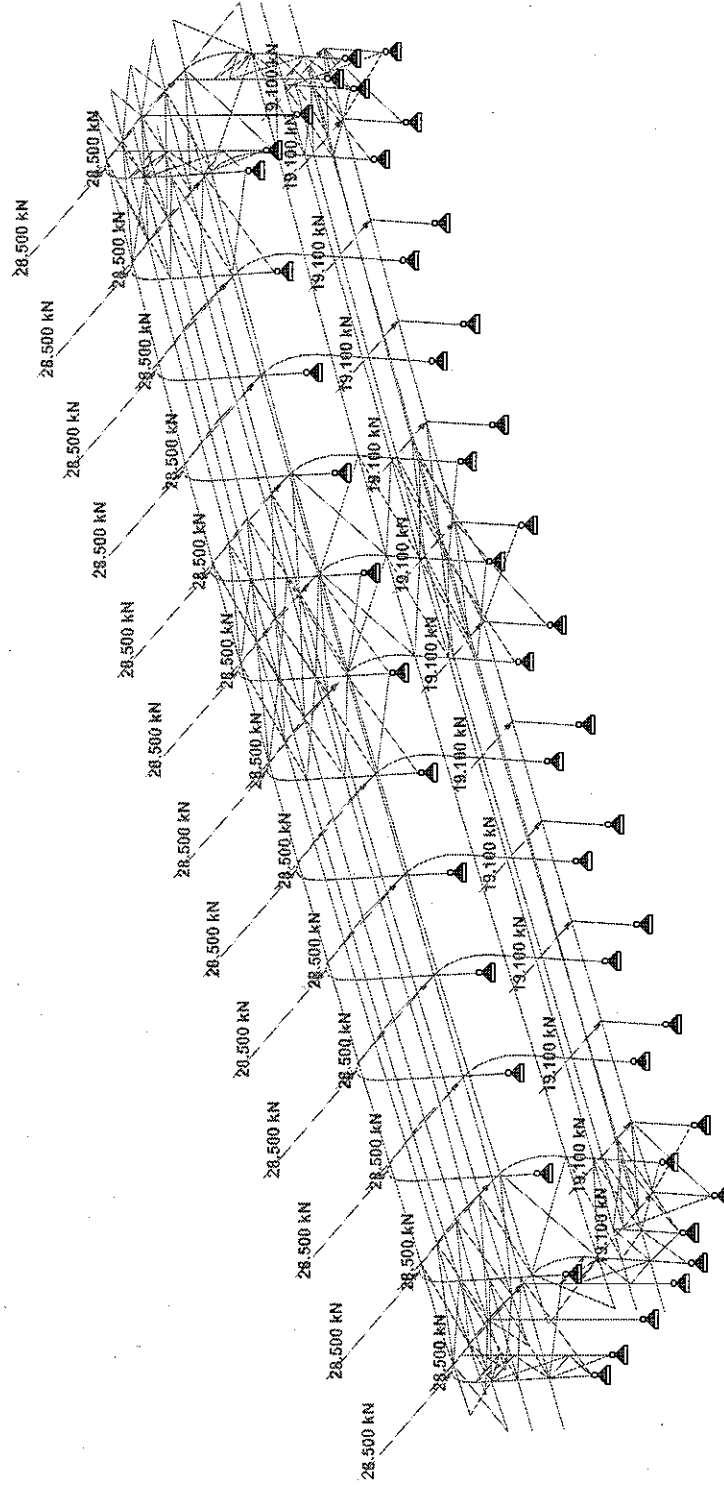
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Load 22

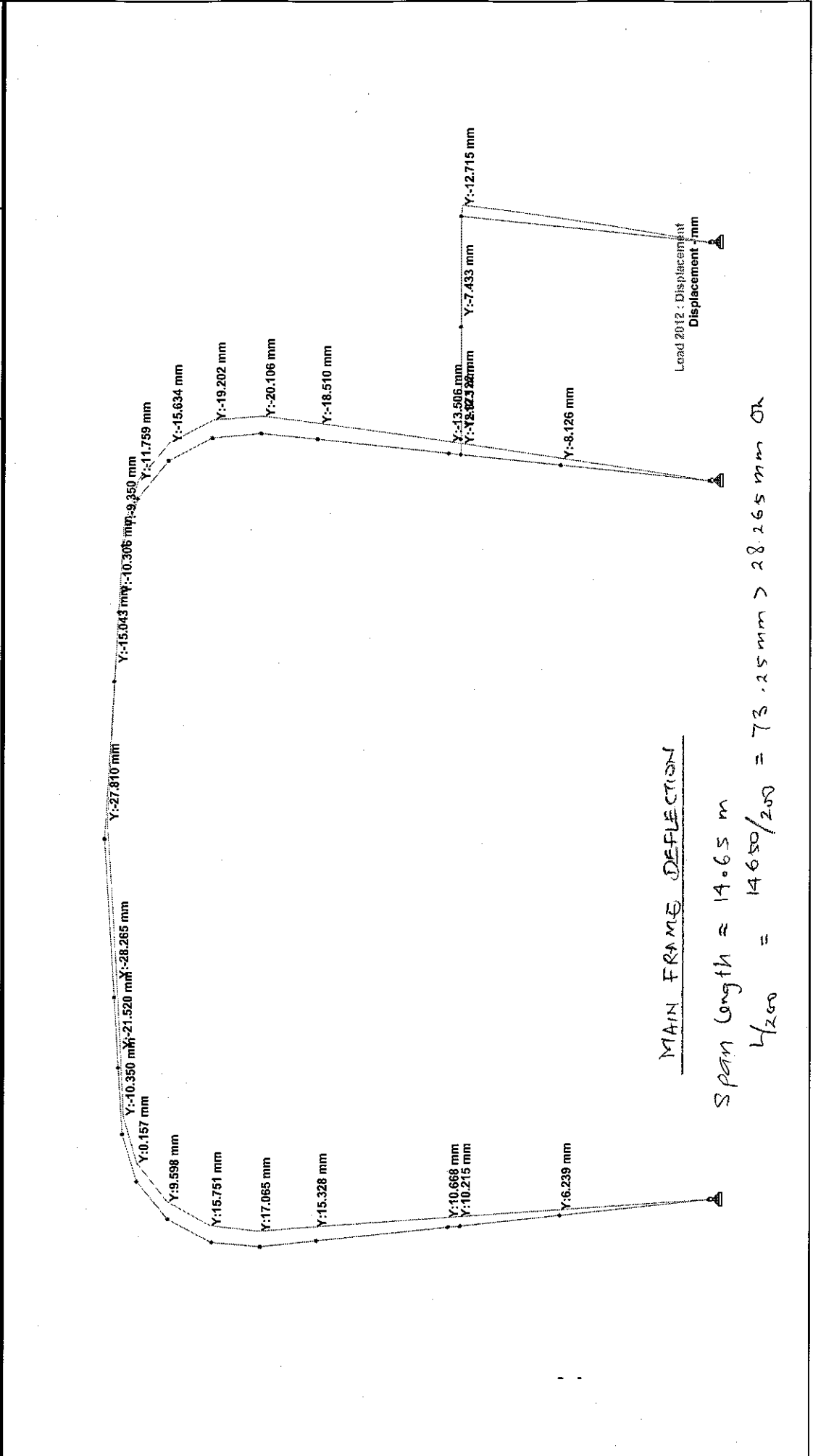


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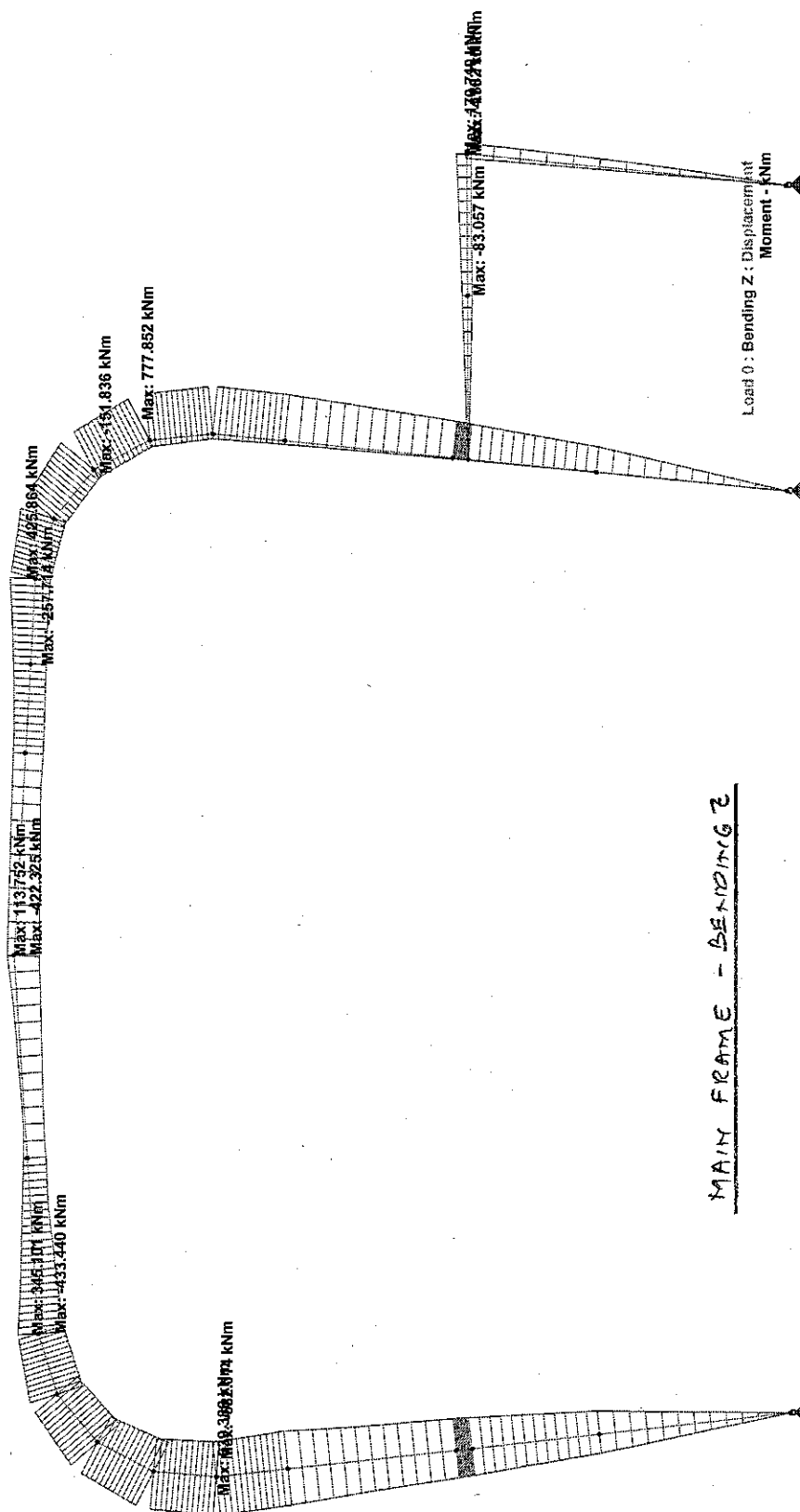


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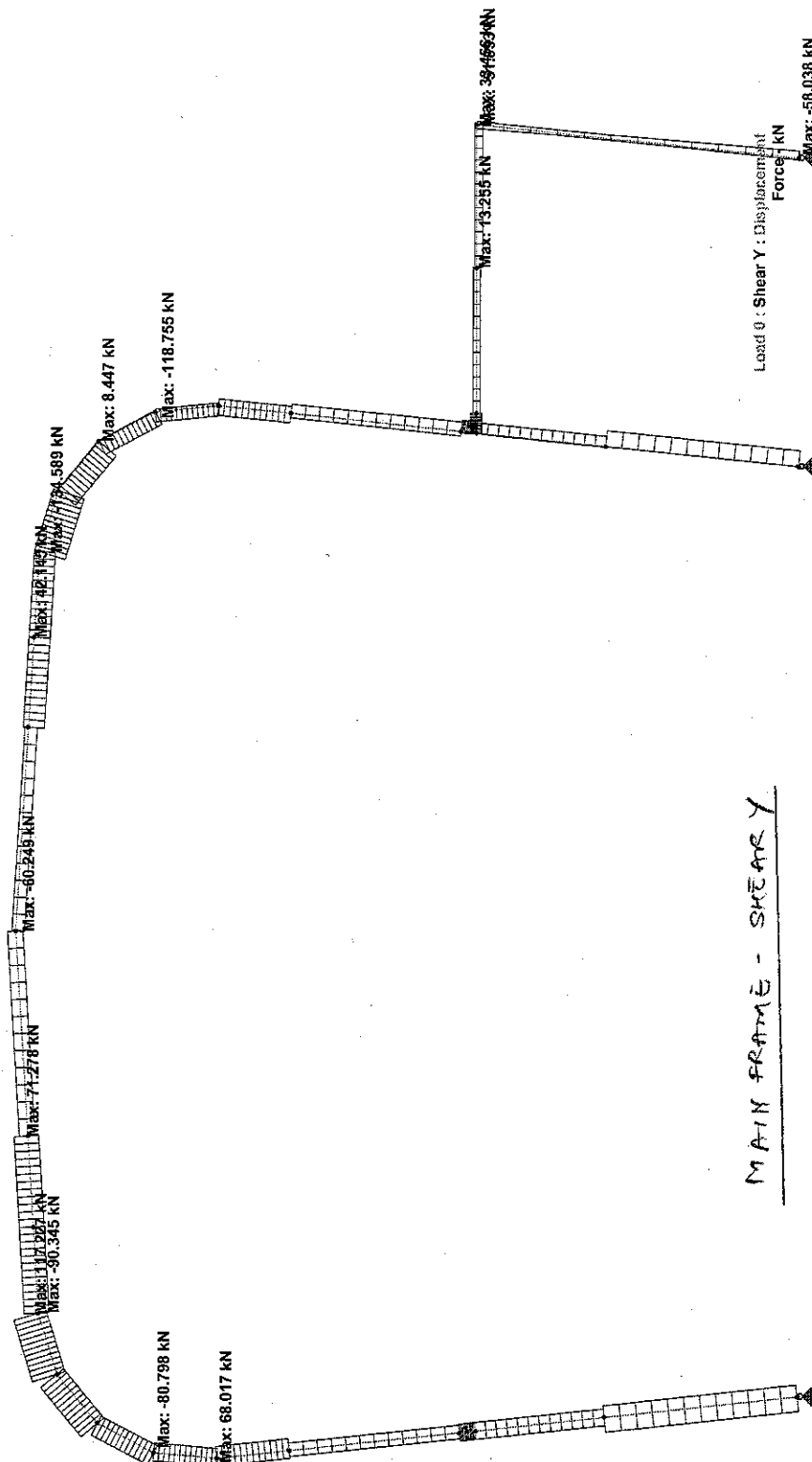
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MAIN FRAME - SHEAR Y



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MAIN HALL - SECONDARY BEAM - BENDING Z



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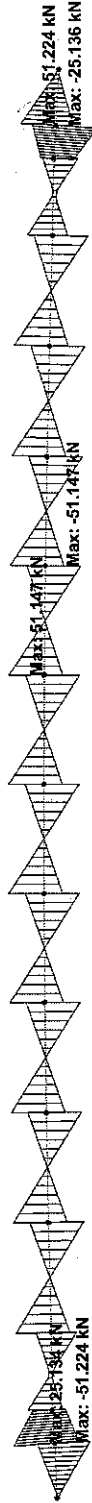
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MAIN HALL - SECONDARY BEAM - SHEAR Y

Load 8: Shear Y
Force - kN



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OFFICE AREA - SECONDARY BEAM - BENDING Z

Load 0 : Bending Z
Moment - kNm



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OFFICE AREA - SECONDARY BEAM - SHEAR

Load 0 : Shear Y
Force - kN

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162.	2125	24.000	0.000	7.715					
163.	2126	24.000	0.000	7.715;	2202	32.000	3.200	-8.051	4
164.	2201	32.000	0.000	-7.715;	2202	32.000	3.200	-8.051	4
165.	2203	32.000	5.151	-8.557;	2204	32.000	5.600	-8.304	
166.	2205	32.000	8.071	-8.364;	2206	32.000	9.271	-8.690	
167.	2207	32.000	10.320	-8.593;	2208	32.000	11.253	-8.103	
168.	2209	32.000	11.329	-7.603;	2210	32.000	12.739	-6.287	
169.	2211	32.000	11.329	-7.603;	2212	32.000	12.414	-6.287	
170.	2213	32.000	11.329	-7.603;	2214	32.000	12.414	-6.287	
171.	2215	32.000	11.329	-7.603;	2216	32.000	12.414	-6.287	
172.	2217	32.000	11.329	-7.603;	2218	32.000	11.529	8.787	
173.	2219	32.000	11.329	-7.603;	2220	32.000	9.271	8.690	
174.	2221	32.000	8.071	8.564;	2222	32.000	5.600	8.304	
175.	2223	32.000	5.151	8.267;	2224	32.000	3.200	8.051	
176.	2225	32.000	0.000	7.715;					
177.	2227	32.000	0.000	7.715;	2302	40.000	3.200	-8.051	
178.	2303	40.000	0.000	5.033	-8.244;	2304	40.000	5.600	-8.304
179.	2305	40.000	7.859	-8.541;	2306	40.000	9.059	-8.667	
180.	2307	40.000	10.108	-8.571;	2308	40.000	11.040	-8.080	
181.	2309	40.000	11.714	-7.270;	2310	40.000	12.027	-6.264	
182.	2311	40.000	12.110	-4.900;	2312	40.000	12.201	-3.400	
183.	2313	40.000	12.407	0.000;	2314	40.000	12.027	6.264	
184.	2315	40.000	12.110	4.900;	2316	40.000	11.040	8.080	
185.	2317	40.000	11.714	7.270;	2318	40.000	9.059	8.667	
186.	2319	40.000	10.108	8.541;	2320	40.000	5.600	8.304	
187.	2321	40.000	7.859	8.244;	2322	40.000	3.200	8.051	
188.	2323	40.000	5.033	8.244;	2324	40.000	3.200	8.051	
189.	2325	40.000	0.000	7.715;					
190.	2327	40.000	0.000	7.715;	2402	45.600	5.600	-8.304	
191.	2329	45.600	3.200	-8.051;	2404	45.600	11.949	6.264	
192.	2405	45.600	7.781	-8.540;	2410	45.600	12.407	0.000	
193.	2407	45.600	12.122	-3.400;	2413	45.600	12.407	0.000	
194.	2415	45.600	12.122	-3.400;	2416	45.600	11.949	6.264	
195.	2417	45.600	7.781	8.540;	2422	45.600	5.600	8.304	
196.	2421	45.600	4.926	8.237;	2424	45.600	3.200	8.051	
197.	2423	45.600	0.000	7.715;					
198.	2425	45.600	0.000	7.715;	2502	48.000	3.200	-8.051	
199.	2503	48.000	0.000	8.233;	2504	48.000	5.600	-8.304	
200.	2505	48.000	4.926	-8.233;	2506	48.000	8.080	-8.640	
201.	2507	48.000	7.600	-8.514;	2508	48.000	10.781	-8.053	
202.	2509	48.000	9.849	-8.543;	2508	48.000	11.939	-3.400	
203.	2511	48							

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UNITITLE	549. LOAD 1 SELFWEIGHT	550. SELFWEIGHT Y -1	551. SELFWEIGHT Y -1	552. LOAD 2 SOIL	553. MEMBER LOAD	554. *MAIN HALL	555. 4001 TO 4016 4401 TO 4416 UNI GY -4.62	556. 4101 TO 4116 4301 TO 4316 UNI GY -3.16	STAD SPACE
557. 4201 TO 4216 UNI GY -3.41	558. 5001 TO 5016 5501 TO 5516 UNI GY -2.19	559. 5101 TO 5116 5401 TO 5416 UNI GY -1.27	560. 5201 TO 5216 5301 TO 5316 UNI GY -1.82	561. *OFFICE	562. 6001 TO 6012 UNI GY 1.53	563. 6101 TO 6112 UNI GY 2.33	564. 6201 TO 6212 UNI GY -3.88	565. LOAD 3 LL	566. MEMBER LOAD
567. *MAIN HALL	568. 4001 TO 4016 4401 TO 4416 UNI GY -2.77	569. 4101 TO 4116 4301 TO 4316 UNI GY -1.90	570. 4201 TO 4216 UNI GY -2.05	571. *OFFICE	572. 6001 TO 6012 UNI GY -0.92	573. 6101 TO 6112 UNI GY -1.43	574. 6201 TO 6212 UNI GY -0.71	575. *AIRCON LOAD	576. 4201 TO 4213 CON GY -5
577. 4203 CON GY -5	578. 4203 CON GY -5	579. 4214 CON GY -5	580. 6101 TO 6112 CON GY -5	581. *CRANE LOAD	582. JOINT LOAD	583. 1804 FY -236.8	584. 1804 FZ 35.5	585. LOAD 11 WL1	586. MEMBER LOAD
587. *MAIN HALL	588. 4001 TO 4016 UNI GY 3.88	589. 4101 TO 4116 UNI GY 1.85	590. 4201 TO 4216 UNI GY 0.74	591. 4401 TO 4416 UNI GY 1.11	592. 4301 TO 4316 UNI GY 1.11	593. 5001 TO 5016 UNI GZ 6.18	594. 5101 TO 5116 UNI GZ 6.18	595. 5201 TO 5216 UNI GZ 5.13	596. 5301 TO 5316 UNI GZ 5.13
597. 5401 TO 5416 UNI GZ 0.91	598. 5501 TO 5516 UNI GZ 0.63	599. 5601 TO 5612 UNI GZ 1.10	600. *OFFICE	601. 6001 TO 6012 UNI GY 1.32	602. 6101 TO 6112 UNI GY 1.62	603. 6201 TO 6212 UNI GY 0.59	604. 1233 1433 1533 1633 1733 1833 1933 2033 2133 -	605. LOAD 12 WL2	606. MEMBER LOAD
607. 4001 TO 4016 UNI GY 6.75	608. 4101 TO 4116 UNI GY 4.61	609. 4201 TO 4216 UNI GY 3.34	610. 4301 TO 4316 UNI GY 1.58	611. 4401 TO 4416 UNI GY 2.31	612. 5001 TO 5016 UNI GZ 3.46	613. 5101 TO 5116 UNI GZ 3.46	614. 5201 TO 5216 UNI GZ 3.46	615. 5301 TO 5316 UNI GZ 3.46	616. 5401 TO 5416 UNI GZ 2.00
617. 5501 TO 5516 UNI GZ 3.81	618. 5601 TO 5612 UNI GY 2.26	619. 5701 TO 5712 UNI GY 2.36	620. 5801 TO 5812 UNI GY 0.59	621. *OFFICE	622. 6001 TO 6012 UNI GY 2.26	623. 6101 TO 6112 UNI GY 2.36	624. 6201 TO 6212 UNI GY 0.59	625. 6301 TO 6312 UNI GY 0.59	626. 6401 TO 6412 UNI GY 0.59
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636. 5001 TO 5016 UNI GZ -2.72
637. 5101 TO 5116 UNI GZ -1.57
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639. 5301 TO 5316 UNI GZ 2.26
640. 5401 TO 5416 UNI GZ 1.57
641. 5501 TO 5516 UNI GZ 2.72
642. *OFFICE
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644. 6101 TO 6112 UNI GY -1.48
645. 6201 TO 6212 UNI GY -0.73
646. 1233 2433 UNI GZ 2.48
647. 1333 1433 1533 1633 1733 1833 1933 2033 2133 -
648. 2233 2333 UNI GZ 4.96
650. LOAD 14 W/L4 (INTERNAL PRESSURE, WIND ANGLE 90)
651. MEMBER LOAD
652. 4001 TO 4016 UNI GY 5.17
653. 4101 TO 4116 UNI GY 3.54
654. 4201 TO 4216 UNI GY 3.82
655. 4301 TO 4316 UNI GY 3.54
656. 4401 TO 4416 UNI GY 5.17
657. 5001 TO 5016 UNI GZ -3.43
658. 5101 TO 5116 UNI GZ -3.43
659. 5201 TO 5216 UNI GZ 4.51
660. 5301 TO 5316 UNI GZ 4.51
661. 5401 TO 5416 UNI GZ 3.14
662. 5501 TO 5516 UNI GZ 5.43
663. *OFFICE
664. 6001 TO 6012 UNI GY 1.71
665. 6101 TO 6112 UNI GY 2.67
666. 6201 TO 6212 UNI GY 1.32
667. 1233 2433 UNI GZ 4.96
668. 1333 1433 1533 1633 1733 1833 1933 2033 2133 -
669. 2233 2333 UNI GZ 9.92
671. 2433 2533 UNI GZ 9.92
672. TOWN LOADING (EARTHQUAKE LOAD)
673. 1109 1115 1309 1315 1409 1415 1509 1515 1609 1615 1709 1715 -
674. 1809 1815 1909 1915 2009 2015 2109 2115 2209 2215 2309 2315 -
STAAD SPACE
675. 2509 2515 FX 28.5
676. 1233 1333 1433 1533 1633 1733 1833 1933 2033 2133 2233 2333 -
677. 2433 FX 19.1
679. LOAD 22 EQ1 (EARTHQUAKE LOAD)
680. JOINT LOAD
681. 1009 1015
682. 1809 1815 1909 1915 2009 2015 2109 2115 2209 2215 2309 2315 -
683. 2509 2515 Fz 28.5
684. 1333 1433 1533 1633 1733 1833 1933 2033 2133 2233 2333 -
687. LOAD COMB 1001 1.40L + 1.6LL
688. 1.4 2 1 4 1 1.6DL + 1.4DL + 1.6LL
689. LOAD COMB 1011 1.2DL + 1.2LL + 1.2WL1
691. 1.1 2 1 2 3 1.2 1.1 1.2
692. LOAD COMB 1111 1.4DL + 1.4WL1
693. 1.1 4 2 1 4 1 1.4
694. LOAD COMB 1211 1.0DL + 1.4WL1
695. 1.1 0 2 1 0 1 1.4
696. LOAD COMB 1012 1.2DL + 1.2LL + 1.2WL2
697. 1.1 2 2 1 2 3 1.2 1.2 1.2
698. LOAD COMB 1112 1.4DL + 1.4WL2
700. 1.1 4 2 1 4 1 2 1.4
701. LOAD COMB 1212 1.0DL + 1.4WL2
702. 1.1 0 2 1 0 1 1.4
703. LOAD COMB 1013 1.4
704. 1.1 2 1 2 3 1.2 1.2 1.2
705. LOAD COMB 1113 1.4DL + 1.4WL3
706. LOAD COMB 1213 1.0DL + 1.4WL3
707. 1.1 0 2 1 0 1 3 1.4
708. LOAD COMB 1014 1.2DL + 1.2LL + 1.2WL4
709. 1.1 2 2 1 2 3 1.2 1.2 1.2
710. LOAD COMB 1114 1.4DL + 1.4WL4
711. 1.1 4 2 1 4 1 4
712. LOAD COMB 1214 1.0DL + 1.4WL4
713. 1.1 0 2 1 0 1 4 1.4
716. LOAD COMB 1021 1.2DL + 1.0EL1 + 1.0EQ1
717. LOAD COMB 1121 0.5 21 1.0
718. 1.1 2 2 1 2 3 1.2 1.2 1.2
719. LOAD COMB 1221 1.2DL + 1.0EL1 + 1.0EQ2
720. 1.1 2 2 1 2 3 0.5 21 1.0
721. LOAD COMB 1321 1.2DL + 1.0EL1 - 1.0EQ2
722. 1.1 2 2 1 2 3 0.5 22 -1.0
723. LOAD COMB 1221 0.9DL + 1.0EQ1
724. 1.0 9 2 0 9 21 1.0
725. LOAD COMB 1222 0.9DL + 1.0EQ2

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Untitled

726. 1 0.9 2 0.9 22 1.0
727. LOAD COMB 1321 0.9DL - 1.0EQ1
728. 1 0.9 2 0.9 21 -1.0
729. LOAD COMB 1322 0.9DL - 1.0EQ2
730. 1 0.9 2 0.9 22 -1.0
731. *SLS
732. LOAD COMB 2001 1.0DL + 1.0LL
733. 1 0.9 2 0.9 21 1.0
734. LOAD COMB 2002 1.0DL + 0.5LL
* STAAD SPACE
735. 1 1.0 2 1.0 3 0.5
736. LOAD COMB 2011 DL + LL + WL1
737. 1 1.0 2 1.0 3 1.0 11 1.0
738. LOAD COMB 2012 DL + LL + WL2
739. 1 1.0 2 1.0 3 1.0 12 1.0
740. LOAD COMB 2013 DL + LL + WL3
741. 1 1.0 2 1.0 3 1.0 13 1.0
742. LOAD COMB 2014 DL + LL + WL4
743. 1 1.0 2 1.0 3 1.0 14 1.0
745. PERFORM ANALYSIS

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Untitled

752. PARAMETER 1
753. CODE BS5950
754. PY 345000 ALL
755. CHECK CODE ALL
STEEL DESIGN

STAAD PRO CODE CHECKING - (BSI)

PROGRAM CODE REVISION V2.9_5950-1_2000

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ALL UNITS ARE - KN METER (UNLESS OTHERWISE NOTED)

MEMBER	TABLE	RESULT/ FX	CRITICAL COND/ MY	RATIO/ MZ	LOADING/ LOCATION
1101	TAPERED	PASS	BS-4.7 (C)	0.208	2011
1102	TAPERED	111.40 C	BS-4.7 (C)	0.326	2013
1103	TAPERED	60.10 C	BS-4.7 (C)	0.365	2011
1104	TAPERED	59.31 C	BS-4.7 (C)	0.377	2012
1105	TAPERED	48.94 T	BS-4.8, 2.2	0.358	2012
1106	TAPERED	62.44 T	BS-4.8, 2.2	0.346	2012
1107	TAPERED	61.13 T	BS-4.8, 2.2	0.343	2012
1108	TAPERED	45.93 T	BS-4.8, 2.2	0.325	2012
1109	TAPERED	24.46 T	BS-4.8, 2.2	0.289	2001
1110	TAPERED	4.07 T	BS-4.8, 2.2	0.175	2012
1111	TAPERED	12.32 T	BS-4.8, 2.2	0.217	2012
1112	TAPERED	33.38 T	BS-4.8, 2.2	0.159	2013
1113	TAPERED	96.03 T	BS-4.8, 2.2	0.082	2013
1114	TAPERED	95.52 T	BS-4.8, 2.2	0.178	2012
1115	TAPERED	19.26 T	BS-4.7 (C)	0.305	2011
1116	TAPERED	17.87 C	BS-4.7 (C)	0.326	2011
1117	TAPERED	72.68 C	BS-4.7 (C)	0.394	2011
1118	TAPERED	117.66 C	BS-4.7 (C)	0.404	2011
1119	TAPERED	145.35 C	BS-4.7 (C)	0.404	2011
1120	TAPERED	150.18 C	BS-4.7 (C)	0.371	2012
1121	TAPERED	125.35 C	BS-4.7 (C)	0.399	2012
1122	TAPERED	208.02 C	BS-4.7 (C)	0.408	2012
1123	TAPERED	255.12 C	BS-4.7 (C)	0.349	2012
1124	TAPERED	255.90 C	BS-4.7 (C)	0.304	2012
1125	TAPERED	305.36 C	BS-4.7 (C)	0.269	2012
1222	TAPERED	PASS	BS-4.7 (C)	0.109	2013
1223	TAPERED	16.89 C	BS-4.7 (C)	0.257	2013

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1223	TAPERED	PASS	BS-4.7 (C)	0.113	2013
1224	TAPERED	26.36 C	BS-4.7 (C)	0.072	2012
1301	TAPERED	23.42 C	BS-4.7 (C)	0.229	2013
1302	TAPERED	292.02 C	BS-4.7 (C)	0.256	2013
1303	TAPERED	231.04 C	BS-4.7 (C)	0.274	2013
1304	TAPERED	230.36 C	BS-4.7 (C)	0.313	2013
1305	TAPERED	176.13 C	BS-4.7 (C)	0.346	2013
1306	TAPERED	158.63 C	BS-4.7 (C)	0.346	2013
1307	TAPERED	160.80 C	BS-4.7 (C)	0.344	2013
1308	TAPERED	149.08 C	BS-4.7 (C)	0.310	2013
1309	TAPERED	114.00 C	BS-4.7 (C)	0.226	2012
1310	TAPERED	49.25 C	BS-4.7 (C)	0.326	2012
1311	TAPERED	29.42 C	BS-4.7 (C)	0.307	2013
1312	TAPERED	37.58 C	BS-4.7 (C)	0.446	2013
1313	TAPERED	42.72 C	BS-4.7 (C)	0.441	2013
1314	TAPERED	42.75 C	BS-4.7 (C)	0.441	2013
1315	TAPERED	37.47 C	BS-4.7 (C)	0.443	2011
1316	TAPERED	45.93 C	BS-4.7 (C)	0.403	2011
1317	TAPERED	70.12 C	BS-4.7 (C)	0.455	2011
1318	TAPERED	111.33 C	BS-4.7 (C)	0.460	2011
1319	TAPERED	135.26 C	BS-4.7 (C)	0.455	2011
1320	TAPERED	139.82 C	BS-4.7 (C)	0.419	2012
1321	TAPERED	118.69 C	BS-4.7 (C)	0.419	2012
1322	TAPERED	155.35 C	BS-4.7 (C)	0.423	2012
1323	TAPERED	203.88 C	BS-4.7 (C)	0.380	2012
1324	TAPERED	198.48 C	BS-4.7 (C)	0.372	2012
1401	TAPERED	254.14 C	BS-4.7 (C)	0.292	2013
1402	TAPERED	367.12 C	BS-4.7 (C)	0.344	2013
1403	TAPERED	303.39 C	BS-4.7 (C)	0.358	2013
1404	TAPERED	302.92 C	BS-4.7 (C)	0.428	2013
1405	TAPERED	245.08 C	BS-4.7 (C)	0.418	2013
1406	TAPERED	184.79 C	BS-4.7 (C)	0.432	2013
1407	TAPERED	182.65 C	BS-4.7 (C)	0.427	2013
1408	TAPERED	160.13 C	BS-4.7 (C)	0.368	2013
1409	TAPERED	112.27 C	BS-4.7 (C)	0.241	2001
1410	TAPERED	52.51 C	BS-4.7 (C)	0.191	2001
1411	TAPERED	17.22 C	BS-4.7 (C)	0.369	2013
1412	TAPERED	0.27 T	BS-4.7 (C)	0.224	2013

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ALL UNITS ARE - KN METER (UNLESS OTHERWISE NOTED)

MEMBER	TABLE	RESULT/ FX	CRITICAL COND/ MY	RATIO/ MZ	LOADING/ LOCATION
1321	TAPERED	PASS	BS-4.7 (C)	0.423	2012
1322	TAPERED	155.35 C	BS-4.7 (C)	0.380	2012
1323	TAPERED	203.88 C	BS-4.7 (C)	0.372	2012
1324	TAPERED	198.48 C	BS-4.7 (C)	0.292	2013
1401	TAPERED	254.14 C	BS-4.7 (C)	0.344	2013
1402	TAPERED	367.12 C	BS-4.7 (C)	0.358	2013
1403	TAPERED	303.39 C	BS-4.7 (C)	0.428	2013
1404	TAPERED	302.92 C	BS-4.7 (C)	0.418	2013
1405	TAPERED	245.08 C	BS-4.7 (C)	0.432	2013
1406	TAPERED	184.79 C	BS-4.7 (C)	0.427	2013
1407	TAPERED	182.65 C	BS-4.7 (C)	0.368	2013
1408	TAPERED	160.13 C	BS-4.7 (C)	0.241	2001
1409	TAPERED	112.27 C	BS-4.7 (C)	0.191	2001
1410	TAPERED	52.51 C	BS-4.7 (C)	0.369	2013
1411	TAPERED	17.22 C	BS-4.7 (C)	0.224	2013

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1412	TAPERED	PASS	BS-4.3.6	0.579	2013	Untitled
1413	TAPERED	4.01 T	-0.09	338.28	3.41	0.531
1414	TAPERED	PASS	BS-4.3.6	0.578	2013	0.00
1415	TAPERED	3.69 T	0.09	338.28	0.00	0.489
1416	TAPERED	PASS	BS-4.3.6	0.341	2011	0.00
1417	TAPERED	37.32 C	-0.21	168.38	0.00	0.497
1418	TAPERED	PASS	BS-4.7 (C)	0.483	2011	0.00
1419	TAPERED	37.44 C	-0.01	293.20	1.39	-600.35
1420	TAPERED	PASS	BS-4.7 (C)	0.444	2011	0.00
1421	TAPERED	74.97 C	-0.02	433.32	1.05	0.430
1422	TAPERED	PASS	BS-4.7 (C)	0.505	2011	0.00
1423	TAPERED	120.38 C	-0.03	532.59	1.05	0.427
1424	TAPERED	PASS	BS-4.7 (C)	0.535	2011	0.00
1425	TAPERED	148.43 C	0.02	578.03	0.00	0.342
1426	TAPERED	PASS	BS-4.7 (C)	0.506	2011	0.00
1427	TAPERED	154.35 C	0.02	578.03	0.00	0.342

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§ STAAD SPACE

ALL UNITS ARE - KN METE (UNLESS OTHERWISE NOTED)

MEMBER	TABLE	RESULT/ FX	CRITICAL COND/ MY	RATIO/ NZ	LOADING/ LOCATION
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1420	TAPERED	PASS	BS-4.7 (C)	0.468	2013
1421	TAPERED	189.95 C	-0.04	569.66	0.00
1422	TAPERED	PASS	BS-4.7 (C)	0.475	2013
1423	TAPERED	250.02 C	0.89	549.46	0.00
1424	TAPERED	PASS	BS-4.7 (C)	0.408	2013
1425	TAPERED	309.87 C	1.00	453.45	0.00
1426	TAPERED	PASS	BS-4.7 (C)	0.407	2013
1427	TAPERED	349.57 C	-1.13	433.41	0.00
1428	TAPERED	PASS	BS-4.7 (C)	0.331	2013
1429	TAPERED	414.04 C	-0.27	304.08	0.00
1430	TAPERED	PASS	BS-4.7 (C)	0.295	2013
1431	TAPERED	365.86 C	-0.12	272.88	3.22
1432	TAPERED	PASS	BS-4.7 (C)	0.356	2013
1433	TAPERED	302.34 C	-0.22	385.63	2.05
1434	TAPERED	PASS	BS-4.7 (C)	0.364	2013
1435	TAPERED	299.70 C	-0.30	403.28	0.26
1436	TAPERED	PASS	BS-4.7 (C)	0.434	2013
1437	TAPERED	244.88 C	-0.18	396.12	0.75
1438	TAPERED	PASS	BS-4.7 (C)	0.431	2013
1439	TAPERED	184.48 C	-0.10	507.71	1.21
1440	TAPERED	PASS	BS-4.7 (C)	0.431	2013
1441	TAPERED	181.40 C	0.07	478.67	1.05
1442	TAPERED	PASS	BS-4.7 (C)	0.425	2013
1443	TAPERED	156.35 C	-0.07	478.67	0.00
1444	TAPERED	PASS	BS-4.7 (C)	0.362	2013
1445	TAPERED	108.80 C	-0.02	377.78	0.00
1446	TAPERED	PASS	BS-4.3.6	0.268	2001
1447	TAPERED	49.33 C	-0.03	188.95	1.05
1448	TAPERED	PASS	BS-4.3.6	0.398	2012
1449	TAPERED	3.39 T	0.06	242.93	0.00
1450	TAPERED	PASS	BS-4.3.6	0.383	2013
1451	TAPERED	5.08 T	0.25	233.59	1.50
1452	TAPERED	PASS	BS-4.3.6	0.391	2013
1453	TAPERED	8.82 T	0.07	345.63	3.41
1454	TAPERED	PASS	BS-4.3.6	0.391	2013
1455	TAPERED	8.40 T	0.07	345.63	0.00
1456	TAPERED	PASS	BS-4.3.6	0.346	2012
1457	TAPERED	2.19 C	-0.15	176.77	0.00
1458	TAPERED	PASS	BS-4.7 (C)	0.512	2011
1459	TAPERED	35.37 C	-0.02	311.79	1.41
1460	TAPERED	PASS	BS-4.7 (C)	0.466	2011
1461	TAPERED	73.70 C	-0.02	455.63	1.05
1462	TAPERED	PASS	BS-4.7 (C)	0.527	2011
1463	TAPERED	121.64 C	-0.02	558.62	1.05
1464	TAPERED	PASS	BS-4.7 (C)	0.533	2011
1465	TAPERED	149.72 C	0.01	558.62	0.00

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§ STAAD SPACE

ALL UNITS ARE - KN METE (UNLESS OTHERWISE NOTED)

MEMBER	TABLE	RESULT/ FX	CRITICAL COND/ MY	RATIO/ NZ	LOADING/ LOCATION
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1618	TAPERED	PASS	BS-4.7 (C)	0.548	2011
1619	TAPERED	151.92 C	0.00	575.35	0.00
1620	TAPERED	PASS	BS-4.7 (C)	0.545	2011
1621	TAPERED	159.02 C	-0.01	623.88	0.00
1622	TAPERED	PASS	BS-4.7 (C)	0.503	2012
1623	TAPERED	136.12 C	0.02	581.84	0.00
1624	TAPERED	PASS	BS-4.7 (C)	0.510	2012
1625	TAPERED	195.42 C	-0.04	617.19	0.00
1626	TAPERED	PASS	BS-4.7 (C)	0.416	2012
1627	TAPERED	252.66 C	0.23	498.18	0.00
1628	TAPERED	PASS	BS-4.7 (C)	0.427	2013
1629	TAPERED	350.08 C	-0.30	465.84	0.00
1630	TAPERED	PASS	BS-4.7 (C)	0.340	2013
1631	TAPERED	413.60 C	-0.08	317.81	0.00
1632	TAPERED	PASS	BS-4.7 (C)	0.284	2013
1633	TAPERED	367.04 C	-0.25	257.09	3.22
1634	TAPERED	PASS	BS-4.7 (C)	0.347	2013
1635	TAPERED	303.08 C	-0.55	369.72	2.16
1636	TAPERED	PASS	BS-4.7 (C)	0.351	2013
1637	TAPERED	302.71 C	-0.69	383.02	0.25
1638	TAPERED	PASS	BS-4.7 (C)	0.474	2013
1639	TAPERED	244.10 C	-0.95	480.22	2.84
1640	TAPERED	PASS	BS-4.7 (C)	0.480	2013
1641	TAPERED	183.41 C	0.02	493.11	1.21
1642	TAPERED	PASS	BS-4.7 (C)	0.419	2013
1643	TAPERED	180.30 C	0.02	465.66	1.05
1644	TAPERED	PASS	BS-4.7 (C)	0.414	2013
1645	TAPERED	156.71 C	-0.02	465.66	0.00

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1519	TAPERED	PASS	BS-4.7 (C)	0.579	2011
1520	TAPERED	157.13 C	0.01	607.87	0.00
1521	TAPERED	PASS	BS-4.7 (C)	0.489	2012
1522	TAPERED	134.06 C	-0.01	612.37	0.00
1523	TAPERED	PASS	BS-4.7 (C)	0.497	2012
1524	TAPERED	193.44 C	0.19	600.35	0.00
1525	TAPERED	PASS	BS-4.7 (C)	0.430	2013
1526	TAPERED	308.81 C	0.87	484.29	0.00
1527	TAPERED	PASS	BS-4.7 (C)	0.427	2013
1528	TAPERED	349.30 C	-0.47	464.50	0.00
1529	TAPERED	PASS	BS-4.7 (C)	0.342	2013
1530	TAPERED	412.31 C	-0.03	321.43	0.00
1531	TAPERED	PASS	BS-4.7 (C)	0.287	2013
1532	TAPERED	365.12 C	0.09	263.82	3.22
1533	TAPERED	PASS	BS-4.7 (C)	0.348	2013
1534	TAPERED	301.77 C	0.11	378.85	2.12
1535	TAPERED	PASS	BS-4.7 (C)	0.315	2013
1536	TAPERED	301.39 C	-0.15	391.88	0.29
1537	TAPERED	PASS	BS-4.7 (C)	0.435	2013
1538	TAPERED	243.71 C	-0.47	487.05	2.77
1539	TAPERED	PASS	BS-4.7 (C)	0.414	2013
1540	TAPERED	183.67 C	-0.07	498.89	1.21
1541	TAPERED	PASS	BS-4.7 (C)	0.424	2013
1542	TAPERED	180.79 C	0.05	471.05	1.05
1543	TAPERED	PASS	BS-4.7 (C)	0.419	2013
1544	TAPERED	156.89 C	-0.05	471.05	0.00
1545	TAPERED	PASS	BS-4.7 (C)	0.357	2013
1546	TAPERED	108.32 C	-0.02	372.63	0.00
1547	TAPERED	PASS	BS-4.3.6	0.283	2001
1548	TAPERED	49.76 C	0.02	186.40	1.05
1549	TAPERED	PASS	BS-4.3.6	0.283	2001
1550	TAPERED	2.87 T	0.05	255.77	0.00
1551	TAPERED	PASS	BS-4.3.6	0.389	2013
1552	TAPERED	4.23 T	0.18	236.90	1.50
1553	TAPERED	PASS	BS-4.3.6	0.592	2013
1554	TAPERED	7.97 T	-0.05	345.84	3.41
1555	TAPERED	PASS	BS-4.3.6	0.591	2013
1556	TAPERED	7.47 T	0.05	345.84	0.00
1557	TAPERED	PASS	BS-4.3.6	0.343	2012
1558	TAPERED	2.97 C	-0.11	187.11	0.00
1559	TAPERED	PASS	BS-4.7 (C)	0.533	2011
1560	TAPERED	36.10 C	-0.03	324.50	1.42
1561	TAPERED	PASS	BS-4.7 (C)	0.480	2011
1562	TAPERED	74.83 C	-0.02	470.02	1.05
1563	TAPERED	PASS	BS-4.7 (C)	0.542	2011
1564	TAPERED	122.81 C	-0.01	575.35	1.05

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§ STAAD SPACE

ALL UNITS ARE - KN METE (UNLESS OTHERWISE NOTED)

MEMBER	TABLE	RESULT/ FX	CRITICAL COND/ MY	RATIO/ NZ	LOADING/ LOCATION
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1618	TAPERED	PASS	BS-4.7 (C)	0.548	2011
1619	TAPERED	151.92 C	0.00	575.35	0.00
1620	TAPERED	PASS	BS-4.7 (C)	0.545	2011
1621	TAPERED	159.02 C	-0.01	623.88	0.00
1622	TAPERED	PASS	BS-4.7 (C)	0.503	2012
1623	TAPERED	136.12 C	0.02	581.84	0.00
1624	TAPERED	PASS	BS-4.7 (C)	0.510	2012
1625	TAPERED	195.42 C	-0.04	617.19	0.00
1626	TAPERED	PASS	BS-4.7 (C)	0.416	2012
1627	TAPERED	252.66 C	0.23	498.18	0.00
1628	TAPERED	PASS	BS-4.7 (C)	0.427	2013
1629	TAPERED	350.08 C	-0.30	465.84	0.00
1630	TAPERED	PASS	BS-4.7 (C)	0.340	2013
1631	TAPERED	413.60 C	-0.08	317.81	0.00
1632	TAPERED	PASS	BS-4.7 (C)	0.284	2013
1633	TAPERED	367.04 C	-0.25	257.09	3.22
1634	TAPERED	PASS	BS-4.7 (C)	0.347	2013
1635	TAPERED	303.08 C	-0.55	369.72	2.16
1636	TAPERED	PASS	BS-4.7 (C)	0.351	2013
1637	TAPERED	302.71 C	-0.69	383.02	0.25
1638	TAPERED	PASS	BS-4.7 (C)	0.474	2013
1639	TAPERED	244.10 C	-0.95	480.22	2.84
1640	TAPERED	PASS	BS-4.7 (C)	0.480	2013
1641	TAPERED	183.41 C	0.02	493.11	1.21
1642	TAPERED	PASS	BS-4.7 (C)	0.419	2013
1643	TAPERED	180.30 C	0.02	465.66	1.05
1644	TAPERED	PASS	BS-4.7 (C)	0.414	2013
1645	TAPERED	156.71 C	-0.02	465.66	0.00

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untitled						
MEMBER	TABLE	RESULT/ FX	CRITICAL COND/ MY	RATIO/ MZ	LOADING/ LOCATION	
METE (UNLESS OTHERWISE NOTED)						
1708	TAPERED	PASS	BS-4.7 (C)	0.353	2013	
1709	TAPERED	108.48 C	-0.01	368.14	0.00	
1710	TAPERED	PASS	BS-4.3.6	0.289	2001	
1710	TAPERED	49.92 C	0.00	183.91	1.05	
1711	TAPERED	PASS	BS-4.3.6	0.429	2012	
1711	TAPERED	2.69 T	0.04	262.05	0.00	
1712	TAPERED	3.45 T	0.00	9.393	2013	
1712	TAPERED	PASS	BS-4.3.6	0.5918	2013.50	
1713	TAPERED	7.50 T	-0.03	346.22	3.41	
1713	TAPERED	PASS	BS-4.3.6	0.592	2013.00	
1714	TAPERED	6.94 T	0.03	346.22	0.00	
1714	TAPERED	PASS	BS-4.3.6	0.341	2012	
1715	TAPERED	3.30 C	-0.07	190.97	0.00	
1715	TAPERED	PASS	BS-4.7 (C)	0.541	2011	
1716	TAPERED	36.35 C	-0.03	-329.76	1.43	
1716	TAPERED	PASS	BS-4.7 (C)	0.486	2011	
1716	TAPERED	75.11 C	-0.02	-476.14	1.05	
§	STAAD SPACE					
					-- PAGE NO. 490	
ALL UNITS ARE - KN						

ALL UNITS ARE - KN METE (UNLESS OTHERWISE NOTED)

MEMBER	TABLE	RESULT/ FX	CRITICAL COND/ MY	RATIO/ MZ	LOADING/ LOCATION
1717	TAPERED	PASS 123.54 C	BS-4.7 (C) 0.00	0.548 -581.88	2011
1718	TAPERED	PASS 152.71 C	BS-4.7 (C) -0.01	0.555 581.88	2011
1719	TAPERED	PASS 159.80 C	BS-4.7 (C) -0.03	0.550 630.42	0.00
1720	TAPERED	PASS 136.90 C	BS-4.7 (C) 0.05	0.508 -636.82	0.00
1721	TAPERED	PASS 196.71 C	BS-4.7 (C) 0.08	0.516 631.88	2012.00
1722	TAPERED	PASS 253.48 C	BS-4.7 (C) 0.05	0.428 -500.12	0.00
1723	TAPERED	PASS 350.50 C	BS-4.7 (C) -0.13	0.428 -467.88	2013.00
1724	TAPERED	PASS 414.02 C	BS-4.7 (C) -0.06	0.340 -317.24	2013.00
1801	TAPERED	PASS 603.72 C	BS-4.7 (C) 0.13	0.335 250.13	2013.22
1802	TAPERED	PASS 540.75 C	BS-4.7 (C) 0.11	0.357 365.40	2013.17
1803	TAPERED	PASS 540.07 C	BS-4.7 (C) 0.19	0.354 368.08	2013.24
1804	TAPERED	PASS 244.48 C	BS-4.7 (C) -0.24	0.418 478.17	2.86
1805	TAPERED	PASS 184.60 C	BS-4.7 (C) 0.01	0.409 492.64	1.21
1806	TAPERED	PASS 182.19 C	BS-4.7 (C) -0.01	0.421 -467.19	2013
1807	TAPERED	PASS 159.02 C	BS-4.7 (C) 0.00	0.416 467.29	2013.00
1808	TAPERED	PASS 110.60 C	BS-4.7 (C) 0.00	0.356 372.83	2013.00
1809	TAPERED	PASS 33.42 C	BS-4.7 (C) -0.05	0.300 267.54	2012.00
1810	TAPERED	PASS 1.08 C	BS-4.3.6 0.05	0.438 267.44	1.05
1811	TAPERED	PASS 1.75 T	BS-4.3.6 -0.02	0.399 240.75	2013.00
1812	TAPERED	PASS 5.89 T	BS-4.3.6 0.00	0.596 348.39	2013.41
1813	TAPERED	PASS 5.95 T	BS-4.3.6 0.00	0.595 345.95	2013.00
1814	TAPERED	PASS 2.99 C	BS-4.3.6 0.00	0.545 184.95	2012.00
1815	TAPERED	PASS 36.23 C	BS-4.7 (C) -0.03	0.544 -331.40	2011.00
STAAD SPACE				1.43	PAGE NO

ALL UNITS ARE - KN METE (UNLESS OTHERWISE NOTED)

MEMBER	TABLE	RESULT/ FX	CRITICAL COND/ MY	RATIO/ MZ	LOADING/ LOCATION

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Untitled

1816	TAPERED	PASS	BS-4.7 (C)	0.490	2011
1817	TAPERED	75.34 C	-0.01	-479.97	1.05
1818	TAPERED	PASS	BS-4.7 (C)	0.554	2011
1818	TAPERED	123.83 C	0.01	-586.23	1.05
1819	TAPERED	PASS	BS-4.7 (C)	0.561	2011
1819	TAPERED	154.09 C	-0.01	586.23	0.00
1820	TAPERED	PASS	BS-4.7 (-0.03)	0.558	2011
1820	TAPERED	161.83 C	0.00	551.00	2012.00
1821	TAPERED	PASS	BS-4.7 (C)	0.510	2012
1821	TAPERED	139.23 C	0.06	641.17	6.00
1822	TAPERED	PASS	BS-4.7 (-0.18)	0.524	2012
1822	TAPERED	199.26 C	0.00	632.87	6.00
1823	TAPERED	PASS	BS-4.7 (-0.11)	0.433	2012
1823	TAPERED	256.89 C	0.00	506.99	0.00
1824	TAPERED	PASS	BS-4.7 (C)	0.431	2013
1824	TAPERED	354.16 C	-0.01	-470.85	0.00
1901	TAPERED	PASS	BS-4.7 (-0.06)	0.342	2013
1901	TAPERED	418.38 C	0.00	-318.65	0.00
1902	TAPERED	PASS	BS-4.7 (C)	0.285	2013
1902	TAPERED	376.10 C	0.01	257.95	3.22
1903	TAPERED	PASS	BS-4.7 (C)	0.348	2013
1903	TAPERED	310.67 C	0.27	370.99	2.16
1904	TAPERED	PASS	BS-4.7 (C)	0.352	2013
1904	TAPERED	310.30 C	0.34	384.42	0.25
1905	TAPERED	PASS	BS-4.7 (C)	0.425	2013
1905	TAPERED	250.01 C	0.51	484.07	2.84
1906	TAPERED	PASS	BS-4.7 (C)	0.414	2013
1906	TAPERED	186.58 C	0.03	499.08	1.21
1907	TAPERED	PASS	BS-4.7 (C)	0.426	2013
1907	TAPERED	184.17 C	-0.02	-473.35	1.05
1908	TAPERED	PASS	BS-4.7 (C)	0.422	2013
1908	TAPERED	160.90 C	0.02	473.35	0.00
1909	TAPERED	PASS	BS-4.7 (C)	0.361	2013
1909	TAPERED	112.58 C	0.01	97.86	2012.00
1910	TAPERED	PASS	BS-4.7 (-0.01)	0.324	2012
1910	TAPERED	66.68 C	0.00	273.17	0.05
1911	TAPERED	PASS	BS-4.3, 6	0.458	2012
1911	TAPERED	1.80 C	0.01	273.67	6.00
1912	TAPERED	PASS	BS-4.3, 6	0.418	2013
1912	TAPERED	1.13 T	0.01	242.43	0.00
1913	TAPERED	PASS	BS-4.3, 6	0.601	2013
1913	TAPERED	4.95 T	0.02	351.94	3.41
1914	TAPERED	PASS	BS-4.3, 6	0.600	2012
1914	TAPERED	4.88 T	-0.02	351.94	0.00
1914	TAPERED	PASS	BS-4.3, 6	0.348	2012
1914	TAPERED	5.12 C	0.00	191.00	0.00
1914	TAPERED	5.12 C	0.00	191.00	0.00
1914	TAPERED	5.12 C	0.00	191.00	0.00
1914	TAPERED	5.12 C	0.00	191.00	0.00
1914	TAPERED	5.12 C	0.00	191.00	0.00
1914	TAPERED	5.12 C	0.00	191.00	0.00
1914	TAPERED	5.12 C	0.00	191.00	0.00
1914	TAPERED	5.12 C	0.00	191.00	0.00
1914	TAPERED	5.12 C	0.00	191.00	0.00
1914	TAPERED	5.12 C	0.00	191.00	0.00
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1914	TAPERED	5.12 C	0.00	191.00	0.00
1914	TAPERED	5.12 C	0.00	191.00	0.00
1914	TAPERED	5.12 C	0.00	191.00	0.00
1914	TAPERED	5.12 C	0.00	191.00	0.00
1914	TAPERED	5.12 C	0.00	191.00	0.00
1914	TAPERED	5.12 C	0.00	191.00	0.00
1914	TAPERED	5.12 C	0.00	191.00	0.00
1914	TAPERED	5.12 C	0.00	191.00	0.00
1914	TAPERED	5.12 C	0.00	191.00	0.00
1914	TAPERED	5.12 C	0.00	191.00	0.00
1914	TAPERED	5.12 C	0.00	191.00	0.00
1914	TAPERED	5.12 C	0.00	191.00	0.00
1914	TAPERED	5.12 C	0.00	191.00	0.00
1914	TAPERED	5.12 C	0.00	191.00	0.00
1914	TAPERED	5.12 C	0.00	191.00	0.00
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1914	TAPERED	5.12 C	0.00	191.00	0.00
1914	TAPERED	5.12 C	0.00	191.00	0.00
1914	TAPERED	5.12 C	0.00	191.00	0.00
1914	TAPERED	5.12 C	0.00	191.00	0.00
1914	TAPERED	5.12 C	0.00	191.00	0.00
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1914	TAPERED	5.12 C	0.00	191.00	0.00
1914	TAPERED	5.12 C	0.00	191.00	0.00
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1914	TAPERED	5.12 C	0.00	191.00	0.00
1914	TAPERED	5.12 C	0.00	191.00	0.00
1914	TAPERED	5.12 C	0.00	191.00	0.00
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1914	TAPERED	5.12 C	0.00	191.00	0.00
1914	TAPERED	5.12 C	0.00	191.00	0.00
1914	TAPERED	5.12 C	0.00	191.00	0.00
1914	TAPERED	5.12 C	0.00	191.00	0.00
1914	TAPERED	5.12 C	0.00	191.00	0.00
1914	TAPERED	5.12 C	0.00	191.00	0.00
1914	TAPERED	5.12 C	0.00	191.00	0.00
1914	TAPERED	5.12 C	0.00	191.00	0.00
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1914	TAPERED	5.12 C	0.00	191.00	0.00
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1914	TAPERED	5.12 C	0.00	191.00	0.00
1914	TAPERED	5.12 C	0.00	191.00	0.00
1914	TAPERED	5.12 C	0.00	191.00	0.00
1914	TAPERED	5.12 C	0.00	191.00	0.00
1914	TAPERED	5.12 C	0.00	191.00	0.00
1914	TAPERED	5.12 C	0.00	191.00	0.00
1914	TAPERED	5.12 C	0.00	191.00	0.00
1914	TAPERED	5.12 C	0.00	191.00	0.00
1914	TAPERED	5.12 C	0.00	191.00	0.00
1914	TAPERED	5.12 C	0.00	191.00	0.00
1914	TAPERED	5.12 C	0.00	191.00	0.00
1914	TAPERED	5.12 C	0.00	191.00	0.00
1914	TAPERED				

ALL UNITS ARE - KN METE (UNLESS OTHERWISE NOTED)

MEMBER	TABLE	RESULT/ FX	CRITICAL COND/ MY	RATIO/ MZ	LOADING/ LOCATION
1915	TAPERED	PASS	BS-4.7 (C)	0.549	2011
1916	TAPERED	37.93 C	-0.03 (C)	-33.81	1.43
1917	TAPERED	75.69 C	-0.01 (C)	0.496	2011
1918	TAPERED	125.96 C	-0.00 (C)	-488.37	1.05
1919	TAPERED	136.59 C	-0.01 (C)	0.562	2011
1920	TAPERED	164.56 C	-0.02 (C)	-598.45	1.05
1921	TAPERED	142.10 C	0.05 (C)	0.568	2011
1922	TAPERED	203.04 C	-0.32 (C)	0.566	0.00
1923	TAPERED	261.35 C	-0.31 (C)	0.525	2011
1924	TAPERED	357.65 C	0.06 (C)	0.533	0.00
2001	TAPERED	472.42 C	-0.06 (C)	-642.25	0.00
2002	TAPERED	368.05 C	-0.15 (C)	-509.74	0.00
2003	TAPERED	304.08 C	-0.15 (C)	0.437	2013
				0.437	0.00
				0.437	0.00
				0.290	2013
				0.352	2013
				0.352	3.22
				0.359	2.12
				0.359	2.12
				396.24	0.29

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MEMBER	TABLE	RESULT/ FX	CRITICAL COND/		RATIO/ MZ	LOADING/ LOCATION	untitled				LOCATION	
			NY	FX			BS-4.7	(C)	0.426	2.77		
STAAD SPACE	TAPERED	PASS	245.56 C	0.07	491.12	2.77						
	TAPERED	PASS	184.81 C	0.07	502.36	1.21						
	TAPERED	PASS	181.61 C	-0.05	-472.68	1.05						
	TAPERED	PASS	156.98 C	0.05	472.68	0.00						
	TAPERED	PASS	107.66 C	0.01	372.22	0.00						
	TAPERED	PASS	24.46 C	0.06	276.02	1.05						
	TAPERED	PASS	7.08 T	-0.07	276.02	0.00						
	TAPERED	PASS	5.60 T	0.03	241.70	0.00						
	TAPERED	PASS	8.65 T	0.04	349.38	3.41						
	TAPERED	PASS	7.21 T	-0.04	349.38	0.00						
							--	PAGE NO.	493			
	ALL UNITS ARE - KN											
	NETE (UNLESS OTHERWISE NOTED)											
MEMBER	TABLE	RESULT/ FX	CRITICAL COND/ NY		RATIO/ MZ	LOADING/ LOCATION						
2004	TAPERED	PASS	245.56 C	0.07	491.12	2.77						
2005	TAPERED	PASS	184.81 C	0.07	502.36	1.21						
2006	TAPERED	PASS	181.61 C	-0.05	-472.68	1.05						
2007	TAPERED	PASS	156.98 C	0.05	472.68	0.00						
2008	TAPERED	PASS	107.66 C	0.01	372.22	0.00						
2009	TAPERED	PASS	24.46 C	0.06	276.02	1.05						
2010	TAPERED	PASS	7.08 T	-0.07	276.02	0.00						
2011	TAPERED	PASS	5.60 T	0.03	241.70	0.00						
2012	TAPERED	PASS	8.65 T	0.04	349.38	3.41						
2013	TAPERED	PASS	7.21 T	-0.04	349.38	0.00						
									</			

2014	TAPERED	PASS	BS-4.3.6	0.343	2012	303.07 C	0.343	2012	303.07 C	0.343	2012	303.07 C	0.343	2012	303.07 C	0.343	2012	303.07 C	0.343	2012	303.07 C	0.343	2012	303.07 C	0.343	2012	303.07 C	0.343	2012	303.07 C	0.343	2012	303.07 C	0.343	2012	303.07 C	0.343	2012	303.07 C	0.343	2012	303.07 C	0.343	2012	303.07 C	0.343	2012	303.07 C	0.343	2012	303.07 C	0.343	2012	303.07 C	0.343	2012	303.07 C	0.343	2012	303.07 C	0.343	2012	303.07 C	0.343	2012	303.07 C	0.343	2012	303.07 C	0.343	2012	303.07 C	0.343	2012	303.07 C	0.343	2012	303.07 C	0.343	2012	303.07 C	0.343	2012	303.07 C	0.343	2012	303.07 C	0.343	2012	303.07 C	0.343	2012	303.07 C	0.343	2012	303.07 C	0.343	2012	303.07 C	0.343	2012	303.07 C	0.343	2012	303.07 C	0.343	2012	303.07 C	0.343	2012	303.07 C	0.343	2012	303.07 C	0.343	2012	303.07 C	0.343	2012	303.07 C	0.343	2012	303.07 C	0.343	2012	303.07 C	0.343	2012	303.07 C	0.343	2012	303.07 C	0.343	2012	303.07 C	0.343	2012	303.07 C	0.343	2012	303.07 C	0.343	2012	303.07 C	0.343	2012	303.07 C	0.343	2012	303.07 C	0.343	2012	303.07 C	0.343	2012	303.07 C	0.343	2012	303.07 C	0.343	2012	303.07 C	0.343	2012	303.07 C	0.343	2012	303.07 C	0.343	2012	303.07 C	0.343	2012	303.07 C	0.343	2012	303.07 C	0.343	2012	303.07 C	0.343	2012	303.07 C	0.343	2012	303.07 C	0.343	2012	303.07 C	0.343	2012	303.07 C	0.343	2012	303.07 C	0.343	2012	303.07 C	0.343	2012	303.07 C	0.343	2012	303.07 C	0.343	2012	303.07 C	0.343	2012	303.07 C	0.343	2012	303.07 C	0.343	2012	303.07 C	0.343	2012	303.07 C	0.343	2012	303.07 C	0.343	2012	303.07 C	0.343	2012	303.07 C	0.343	2012	303.07 C	0.343	2012	303.07 C	0.343	2012	303.07 C	0.343	2012	303.07 C	0.343	2012	303.07 C	0.343	2012	303.07 C	0.343	2012	303.07 C	0.343	2012	303.07 C	0.343	2012	303.07 C	0.343	2012	303.07 C	0.343	2012	303.07 C	0.343	2012	303.07 C	0.343	2012	303.07 C	0.343	2012	303.07 C	0.343	2012	303.07 C	0.343	2012	303.07 C	0.343	2012	303.07 C	0.343	2012	303.07 C	0.343	2012	303.07 C	0.343	2012	303.07 C	0.343	2012	303.07 C	0.343	2012	303.07 C	0.343	2012	303.07 C	0.343	2012	303.07 C	0.343	2012	303.07 C	0.343	2012	303.07 C	0.343	2012	303.07 C	0.343	2012	303.07 C	0.343	2012	303.07 C	0.343	2012	303.07 C	0.343	2012	303.07 C	0.343	2012	303.07 C	0.343	2012	303.07 C	0.343	2012	303.07 C	0.343	2012	303.07 C	0.343	2012	303.07 C	0.343	2012	303.07 C	0.343	2012	303.07 C	0.343	2012	303.07 C	0.343	2012	303.07 C	0.343	2012	303.07 C	0.343	2012	303.07 C	0.343	2012	303.07 C	0.343	2012	303.07 C	0.343	2012	303.07 C	0.343	2012	303.07 C	0.343	2012	303.07 C	0.343	2012	303.07 C	0.343	2012	303.07 C	0.343	2012	303.07 C	0.343	2012	303.07 C	0.343	2012	303.07 C	0.343	2012	303.07 C	0.343	2012	303.07 C	0.343	2012	303.07 C	0.343	2012	303.07 C	0.343	2012	303.07 C	0.343	2012	303.07 C	0.343	2012	303.07 C	0.343	2012	303.07 C	0.343	2012	303.07 C	0.343	2012	303.07 C	0.343	2012	303.07 C	0.343	2012	303.07 C	0.343	2012	303.07 C	0.343	2012	303.07 C	0.343	2012	303.07 C	0.343	2012	303.07 C	0.343	2012	303.07 C	0.343	2012	303.07 C	0.343	2012	303.07 C	0.343	2012	303.07 C	0.343	2012	303.07 C	0.343	2012	303.07 C	0.343	2012	303.07 C	0.343	2012	303.07 C	0.343	2012	303.07 C	0.343	2012	303.07 C	0.343	2012	303.07 C	0.343	2012	303.07 C	0.343	2012	303.07 C	0.343	2012	303.07 C	0.343	2012	303.07 C	0.343	2012	303.07 C	0.343	2012	303.07 C	0.343	2012	303.07 C	0.343	2012	303.07 C	0.343	2012	303.07 C	0.343	2012	303.07 C	0.343	2012	303.07 C	0.343	2012	303.07 C	0.343	2012	303.07 C	0.343	2012	303.07 C	0.343	2012	303.07 C	0.343	2012	303.07 C	0.343	2012	303.07 C	0.343	2012	303.07 C	0.343	2012	303.07 C	0.343	2012	303.07 C	0.343	2012	303.07 C	0.343	2012	303.07 C	0.343	2012	303.07 C	0.343	2012	303.07 C	0.343	2012	303.07 C	0.343	2012	303.07 C	0.343	2012	303.07 C	0.343	2012	303.07 C	0.343	2012	303.07 C	0.343	2012	303.07 C	0.343	2012	303.07 C	0.343	2012	303.07 C	0.343	2012	303.07 C	0.343	2012	303.07 C	0.343	2012	303.07 C	0.343	2012	303.07 C	0.343	2012	303.07 C	0.343	2012	303.07 C	0.343	2012	303.07 C	0.343	2012	303.07 C	0.343	2012	303.07 C	0.343	2012	303.07 C	0.343	2012	303.07 C	0.343	2012	303.07 C	0.343	2012	303.07 C	0.343	2012	303.07 C	0.343	2012	303.07 C	0.343	2012	303.07 C	0.343	2012	303.07 C	0.343	2012	303.07 C	0.343	2012	303.07 C	0.343	2012	303.07 C	0.343	2012	303.07 C	0.343	2012	303.07 C	0.343	2012	303.07 C	0.343	2012	303.07 C	0.343	2012	303.07 C	0.343	2012	303.07 C	0.343	2012	303.07 C	0.343	2012	303.07 C	0.343	2012	303.07 C	0.343	2012	303.07 C	0.343	2012	303.07 C	0.343	2012	303.07 C	0.343	2012	303.07 C	0.343	2012	303.07 C	0.343	2012	303.07 C	0.343	2012	303.07 C	0.343	2012	303.07 C	0.343	2012	303.07 C	0.343	2012	303.07 C	0.343	2012	303.07 C	0.343	2012	303.07 C	0.343	2012	303.07 C	0.343	2012	303.07 C	0.343	2012	303.07 C	0.343	2012	303.07 C	0.343	2012	303.07 C	0.343	2012	303.07 C	0.343	2012	303.07 C	0.343	2012	303.07 C	0.343	2012	303.07 C	0.343	2012	303.07 C	0.343	2012	303.07 C	0.343	2012	303.07 C	0.343	2012	303.07 C	0.343	2012	303.07 C	0.343	2012	303.07 C	0.343	2012	303.07 C	0.343	2012	303.07 C	0.343	2012	303.07 C	0.343	2012	303.07 C	0.343	2012	303.07 C	0.343	2012	303.07 C	0.343	2012	303.07 C	0.343	2012	303.07 C	0.343	2012	303.07 C	0.343	2012	303.07 C	0.343	2012	303.07 C	0.343	2012	303.07 C	0.343	2012	303.07 C	0.343	2012	303.07 C	0.343	2012	303.07 C	0.343	2012	303.07 C	0.343	2012	303.07 C	0.343	2012	303.07 C	0.343	2012	303.07 C	0.343	2012	303.07 C	0.343	2012	303.07 C	0.343	2012	303.07 C	0.343	2012	303.07 C	0.343	2012	303.07 C	0.343	2012	303.07 C	0.343	2012	303.07 C	0.343	2012	303.07 C	0.343	2012	303.07 C	0.343	2012	303.07 C	0.343	2012	303.07 C	0.343	2012	303.07 C	0.343	2012	303.07 C	0.343	2012	303.07 C	0.343	2012	303.07 C	0.343	2012	303.07 C	0.343	2012	303.07 C	0.3
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ALL UNITS ARE - KN										METE (UNLESS OTHERWISE NOTED)									
MEMBER	TABLE	RESULT/ FX	CRITICAL COND/ NY	RATIO/ MZ	LOADING/ LOCATION	MEMBER	TABLE	RESULT/ FX	CRITICAL COND/ NY	RATIO/ MZ	LOADING/ LOCATION								
2101	TAPERED	PASS	BS-4.7 (C)	0.294	2013	2101	TAPERED	PASS	BS-4.7 (C)	0.294	2013								
2102	TAPERED	365.89 C	-0.12	271.36	3.22	2102	TAPERED	3.89 T	0.09	338.21	2013								
2103	TAPERED	PASS	BS-4.7 (C)	0.353	2013	2103	TAPERED	PASS	BS-4.7 (C)	0.353	2013								
2104	TAPERED	302.37 C	-0.06	383.35	2.05	2104	TAPERED	3.38 T	-0.09	338.21	2011								
2105	TAPERED	PASS	BS-4.7 (C)	0.363	2013	2105	TAPERED	37.425 C	0.139	170.66	2011								
2106	TAPERED	302.115 C	-0.08	603.12	0.36	2106	TAPERED	37.425 C	0.139	170.66	2011								
2107	TAPERED	PASS	BS-4.7 (C)	0.450	2013	2107	TAPERED	37.425 C	0.139	170.66	2011								
2108	TAPERED	244.53 C	0.37 (C)	494.185	0.65	2108	TAPERED	37.425 C	0.139	170.66	2011								
2109	TAPERED	PASS	BS-4.7 (C)	0.416	2013	2109	TAPERED	37.425 C	0.139	170.66	2011								
2110	TAPERED	184.19 C	0.11	502.16	0.21	2110	TAPERED	37.425 C	0.139	170.66	2011								
2111	TAPERED	PASS	BS-4.7 (C)	0.426	2013	2111	TAPERED	37.425 C	0.139	170.66	2011								
2112	TAPERED	181.05 C	-0.07 (C)	478.31	1.05	2112	TAPERED	37.425 C	0.139	170.66	2011								
2113	TAPERED	PASS	BS-4.7 (C)	0.433	2013	2113	TAPERED	37.425 C	0.139	170.66	2011								
2114	TAPERED	156.12 C	0.07 (C)	478.31	0.00	2114	TAPERED	37.425 C	0.139	170.66	2011								
2115	TAPERED	PASS	BS-4.7 (C)	0.360	2013	2115	TAPERED	37.425 C	0.139	170.66	2011								
2116	TAPERED	108.35 C	0.02	373.29	0.00	2116	TAPERED	37.425 C	0.139	170.66	2011								
2117	TAPERED	PASS	BS-4.7 (C)	0.273	2001	2117	TAPERED	75.30 C	0.00	435.20	2011								
2118	TAPERED	49.02 C	0.04	187.02	1.05	2118	TAPERED	121.38 C	0.04	536.61	2011								
2119	TAPERED	PASS	BS-4.7 (C)	0.406	2012	2119	TAPERED	121.38 C	0.04	536.61	2011								
2120	TAPERED	3.78 T	-0.08	247.87	0.00	2120	TAPERED	148.85 C	-0.04	536.61	2011								
2121	TAPERED	PASS	BS-4.7 (C)	0.386	2013	2121	TAPERED	148.85 C	-0.04	536.61	2011								
2122	TAPERED	5.41 T	-0.20	234.92	1.50	2122	TAPERED	154.72 C	-0.07	581.01	2013								
2123	TAPERED	PASS	BS-4.7 (C)	0.591	2013	2123	TAPERED	154.72 C	-0.07	581.01	2013								
2124	TAPERED	9.14 T	0.06	345.39	3.41	2124	TAPERED	154.72 C	-0.07	581.01	2013								
2125	TAPERED	PASS	BS-4.7 (C)	0.591	2013	2125	TAPERED	190.11 C	-0.01	570.69	2013								
2126	TAPERED	154.72 C	-0.07 (C)	581.01	0.00	2126	TAPERED	190.11 C	-0.01	570.69	2013								
2127	TAPERED	PASS	BS-4.7 (C)	0.469	2013	2127	TAPERED	190.11 C	-0.01	570.69	2013								
2128	TAPERED	190.11 C	-0.01	570.69	0.00	2128	TAPERED	250.29 C	-1.08	550.38	2013								
2129	TAPERED	PASS	BS-4.7 (C)	0.477	2013	2129	TAPERED	250.29 C	-1.08	550.38	2013								
2130	TAPERED	250.29 C	-1.08	550.38	0.00	2130	TAPERED	308.76 C	-1.07	456.90	2013								
2131	TAPERED	PASS	BS-4.7 (C)	0.411	2013	2131	TAPERED	308.76 C	-1.07	456.90	2013								
2132	TAPERED	308.76 C	-1.07	456.90	0.00	2132	TAPERED	349.35 C	1.15	433.38	2013								
2133	TAPERED	PASS	BS-4.7 (C)	0.407	2013	2133	TAPERED	349.35 C	1.15	433.38	2013								
2134	TAPERED	349.35 C	1.15	433.38	0.00	2134	TAPERED	349.35 C	1.15	433.38	2013								

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MEMBER	TABLE	RESULT/ FX	CRITICAL COND/ MY	RATIO/ MZ	LOADING/ LOCATION				
2224	TAPERED	PASS 414.02 C	BS-4.7 (C) 0.14	0.330 -303.97	2013 0.00				
2301	TAPERED	PASS 294.13 C	BS-4.7 (C) 0.44	0.229 205.72	2013 3.22				
2302	TAPERED	PASS 232.65 C	BS-4.7 (C) 0.59	0.255 269.70	2013 1.84				
2303	TAPERED	PASS 231.97 C	BS-4.7 (C) 1.19	0.273 291.89	2013 0.57				
2304	TAPERED	PASS 177.11 C	BS-4.7 (C) 2.05	0.309 342.84	2013 2.27				
2305	TAPERED	PASS 158.15 C	BS-4.7 (C) 0.04	0.340 360.00	2013 1.21				
2306	TAPERED	PASS 160.58 C	BS-4.7 (C) -0.03	0.340 381.00	2013 1.05				
2307	TAPERED	PASS 148.68 C	BS-4.7 (C) 0.01	0.343 381.00	2013 0.00				
2308	TAPERED	PASS 113.49 C	BS-4.7 (C) -0.01	0.309 317.38	2013 0.00				
2309	TAPERED	PASS 48.70 C	BS-4.7 (C) 0.02	0.229 199.57	2012 1.05				
2310	TAPERED	PASS 28.32 C	BS-4.7 (C) -0.02	0.330 -199.57	2012 0.00				
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MEMBER	TABLE	RESULT/ FX	CRITICAL COND/ MY	RATIO/ MZ	LOADING/ LOCATION				
2311	TAPERED	PASS 36.69 C	BS-4.7 (C) -0.43	0.307 180.56	2013 1.50	2517	TAPERED	PASS 116.38 C	BS-4.7 (C) -403.34
2312	TAPERED	PASS 41.85 C	BS-4.7 (C) 0.11	0.446 264.47	2013 3.41	2518	TAPERED	PASS 144.70 C	BS-4.7 (C) -454.04
2313	TAPERED	PASS 42.11 C	BS-4.7 (C) -0.11	0.446 264.47	2013 0.00	2519	TAPERED	PASS 149.83 C	BS-4.7 (C) -454.04
2314	TAPERED	PASS 37.23 C	BS-4.7 (C) 0.36	0.279 163.35	2013 0.00	2520	TAPERED	PASS 125.04 C	BS-4.7 (C) -457.47
2315	TAPERED	PASS 46.78 C	BS-4.7 (C) 0.00	0.448 269.58	2011 1.37	2521	TAPERED	PASS 210.49 C	BS-4.7 (C) -441.17
2316	TAPERED	PASS 69.65 C	BS-4.7 (C) 0.01	0.404 393.44	2011 1.05	2522	TAPERED	PASS 258.33 C	BS-4.7 (C) -385.26
2317	TAPERED	PASS 110.23 C	BS-4.7 (C) 0.02	0.455 480.81	2011 1.05	2523	TAPERED	PASS 259.13 C	BS-4.7 (C) -385.26
2318	TAPERED	PASS 133.71 C	BS-4.7 (C) -0.01	0.460 480.81	2011 0.00	2524	TAPERED	PASS 309.05 C	BS-4.7 (C) -421.00
2319	TAPERED	PASS 138.04 C	BS-4.7 (C) -0.02	0.454 517.89	2011 0.00	4001	TAPERED	PASS 1.94 T	BS-4.3.6 0.539
2320	TAPERED	PASS 116.65 C	BS-4.7 (C) 0.05	0.417 521.17	2012 0.00	4002	TAPERED	PASS 86.71 C	BS-4.7 (C) 0.465
2321	TAPERED	PASS 152.17 C	BS-4.7 (C) -2.42	0.425 500.54	2012 0.00	4003	TAPERED	PASS 82.71 C	BS-4.7 (C) 0.395
2322	TAPERED	PASS 200.47 C	BS-4.7 (C) 4.97	0.380 411.19	2012 0.00	4004	TAPERED	PASS 36.72 C	BS-4.7 (C) 0.313
2323	TAPERED	PASS 195.01 C	BS-4.7 (C) 0.87	0.327 472.89	2012 0.00	4005	TAPERED	PASS 36.08 C	BS-4.7 (C) 0.511
2324	TAPERED	PASS 250.72 C	BS-4.7 (C) 1.33	0.277 276.41	2013 0.00	9	STAAD SPACE		-88.26
2422	TAPERED	PASS 16.78 C	BS-4.7 (C) -0.85	0.114 23.92	2013 0.68				
2423	TAPERED	PASS 26.17 C	BS-4.7 (C) 0.85	0.114 23.92	2013 0.00				
2424	TAPERED	PASS 22.94 C	BS-4.7 (C) -0.85	0.072 -14.73	2012 0.00				
2501	TAPERED	PASS 107.50 C	BS-4.7 (C) -2.78	0.298 -337.74	2011 3.22				
2502	TAPERED	PASS 57.48 C	BS-4.7 (C) -2.37	0.326 -396.71	2011 1.74				
2503	TAPERED	PASS 56.69 C	BS-4.7 (C) -5.77	0.366 -423.03	2011 0.68				
2504	TAPERED	PASS 50.38 T	BS-4.8.2.2 3.21	0.372 -45.89	2012 2.01				
2505	TAPERED	PASS 62.91 T	BS-4.8.2.2 0.44	0.372 45.89	2012 0.00				
2506	TAPERED	PASS 61.57 T	BS-4.8.2.2 -0.03	0.342 407.55	2012 1.05				
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MEMBER	TABLE	RESULT/ FX	CRITICAL COND/ MY	RATIO/ MZ	LOADING/ LOCATION				
4006	TAPERED	PASS	BS-4.3.6	0.00	85.96			Page 10	0.00
4016	TAPERED	1.94 T							

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ALL UNITS ARE - KN METE (UNLESS OTHERWISE NOTED)

MEMBER	TABLE	RESULT/ FX	CRITICAL COND/ MY	RATIO/ MZ	LOADING/ LOCATION
4304	TAPERED	PASS 26.30 C	BS-4.7 (C) 0.00	0.391 -66.63	2013 4.00
4305	TAPERED	PASS 26.52 C	BS-4.7 (C) 0.00	0.392 -66.62	2013 4.00
4306	TAPERED	PASS 26.70 C	BS-4.7 (C) 0.00	0.392 -66.61	2013 4.00
4307	TAPERED	PASS 26.88 C	BS-4.7 (C) 0.00	0.393 -66.61	2013 4.00
4308	TAPERED	PASS 26.99 C	BS-4.7 (C) 0.00	0.393 -66.61	2013 4.00
4309	TAPERED	PASS 27.09 C	BS-4.7 (C) 0.00	0.394 -66.61	2013 4.00
4310	TAPERED	PASS 27.19 C	BS-4.7 (C) 0.00	0.394 -66.61	2013 4.00
4311	TAPERED	PASS 27.29 C	BS-4.7 (C) 0.00	0.394 -66.61	2013 4.00
4312	TAPERED	PASS 27.39 C	BS-4.7 (C) 0.00	0.394 -66.61	2013 4.00
4313	TAPERED	PASS 27.49 C	BS-4.7 (C) 0.00	0.394 -66.61	2013 4.00
4314	TAPERED	PASS 27.59 C	BS-4.7 (C) 0.00	0.394 -66.61	2013 4.00
4315	TAPERED	PASS 27.69 C	BS-4.7 (C) 0.00	0.394 -66.61	2013 4.00
4316	TAPERED	PASS 27.79 C	BS-4.7 (C) 0.00	0.394 -66.61	2013 4.00
4401	TAPERED	PASS 27.89 C	BS-4.7 (C) 0.00	0.394 -66.61	2013 4.00
4402	TAPERED	PASS 27.99 C	BS-4.7 (C) 0.00	0.394 -66.61	2013 4.00
4403	TAPERED	PASS 28.09 C	BS-4.7 (C) 0.00	0.394 -66.61	2013 4.00
4404	TAPERED	PASS 28.19 C	BS-4.7 (C) 0.00	0.394 -66.61	2013 4.00
4405	TAPERED	PASS 28.29 C	BS-4.7 (C) 0.00	0.394 -66.61	2013 4.00
4406	TAPERED	PASS 28.39 C	BS-4.7 (C) 0.00	0.394 -66.61	2013 4.00
4407	TAPERED	PASS 28.49 C	BS-4.7 (C) 0.00	0.394 -66.61	2013 4.00
4408	TAPERED	PASS 28.59 C	BS-4.7 (C) 0.00	0.394 -66.61	2013 4.00
4409	TAPERED	PASS 28.69 C	BS-4.7 (C) 0.00	0.394 -66.61	2013 4.00
4410	TAPERED	PASS 28.79 C	BS-4.7 (C) 0.00	0.394 -66.61	2013 4.00
STAAD SPACE					

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Untitled

ALL UNITS ARE - KN METE (UNLESS OTHERWISE NOTED)

MEMBER	TABLE	RESULT/ FX	CRITICAL COND/ MY	RATIO/ MZ	LOADING/ LOCATION
4101	TAPERED	PASS 1.20 T	BS-4.3.6 -0.54	0.376 59.92	2013 4.00
4102	TAPERED	PASS 50.76 C	BS-4.7 (C) -0.22	0.290 60.06	2013 0.00
4103	TAPERED	PASS 47.94 C	BS-4.7 (C) -0.08	0.251 30.56	2013 2.80
4104	TAPERED	PASS 29.06 C	BS-4.7 (C) 0.00	0.374 -59.43	2013 4.00
4105	TAPERED	PASS 29.32 C	BS-4.7 (C) 0.00	0.375 -59.42	2013 4.00
4106	TAPERED	PASS 29.58 C	BS-4.7 (C) 0.00	0.376 -59.41	2013 4.00
4107	TAPERED	PASS 29.84 C	BS-4.7 (C) 0.00	0.376 -59.41	2013 4.00
4108	TAPERED	PASS 29.73 C	BS-4.7 (C) 0.00	0.377 -59.41	2013 4.00
4109	TAPERED	PASS 30.55 C	BS-4.7 (C) 0.00	0.380 -59.41	2013 4.00
4110	TAPERED	PASS 30.28 C	BS-4.7 (C) 0.00	0.379 -59.41	2013 4.00
4111	TAPERED	PASS 27.84 C	BS-4.7 (C) 0.00	0.368 -59.41	2013 4.00
4112	TAPERED	PASS 27.67 C	BS-4.7 (C) 0.00	0.367 -59.42	2013 4.00
STAAD SPACE					

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Untitled

ALL UNITS ARE - KN METE (UNLESS OTHERWISE NOTED)

MEMBER	TABLE	RESULT/ FX	CRITICAL COND/ MY	RATIO/ MZ	LOADING/ LOCATION
4411	TAPERED	PASS 30.21 C	BS-4.7 (C) 0.00	0.455 -78.05	2013 4.00
4412	TAPERED	PASS 30.26 C	BS-4.7 (C) 0.00	0.455 -78.06	2013 4.00
4413	TAPERED	PASS 30.34 C	BS-4.7 (C) 0.00	0.456 -78.07	2013 4.00
4414	TAPERED	PASS 73.46 C	BS-4.7 (C) -0.13	0.356 -39.61	2013 2.80
4415	TAPERED	PASS 77.12 C	BS-4.7 (C) -1.61	0.424 -78.94	2013 2.41
4416	TAPERED	PASS 1.78 T	BS-4.3.6 0.00	0.493 78.75	2013 0.00
5001	TAPERED	PASS 0.01 T	BS-4.8.2.2 0.00	0.078 49.44	2011 4.00
5002	TAPERED	PASS 6.08 C	BS-4.7 (C) -53.61	0.107 -88.56	2011 0.00
5003	TAPERED	PASS 2.63 C	BS-4.7 (C) -36.06	0.063 50.19	2011 5.60
5004	TAPERED	PASS 2.63 C	BS-4.7 (C) 38.61	0.063 -50.46	2011 0.00
STAAD SPACE					

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ALL UNITS ARE - KN METE (UNLESS OTHERWISE NOTED)

MEMBER	TABLE	RESULT/ FX	CRITICAL COND/ NY	RATIO/ MZ	LOADING/ LOCATION
5209	TAPERED	PASS	BS-4.7 (C)	0.054	2011
5210	TAPERED	56.95 C	BS-4.7 (C)	32.46	8.00
5211	TAPERED	54.63 C	BS-4.7 (C)	0.054	2011
5212	TAPERED	26.28 T	BS-4.8.2.2	-32.43	0.00
5213	TAPERED	12.13 T	BS-4.8.2.2	0.050	2014
5214	TAPERED	11.54 T	BS-4.8.2.2	30.15	0.00
5215	TAPERED	75.46 C	BS-4.7 (C)	0.051	2011
5216	TAPERED	78.23 C	BS-4.7 (C)	3.91	8.00
5301	TAPERED	7.83 C	BS-4.7 (C)	0.072	2011
5302	TAPERED	2.57 C	BS-4.7 (C)	62.15	8.00
5303	TAPERED	20.86 C	BS-4.7 (C)	0.075	2011
5304	TAPERED	7.09 C	BS-4.7 (C)	-61.95	0.00
5305	TAPERED	41.62 T	BS-4.8.2.2	0.067	2011
5306	TAPERED	33.27 T	BS-4.8.2.2	61.63	2.41
5307	TAPERED	32.94 T	BS-4.8.2.2	-61.53	0.00
5308	TAPERED	32.71 T	BS-4.8.2.2	0.042	2012
5309	TAPERED	32.56 T	BS-4.8.2.2	0.040	2014
5310	TAPERED	19.61 T	BS-4.8.2.2	0.043	2012
5311	TAPERED	19.24 T	BS-4.8.2.2	21.76	8.00
5312	TAPERED	30.74 T	BS-4.8.2.2	0.045	2014
5313	TAPERED	32.09 T	BS-4.8.2.2	24.19	8.00
5314	TAPERED	44.45 T	BS-4.8.2.2	-24.21	0.00
5315	TAPERED	3.46 C	BS-4.7 (C)	0.044	2014
5316	TAPERED	22.29 C	BS-4.7 (C)	0.043	2014

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ALL UNITS ARE - KN METE (UNLESS OTHERWISE NOTED)

MEMBER	TABLE	RESULT/ FX	CRITICAL COND/ NY	RATIO/ MZ	LOADING/ LOCATION
5316	TAPERED	PASS	BS-4.7 (C)	0.042	2012
5401	TAPERED	2.57 C	BS-4.7 (C)	-16.96	0.00
5402	TAPERED	0.02 C	BS-4.7 (C)	0.054	2014
5403	TAPERED	1.84 T	BS-4.2.3 (V)	25.12	4.00
5404	TAPERED	0.46 C	BS-4.7 (C)	-26.36	0.00
5405	TAPERED	0.30 C	BS-4.7 (C)	38.85	5.60
5406	TAPERED	6.10 C	BS-4.7 (C)	-38.05	0.00
5407	TAPERED	8.33 C	BS-4.7 (C)	0.034	2014
5408	TAPERED	9.77 C	BS-4.7 (C)	14.52	8.00
5409	TAPERED	10.31 C	BS-4.7 (C)	14.90	8.00
5410	TAPERED	10.31 C	BS-4.7 (C)	0.031	2014
5411	TAPERED	10.31 C	BS-4.7 (C)	15.16	8.00

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ALL UNITS ARE - KN METE (UNLESS OTHERWISE NOTED)

MEMBER	TABLE	RESULT/ FX	CRITICAL COND/ NY	RATIO/ MZ	LOADING/ LOCATION
5005	TAPERED	PASS	BS-4.7 (C)	0.055	2011
5006	TAPERED	2.56 C	BS-4.7 (C)	-36.65	0.00
5007	TAPERED	1.82 C	BS-4.7 (C)	0.050	2014
5008	TAPERED	2.09 C	BS-4.7 (C)	30.06	0.00
5009	TAPERED	1.56 C	BS-4.7 (C)	0.051	2011
5010	TAPERED	3.70 C	BS-4.7 (C)	31.87	8.00
5011	TAPERED	4.34 C	BS-4.7 (C)	0.052	2011
5012	TAPERED	2.66 C	BS-4.7 (C)	-31.97	0.00
5013	TAPERED	2.40 C	BS-4.7 (C)	0.053	2011
5014	TAPERED	5.22 C	BS-4.7 (C)	34.39	8.00
5015	TAPERED	5.22 C	BS-4.7 (C)	0.053	2011
5016	TAPERED	0.01 T	BS-4.7 (C)	-34.38	0.00
5017	TAPERED	0.02 T	BS-4.7 (C)	0.050	2014
5018	TAPERED	0.02 T	BS-4.7 (C)	30.00	0.00
5019	TAPERED	0.02 T	BS-4.7 (C)	30.77	8.00
5020	TAPERED	0.02 T	BS-4.7 (C)	0.063	2011
5021	TAPERED	0.02 T	BS-4.7 (C)	50.86	8.00
5022	TAPERED	0.02 T	BS-4.7 (C)	0.062	2011
5023	TAPERED	0.02 T	BS-4.7 (C)	-50.68	0.00
5024	TAPERED	0.02 T	BS-4.7 (C)	0.071	2011
5025	TAPERED	0.02 T	BS-4.7 (C)	48.49	2.40
5026	TAPERED	0.02 T	BS-4.7 (C)	0.078	2011
5027	TAPERED	0.02 T	BS-4.7 (C)	-49.44	0.00
5028	TAPERED	0.02 T	BS-4.7 (C)	0.057	2011
5029	TAPERED	0.02 T	BS-4.7 (C)	28.56	4.00

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ALL UNITS ARE - KN METE (UNLESS OTHERWISE NOTED)

MEMBER	TABLE	RESULT/ FX	CRITICAL COND/ NY	RATIO/ MZ	LOADING/ LOCATION
5102	TAPERED	PASS	BS-4.8.2.2	0.052	2011
5103	TAPERED	0.18 T	BS-4.8.2.2	-27.78	0.00
5104	TAPERED	0.18 T	BS-4.8.2.2	0.058	2011
5105	TAPERED	4.42 T	BS-4.8.2.2	50.37	5.60
5106	TAPERED	4.81 T	BS-4.8.2.2	0.060	2011
5107	TAPERED	6.05 T	BS-4.8.2.2	-50.67	0.00
5108	TAPERED	6.39 T	BS-4.8.2.2	0.041	2011
5109	TAPERED	6.35 T	BS-4.8.2.2	-22.45	0.00
5110	TAPERED	0.52 T	BS-4.8.2.2	0.039	2014
5111	TAPERED	0.85 T	BS-4.8.2.2	18.31	0.00
5112	TAPERED	5.35 T	BS-4.8.2.2	0.041	2014
5113	TAPERED	3.05 T	BS-4.8.2.2	0.039	2014
5114	TAPERED	3.15 T	BS-4.8.2.2	0.041	2011
5115	TAPERED	0.29 T	BS-4.8.2.2	-21.50	8.00
5116	TAPERED	0.29 T	BS-4.8.2.2	0.041	2011
5201	TAPERED	0.02 T	BS-4.7 (C)	22.01	8.00
5202	TAPERED	7.84 C	BS-4.7 (C)	-21.85	0.00
5203	TAPERED	79.88 C	BS-4.7 (C)	0.039	2014
5204	TAPERED	77.10 C	BS-4.7 (C)	20.32	0.00
5205	TAPERED	10.19 T	BS-4.8.2.2	0.041	2011
5206	TAPERED	10.21 T	BS-4.8.2.2	23.22	8.00
5207	TAPERED	24.09 T	BS-4.8.2.2	0.060	2011
5208	TAPERED	25.53 T	BS-4.8.2.2	51.21	8.00
5209	TAPERED	24.67 T	BS-4.8.2.2	0.059	2011
5210	TAPERED	24.67 T	BS-4.8.2.2	0.056	2011
5211	TAPERED	24.67 T	BS-4.8.2.2	0.057	2011
5212	TAPERED	24.67 T	BS-4.8.2.2	-28.56	0.00
5213	TAPERED	24.67 T	BS-4.7 (C)	0.063	2011
5214	TAPERED	24.67 T	BS-4.7 (C)	61.55	4.00
5215	TAPERED	24.67 T	BS-4.7 (C)	0.067	2011
5216	TAPERED	24.67 T	BS-4.7 (C)	-61.72	0.00
5217	TAPERED	24.67 T	BS-4.7 (C)	0.074	2011
5218	TAPERED	24.67 T	BS-4.7 (C)	61.08	5.60
5219	TAPERED	24.67 T	BS-4.8.2.2	0.072	2011
5220	TAPERED	24.67 T	BS-4.8.2.2	-61.37	0.00
5221	TAPERED	24.67 T	BS-4.8.2.2	0.050	2011
5222	TAPERED	24.67 T	BS-4.8.2.2	-31.75	0.00
5223	TAPERED	24.67 T	BS-4.8.2.2	0.047	2014
5224	TAPERED	24.67 T	BS-4.8.2.2	26.71	0.00
5225	TAPERED	24.67 T	BS-4.8.2.2	-26.71	8.00
5226	TAPERED	24.67 T	BS-4.8.2.2	0.047	2014
5227	TAPERED	24.67 T	BS-4.8.2.2	25.67	0.00

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6206 TAPERED PASS 18.53 T BS-4.3.6 0.00 Untitled 0.206 2001 4.00
‡ STAAD SPACE 39.96 39.96 -- PAGE NO. 509

ALL UNITS ARE - KN METE (UNLESS OTHERWISE NOTED)

MEMBER	TABLE	RESULT/ FX	CRITICAL MY	COND/ MY	RATIO/ MZ	LOADING/ LOCATION
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6207	TAPERED	PASS	17.18 T	BS-4.3.6	0.203	2001 4.00
6208	TAPERED	PASS	17.24 T	BS-4.3.6	0.203	2001 4.00
6209	TAPERED	PASS	17.24 T	BS-4.3.6	0.203	2001 4.00
6210	TAPERED	PASS	18.61 T	BS-4.3.6	0.203	2001 4.00
6211	TAPERED	PASS	18.54 T	BS-4.3.6	0.203	2001 4.00
6212	TAPERED	PASS	18.42 T	BS-4.7 (C)	0.205	2001 4.00
110	TAPERED	PASS	14.05 C	BS-4.7 (C)	0.128	2013 2.80
111	TAPERED	PASS	161.43 C	BS-4.7 (C)	0.354	2011 3.20
112	TAPERED	PASS	153.93 C	BS-4.7 (C)	0.359	2012 0.00
113	TAPERED	PASS	116.59 C	BS-4.7 (C)	0.258	2012 0.00
120	TAPERED	PASS	113.16 C	BS-4.7 (C)	0.861	2013 0.05
121	TAPERED	PASS	89.01 C	BS-4.7 (C)	0.084	2011 3.20
122	TAPERED	PASS	85.06 C	BS-4.7 (C)	0.165	2011 4.40
123	TAPERED	PASS	83.98 C	BS-4.7 (C)	0.182	2011 1.20
130	TAPERED	PASS	59.93 C	BS-4.7 (C)	0.246	2012 3.35
131	TAPERED	PASS	9.10 T	BS-4.8.2.2	-11.27	2012 3.20
132	TAPERED	PASS	15.22 C	BS-4.7 (C)	0.405	2012 0.00
133	TAPERED	PASS	12.03 C	BS-4.7 (C)	0.409	2012 0.00
210	TAPERED	PASS	186.53 C	BS-4.7 (C)	0.156	2012 0.00
211	TAPERED	PASS	160.82 C	BS-4.7 (C)	0.146	2013 0.05
212	TAPERED	PASS	153.13 C	BS-4.7 (C)	0.359	2012 0.00
213	TAPERED	PASS	115.78 C	BS-4.7 (C)	0.257	2012 0.00
220	TAPERED	PASS	112.36 C	BS-4.7 (C)	0.881	2012 3.05
‡ STAAD SPACE		PASS	88.99 C	BS-4.7 (C)	0.084	2011 3.20

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ALL UNITS ARE - KN METE (UNLESS OTHERWISE NOTED)

MEMBER	TABLE	RESULT/ FX	CRITICAL MY	COND/ MY	RATIO/ MZ	LOADING/ LOCATION
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221	TAPERED	PASS	85.04 C	BS-4.7 (C)	0.166	2011 4.40
222	TAPERED	PASS	83.96 C	BS-4.7 (C)	0.182	2011 1.20
223	TAPERED	PASS	59.92 C	BS-4.7 (C)	0.246	2012 3.35
230	TAPERED	PASS	7.86 T	BS-4.8.2.2	11.40	2012 3.20
231	TAPERED	PASS	16.44 C	BS-4.7 (C)	0.410	2012 0.00
232	TAPERED	PASS	13.32 C	BS-4.7 (C)	0.157	2013 0.00
233	TAPERED	PASS	186.98 C	BS-4.7 (C)	0.147	2013 3.05

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2330 TAPERED PASS 27.14 C BS-4.7 (C) 0.145 2013 0.36
2331 TAPERED PASS 27.14 C BS-4.7 (C) 0.195 2013 2.40
2332 TAPERED PASS 8.35 C BS-4.3.6 0.471 2012 2.33
2333 TAPERED PASS 66.11 C BS-4.7 (C) 0.448 2011 0.00
2430 TAPERED PASS 43.31 T BS-4.8.2.2 0.132 2013 0.36
2431 TAPERED PASS 1.24 C BS-4.7 (C) 0.210 2012 2.40
2432 TAPERED PASS 11.24 C BS-4.3.6 0.470 2012 2.32
2433 TAPERED PASS 47.15 C BS-4.7 (C) 0.151 2013 0.00
6001 TAPERED PASS 38.77 C BS-4.7 (C) 0.151 2013 2.80
6002 TAPERED PASS 29.09 C BS-4.7 (C) 0.255 2013 4.00
6003 TAPERED PASS 22.25 C BS-4.7 (C) 0.224 2013 4.00
6004 TAPERED PASS 17.49 C BS-4.7 (C) 0.203 2013 4.00
6005 TAPERED PASS 14.55 C BS-4.7 (C) 0.190 2013 4.00
6006 TAPERED PASS 13.23 C BS-4.7 (C) 0.184 2013 4.00
6007 TAPERED PASS 13.88 C BS-4.7 (C) 0.187 2013 4.00
‡ STAAD SPACE 30.44 30.44 -- PAGE NO. 508

ALL UNITS ARE - KN METE (UNLESS OTHERWISE NOTED)

MEMBER	TABLE	RESULT/ FX	CRITICAL MY	COND/ MY	RATIO/ MZ	LOADING/ LOCATION
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6008	TAPERED	PASS	15.55 C	BS-4.7 (C)	0.194	2013 4.00
6009	TAPERED	PASS	16.94 C	BS-4.7 (C)	0.201	2013 4.00
6010	TAPERED	PASS	22.84 C	BS-4.7 (C)	0.223	2013 0.00
6011	TAPERED	PASS	29.12 C	BS-4.7 (C)	0.255	2013 4.00
6012	TAPERED	PASS	38.96 C	BS-4.7 (C)	0.162	2013 2.80
6101	TAPERED	PASS	44.13 C	BS-4.7 (C)	0.234	2013 2.80
6102	TAPERED	PASS	2.43 T	BS-4.3.6	0.231	2001 4.00
6103	TAPERED	PASS	1.71 T	BS-4.3.6	0.230	2001 4.00
6104	TAPERED	PASS	1.22 T	BS-4.3.6	0.229	2001 4.00
6105	TAPERED	PASS	0.94 T	BS-4.3.6	0.229	2001 4.00
6106	TAPERED	PASS	0.82 T	BS-4.3.6	0.229	2001 4.00
6107	TAPERED	PASS	1.36 C	BS-4.7 (C)	0.234	2013 0.00
6108	TAPERED	PASS	2.17 C	BS-4.7 (C)	0.237	2013 4.00
6109	TAPERED	PASS	0.75 T	BS-4.3.6	0.229	2001 4.00
6110	TAPERED	PASS	1.28 T	BS-4.3.6	0.229	2001 4.00
6111	TAPERED	PASS	2.04 T	BS-4.3.6	0.230	2001 4.00
6112	TAPERED	PASS	44.96 C	BS-4.7 (C)	0.236	2013 2.80
6201	TAPERED	PASS	13.47 C	BS-4.7 (C)	0.127	2013 2.80
6202	TAPERED	PASS	18.92 T	BS-4.3.6	0.205	2001 4.00
6203	TAPERED	PASS	18.42 T	BS-4.3.6	0.205	2001 4.00
6204	TAPERED	PASS	18.49 T	BS-4.3.6	0.206	2001 4.00
6205	TAPERED	PASS	18.52 T	BS-4.3.6	0.206	2001 4.00

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7306 ST UA90X90X6 PASS 6.74 T 0.018 0.00 2012 0.00
7307 ST UA90X90X6 PASS 0.89 T 0.002 0.00 2014 0.00
§ STAAD SPACE BS-4.6 (T) 0.00 0.00 -- PAGE NO. 512

ALL UNITS ARE - KN METE (UNLESS OTHERWISE NOTED)

MEMBER	TABLE	RESULT/ FX	CRITICAL COND/ MY	RATIO/ M/Z	LOADING/ LOCATION
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7308 ST UA90X90X6	PASS	7.73 T	BS-4.6 (T)	0.021	2012
7309 ST UA90X90X6	PASS	0.07 T	BS-4.6 (T)	0.000	2014
7310 ST UA90X90X6	PASS	6.87 T	BS-4.6 (T)	0.019	2012
7311 ST UA90X90X6	PASS	13.27 T	BS-4.6 (T)	0.036	2014
7312 ST UA90X90X6	PASS	16.37 T	BS-4.6 (T)	0.045	2014
7401 ST UA90X90X6	PASS	103.71 T	BS-4.6 (T)	0.284	2012
7402 ST UA90X90X6	PASS	90.22 T	BS-4.6 (T)	0.247	2012
7403 ST UA90X90X6	PASS	48.07 T	BS-4.6 (T)	0.130	2013
7404 ST UA90X90X6	PASS	23.01 T	BS-4.6 (T)	0.063	2012
7405 ST UA90X90X6	PASS	21.49 T	BS-4.6 (T)	0.059	2013
7406 ST UA90X90X6	PASS	25.77 T	BS-4.6 (T)	0.070	2012
7407 ST UA90X90X6	PASS	0.00	BS-4.6 (T)	0.000	2012
7408 ST UA90X90X6	PASS	21.93 T	BS-4.6 (T)	0.060	2011
7409 ST UA90X90X6	PASS	0.00	BS-4.6 (T)	0.000	2011
7410 ST UA90X90X6	PASS	45.68 T	BS-4.6 (T)	0.125	2013
7411 ST UA90X90X6	PASS	13.68 T	BS-4.6 (T)	0.050	2014
7412 ST UA90X90X6	PASS	90.00 T	BS-4.6 (T)	0.240	2013
7501 ST UA90X90X6	PASS	2.73 T	BS-4.6 (T)	0.007	2012
7502 ST UA90X90X6	PASS	17.64 T	BS-4.6 (T)	0.048	2012
7503 ST UA90X90X6	PASS	5.23 T	BS-4.6 (T)	0.014	2014
7504 ST UA90X90X6	PASS	0.22 T	BS-4.6 (T)	0.001	2013
7505 ST UA90X90X6	PASS	0.22 T	BS-4.6 (T)	0.001	2013
7506 ST UA90X90X6	PASS	5.24 T	BS-4.6 (T)	0.014	2014
§ STAAD SPACE				0.00	0.00

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7001 ST UA90X90X6 PASS 17.65 T 0.048 0.00 2012 6.32
7002 ST UA90X90X6 PASS 2.73 T 0.007 0.00 2012 0.00
7003 ST UA90X90X6 PASS 0.22 T 0.001 0.00 2013 0.00
7004 ST UA90X90X6 PASS 5.01 T 0.014 0.00 2014 5.26
7005 ST UA90X90X6 PASS 5.01 T 0.014 0.00 2014 0.00
7006 ST UA90X90X6 PASS 0.22 T 0.001 0.00 2013 0.00
7007 ST UA90X90X6 PASS 1.00 T 0.003 0.00 2014 0.00
7008 ST UA90X90X6 PASS 9.84 T 0.020 0.00 2014 0.00
7101 ST UA90X90X6 PASS 89.00 T 0.243 0.00 2012 0.00
7102 ST UA90X90X6 PASS 106.47 T 0.291 0.00 2012 9.20
7103 ST UA90X90X6 PASS 23.65 T 0.065 0.00 2012 0.00
7104 ST UA90X90X6 PASS 49.16 T 0.134 0.00 2013 0.00
7105 ST UA90X90X6 PASS 27.44 T 0.075 0.00 2012 0.00
7106 ST UA90X90X6 PASS 20.31 T 0.057 0.00 2013 0.00
7107 ST UA90X90X6 PASS 24.1 T 0.060 0.00 2012 6.69
7108 ST UA90X90X6 PASS 0.00 0.000 0.00 2012 0.22
§ STAAD SPACE BS-4.6 (T) 0.00 0.00 -- PAGE NO. 511

ALL UNITS ARE - KN METE (UNLESS OTHERWISE NOTED)

MEMBER	TABLE	RESULT/ FX	CRITICAL COND/ MY	RATIO/ M/Z	LOADING/ LOCATION
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7109 ST UA90X90X6	PASS	48.23 T	BS-4.6 (T)	0.132	2013
7110 ST UA90X90X6	PASS	0.55 T	BS-4.6 (T)	0.000	2014
7111 ST UA90X90X6	PASS	87.56 T	BS-4.6 (T)	0.239	2013
7112 ST UA90X90X6	PASS	13.09 T	BS-4.6 (T)	0.036	2014
7201 ST UA90X90X6	PASS	91.84 T	BS-4.6 (T)	0.251	2012
7202 ST UA90X90X6	PASS	91.41 T	BS-4.6 (T)	0.250	2012
7203 ST UA90X90X6	PASS	0.00	BS-4.6 (T)	0.000	2012
7204 ST UA90X90X6	PASS	6.63 T	BS-4.6 (T)	0.018	2012
7205 ST UA90X90X6	PASS	1.00 T	BS-4.6 (T)	0.003	2012
7206 ST UA90X90X6	PASS	3.32 T	BS-4.6 (T)	0.000	2013
7207 ST UA90X90X6	PASS	1.44 T	BS-4.6 (T)	0.004	2014
7208 ST UA90X90X6	PASS	2.89 T	BS-4.6 (T)	0.008	2012
7209 ST UA90X90X6	PASS	0.93 T	BS-4.6 (T)	0.003	2014
7210 ST UA90X90X6	PASS	2.66 T	BS-4.6 (T)	0.007	2012
7211 ST UA90X90X6	PASS	15.09 T	BS-4.6 (T)	0.041	2014
7212 ST UA90X90X6	PASS	14.78 T	BS-4.6 (T)	0.040	2014
7301 ST UA90X90X6	PASS	89.71 T	BS-4.6 (T)	0.245	2012
7302 ST UA90X90X6	PASS	94.85 T	BS-4.6 (T)	0.250	2012
7303 ST UA90X90X6	PASS	0.27 T	BS-4.6 (T)	0.001	2011
7304 ST UA90X90X6	PASS	0.74 T	BS-4.6 (T)	0.002	2011
7305 ST UA90X90X6	PASS	0.75 T	BS-4.6 (T)	0.002	2014

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7306 ST UA90X90X6 PASS 6.74 T 0.018 0.00 2012 0.00
7307 ST UA90X90X6 PASS 0.89 T 0.002 0.00 2014 0.00
§ STAAD SPACE BS-4.6 (T) 0.00 0.00 -- PAGE NO. 512

ALL UNITS ARE - KN METE (UNLESS OTHERWISE NOTED)

MEMBER	TABLE	RESULT/ FX	CRITICAL COND/ MY	RATIO/ M/Z	LOADING/ LOCATION
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7308 ST UA90X90X6	PASS	7.73 T	BS-4.6 (T)	0.021	2012
7309 ST UA90X90X6	PASS	0.07 T	BS-4.6 (T)	0.000	2014
7310 ST UA90X90X6	PASS	6.87 T	BS-4.6 (T)	0.019	2012
7311 ST UA90X90X6	PASS	13.27 T	BS-4.6 (T)	0.036	2014
7312 ST UA90X90X6	PASS	16.37 T	BS-4.6 (T)	0.045	2014
7401 ST UA90X90X6	PASS	103.71 T	BS-4.6 (T)	0.284	2012
7402 ST UA90X90X6	PASS	90.22 T	BS-4.6 (T)	0.247	2012
7403 ST UA90X90X6	PASS	48.07 T	BS-4.6 (T)	0.130	2013
7404 ST UA90X90X6	PASS	23.01 T	BS-4.6 (T)	0.063	2012
7405 ST UA90X90X6	PASS	21.49 T	BS-4.6 (T)	0.059	2013
7406 ST UA90X90X6	PASS	25.77 T	BS-4.6 (T)	0.070	2012
7407 ST UA90X90X6	PASS	0.00	BS-4.6 (T)	0.000	2012
7408 ST UA90X90X6	PASS	21.93 T	BS-4.6 (T)	0.060	2011
7409 ST UA90X90X6	PASS	0.00	BS-4.6 (T)	0.000	2011
7410 ST UA90X90X6	PASS	45.68 T	BS-4.6 (T)	0.125	2013
7411 ST UA90X90X6	PASS	13.68 T	BS-4.6 (T)	0.050	2014
7412 ST UA90X90X6	PASS	90.00 T	BS-4.6 (T)	0.240	2013
7501 ST UA90X90X6	PASS	2.73 T	BS-4.6 (T)	0.007	2012
7502 ST UA90X90X6	PASS	17.64 T	BS-4.6 (T)	0.048	2012
7503 ST UA90X90X6	PASS	5.23 T	BS-4.6 (T)	0.014	2014
7504 ST UA90X90X6	PASS	0.22 T	BS-4.6 (T)	0.001	2013
7505 ST UA90X90X6	PASS	0.22 T	BS-4.6 (T)	0.001	2013
7506 ST UA90X90X6	PASS	5.24 T	BS-4.6 (T)	0.014	2014
§ STAAD SPACE				0.00	0.00

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7001 ST UA90X90X6 PASS 17.65 T 0.048 0.00 2012 6.32
7002 ST UA90X90X6 PASS 2.73 T 0.007 0.00 2012 0.00
7003 ST UA90X90X6 PASS 0.22 T 0.001 0.00 2013 0.00
7004 ST UA90X90X6 PASS 5.01 T 0.014 0.00 2014 5.26
7005 ST UA90X90X6 PASS 5.01 T 0.014 0.00 2014 0.00
7006 ST UA90X90X6 PASS 0.22 T 0.001 0.00 2013 0.00
7007 ST UA90X90X6 PASS 1.00 T 0.003 0.00 2014 0.00
7008 ST UA90X90X6 PASS 9.84 T 0.020 0.00 2014 0.00
7101 ST UA90X90X6 PASS 89.00 T 0.243 0.00 2012 0.00
7102 ST UA90X90X6 PASS 106.47 T 0.291 0.00 2012 9.20
7103 ST UA90X90X6 PASS 23.65 T 0.065 0.00 2012 0.00
7104 ST UA90X90X6 PASS 49.16 T 0.134 0.00 2013 0.00
7105 ST UA90X90X6 PASS 27.44 T 0.075 0.00 2012 0.00
7106 ST UA90X90X6 PASS 20.31 T 0.057 0.00 2013 0.00
7107 ST UA90X90X6 PASS 24.1 T 0.060 0.00 2012 6.69
7108 ST UA90X90X6 PASS 0.00 0.000 0.00 2012 0.22
§ STAAD SPACE BS-4.6 (T) 0.00 0.00 -- PAGE NO. 511

ALL UNITS ARE - KN METE (UNLESS OTHERWISE NOTED)

MEMBER	TABLE	RESULT/ FX	CRITICAL COND/ MY	RATIO/ M/Z	LOADING/ LOCATION
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7109 ST UA90X90X6	PASS	48.23 T	BS-4.6 (T)	0.132	2013
7110 ST UA90X90X6	PASS	0.55 T	BS-4.6 (T)	0.000	2014
7111 ST UA90X90X6	PASS	87.56 T	BS-4.6 (T)	0.239	2013
7112 ST UA90X90X6	PASS	13.09 T	BS-4.6 (T)	0.036	2014
7201 ST UA90X90X6	PASS	91.84 T	BS-4.6 (T)	0.251	2012
7202 ST UA90X90X6	PASS	91.41 T	BS-4.6 (T)	0.250	2012
7203 ST UA90X90X6	PASS	0.00	BS-4.6 (T)	0.000	2012
7204 ST UA90X90X6	PASS	6.63 T	BS-4.6 (T)	0.018	2012
7205 ST UA90X90X6	PASS	1.00 T	BS-4.6 (T)	0.003	2012
7206 ST UA90X90X6	PASS	3.32 T	BS-4.6 (T)	0.000	2013
7207 ST UA90X90X6	PASS	1.44 T	BS-4.6 (T)	0.004	2014
7208 ST UA90X90X6	PASS	2.89 T	BS-4.6 (T)	0.008	2012
7209 ST UA90X90X6	PASS	0.93 T	BS-4.6 (T)	0.003	2014
7210 ST UA90X90X6	PASS	2.66 T	BS-4.6 (T)	0.007	2012
7211 ST UA90X90X6	PASS	15.09 T	BS-4.6 (T)	0.041	2014
7212 ST UA90X90X6	PASS	14.78 T	BS-4.6 (T)	0.040	2014
7301 ST UA90X90X6	PASS	89.71 T	BS-4.6 (T)	0.245	2012
7302 ST UA90X90X6	PASS	94.85 T	BS-4.6 (T)	0.250	2012
7303 ST UA90X90X6	PASS	0.27 T	BS-4.6 (T)	0.001	2011
7304 ST UA90X90X6	PASS	0.74 T	BS-4.6 (T)	0.002	2011
7305 ST UA90X90X6	PASS	0.75 T	BS-4.6 (T)	0.002	2014

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9306 ST UA90X90X6 PASS BS-4.6 (T) 0.00 2013 0.00
19.39 T

***** END OF TABULATED RESULT OF DESIGN *****

8101 ST UA90X90X6	PASS	BS-4.6 (T)	0.052	2011
8102 ST UA90X90X6	19.10 T	0.00	0.00	0.00
8103 ST UA90X90X6	PASS	BS-4.6 (T)	0.008	2001
8104 ST UA90X90X6	2.79 T	0.00	0.00	4.49
8105 ST UA90X90X6	PASS	BS-4.6 (T)	0.115	2012
8106 ST UA90X90X6	42.23 T	0.00	0.00	5.41
8107 ST UA90X90X6	PASS	BS-4.6 (T)	0.000	2012
8108 ST UA90X90X6	0.00	0.00	0.00	0.14
8109 ST UA90X90X6	PASS	BS-4.6 (T)	0.005	2001
8110 ST UA90X90X6	1.84 T	0.00	0.00	3.81
8111 ST UA90X90X6	PASS	BS-4.6 (T)	0.000	2001
8112 ST UA90X90X6	0.00	0.00	0.00	0.10
8113 ST UA90X90X6	PASS	BS-4.6 (T)	0.000	2001
8114 ST UA90X90X6	0.00	0.00	0.00	0.00
8115 ST UA90X90X6	PASS	BS-4.6 (T)	0.000	2001
8116 ST UA90X90X6	0.00	0.00	0.00	0.00
9001 ST UA90X90X6	PASS	BS-4.6 (T)	0.000	2011
9002 ST UA90X90X6	54.16 T	0.00	0.148	2013
	PASS	BS-4.6 (T)	0.015	2011
	5.60 T	0.00	0.00	0.00
STAAD SPACE				-- PAGE NO. 517

ALL UNITS ARE - KN METE (UNLESS OTHERWISE NOTED)

MEMBER	TABLE	RESULT/ FX	CRITICAL COND/ MY	RATIO/ MZ	LOADING/ LOCATION
9003 ST UA90X90X6	PASS	BS-4.6 (T)	0.00	0.125	2013
9004 ST UA90X90X6	45.60 T	0.00	0.00	0.00	6.07
9005 ST UA90X90X6	PASS	BS-4.6 (T)	0.00	0.018	2011
9006 ST UA90X90X6	6.46 T	0.00	0.00	0.00	0.00
9101 ST UA90X90X6	PASS	BS-4.6 (T)	0.00	0.051	2013
9102 ST UA90X90X6	18.61 T	0.00	0.00	0.00	0.00
9103 ST UA90X90X6	PASS	BS-4.6 (T)	0.00	0.012	2011
9104 ST UA90X90X6	4.27 T	0.00	0.00	0.00	0.00
9105 ST UA90X90X6	PASS	BS-4.6 (T)	0.00	0.002	2014
9106 ST UA90X90X6	0.77 T	0.00	0.00	0.00	0.00
9107 ST UA90X90X6	PASS	BS-4.6 (T)	0.00	0.001	2014
9108 ST UA90X90X6	0.35 T	0.00	0.00	0.00	0.00
9109 ST UA90X90X6	PASS	BS-4.6 (T)	0.00	0.00	0.00
9110 ST UA90X90X6	2.74 T	0.00	0.00	0.007	2013
9111 ST UA90X90X6	PASS	BS-4.6 (T)	0.00	0.00	0.00
9112 ST UA90X90X6	2.57 T	0.00	0.00	0.001	2014
9113 ST UA90X90X6	PASS	BS-4.6 (T)	0.00	0.00	0.00
9114 ST UA90X90X6	0.54 T	0.00	0.00	0.007	2012
9115 ST UA90X90X6	PASS	BS-4.6 (T)	0.00	0.00	0.00
9116 ST UA90X90X6	2.56 T	0.00	0.00	0.000	2012
9117 ST UA90X90X6	PASS	BS-4.6 (T)	0.00	0.00	0.00
9118 ST UA90X90X6	0.00	0.00	0.00	0.00	0.00
9119 ST UA90X90X6	PASS	BS-4.6 (T)	0.00	0.020	2012
9120 ST UA90X90X6	7.20 T	0.00	0.00	0.00	8.35
9121 ST UA90X90X6	PASS	BS-4.6 (T)	0.00	0.006	2013
9122 ST UA90X90X6	2.10 T	0.00	0.00	0.00	0.00
9123 ST UA90X90X6	PASS	BS-4.6 (T)	0.00	0.019	2012
9124 ST UA90X90X6	6.88 T	0.00	0.00	0.00	8.34
9125 ST UA90X90X6	PASS	BS-4.6 (T)	0.00	0.017	2011
9126 ST UA90X90X6	6.15 T	0.00	0.00	0.00	0.00
9127 ST UA90X90X6	PASS	BS-4.6 (T)	0.00	0.149	2013
9128 ST UA90X90X6	54.54 T	0.00	0.00	0.00	6.09
9129 ST UA90X90X6	PASS	BS-4.6 (T)	0.00	0.020	2011
9130 ST UA90X90X6	7.17 T	0.00	0.00	0.00	0.00
9131 ST UA90X90X6	PASS	BS-4.6 (T)	0.00	0.127	2013
9132 ST UA90X90X6	46.46 T	0.00	0.00	0.00	6.07
9133 ST UA90X90X6	PASS	BS-4.6 (T)	0.00	0.004	2011
9134 ST UA90X90X6	1.36 T	0.00	0.00	0.00	0.00

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UNITED									
JOINT	LOAD	FORCE-X	FORCE-Y	FORCE-Z	MOM-X	MOM-Y	MOM-Z	STAND SPACE	UNITED
1113	1222	-0.10	511.45	-62.39	0.00	0.00	0.00	0.00	0.00
1213	1321	-0.07	388.38	-61.35	0.00	0.00	0.00	0.00	0.00
1014	1322	-0.10	323.29	-82.89	0.00	0.00	0.00	0.00	0.00
1114	1001	-0.11	293.25	-88.31	0.00	0.00	0.00	0.00	0.00
1214	1011	-0.07	170.18	-87.27	0.00	0.00	0.00	0.00	0.00
1021	1211	-1.79	400.48	-8.26	0.00	0.00	0.00	0.00	0.00
1022	1212	-0.09	421.89	-31.55	0.00	0.00	0.00	0.00	0.00
1121	1121	-1.58	397.87	-3.99	0.00	0.00	0.00	0.00	0.00
1122	1122	-0.11	376.45	19.30	0.00	0.00	0.00	0.00	0.00
1221	1221	-1.76	278.21	-4.48	0.00	0.00	0.00	0.00	0.00
STAND SPACE									
SUPPORT REACTIONS -UNIT KN METE STRUCTURE TYPE = SPACE									
JOINT	LOAD	FORCE-X	FORCE-Y	FORCE-Z	MOM-X	MOM-Y	MOM-Z	STAND SPACE	UNITED
1222	1222	-0.06	299.62	-27.77	0.00	0.00	0.00	0.00	0.00
1321	1321	1.61	275.60	-0.21	0.00	0.00	0.00	0.00	0.00
1322	1322	-0.09	254.18	23.08	0.00	0.00	0.00	0.00	0.00
1001	1001	0.05	273.86	18.48	0.00	0.00	0.00	0.00	0.00
1011	1011	0.06	282.82	-102.74	0.00	0.00	0.00	0.00	0.00
1211	1211	0.04	182.87	-139.84	0.00	0.00	0.00	0.00	0.00
1012	1012	0.04	228.77	78.28	0.00	0.00	0.00	0.00	0.00
1112	1112	0.04	199.78	-102.50	0.00	0.00	0.00	0.00	0.00
1212	1212	0.04	86.31	-104.00	0.00	0.00	0.00	0.00	0.00
1013	1013	0.05	451.59	55.14	0.00	0.00	0.00	0.00	0.00
1113	1113	0.03	459.74	53.35	0.00	0.00	0.00	0.00	0.00
1213	1213	0.02	346.27	51.66	0.00	0.00	0.00	0.00	0.00
1014	1014	0.05	284.97	59.78	0.00	0.00	0.00	0.00	0.00
1114	1114	0.02	265.35	58.77	0.00	0.00	0.00	0.00	0.00
1214	1214	0.02	151.88	57.07	0.00	0.00	0.00	0.00	0.00
1021	1021	-2.68	365.26	8.82	0.00	0.00	0.00	0.00	0.00
1022	1022	0.03	321.45	-17.76	0.00	0.00	0.00	0.00	0.00
1121	1121	2.73	363.51	9.20	0.00	0.00	0.00	0.00	0.00
1122	1122	0.02	407.32	35.78	0.00	0.00	0.00	0.00	0.00
1221	1221	-2.70	256.19	3.63	0.00	0.00	0.00	0.00	0.00
1222	1222	0.11	212.36	-24.95	0.00	0.00	0.00	0.00	0.00
1321	1321	0.72	208.44	30.91	0.00	0.00	0.00	0.00	0.00
1322	1322	-0.01	528.53	-17.94	0.00	0.00	0.00	0.00	0.00
1001	1001	-0.02	415.83	-46.55	0.00	0.00	0.00	0.00	0.00
1011	1011	-0.01	400.80	-44.40	0.00	0.00	0.00	0.00	0.00
1111	1111	-0.01	278.59	-42.51	0.00	0.00	0.00	0.00	0.00
1211	1211	-0.02	378.08	-85.48	0.00	0.00	0.00	0.00	0.00
1012	1012	-0.01	356.86	-89.81	0.00	0.00	0.00	0.00	0.00
1112	1112	-0.01	234.56	-87.93	0.00	0.00	0.00	0.00	0.00
1212	1212	-0.01	509.16	-67.30	0.00	0.00	0.00	0.00	0.00
1013	1013	0.00	509.78	-68.60	0.00	0.00	0.00	0.00	0.00
1113	1113	0.00	387.48	-66.71	0.00	0.00	0.00	0.00	0.00
1213	1213	-0.01	322.43	-87.19	0.00	0.00	0.00	0.00	0.00
1014	1014	0.00	291.93	-91.81	0.00	0.00	0.00	0.00	0.00
1114	1114	0.00	169.63	-89.92	0.00	0.00	0.00	0.00	0.00
1214	1214	-1.69	397.32	-9.17	0.00	0.00	0.00	0.00	0.00
1021	1021	0.64	361.47	-5.68	0.00	0.00	0.00	0.00	0.00
1121	1121	-0.03	376.56	19.67	0.00	0.00	0.00	0.00	0.00
1122	1122	-1.68	272.51	-4.22	0.00	0.00	0.00	0.00	0.00
1221	1221	0.02	298.61	-32.93	0.00	0.00	0.00	0.00	0.00
1321	1321	1.68	274.86	-4.28	0.00	0.00	0.00	0.00	0.00
1322	1322	-0.02	250.75	24.43	0.00	0.00	0.00	0.00	0.00
1601	1001	0.03	472.75	16.00	0.00	0.00	0.00	0.00	0.00
1011	1011	0.04	279.93	-105.67	0.00	0.00	0.00	0.00	0.00
1111	1111	0.03	260.70	-133.41	0.00	0.00	0.00	0.00	0.00
STAND SPACE									
SUPPORT REACTIONS -UNIT KN METE STRUCTURE TYPE = SPACE									
JOINT	LOAD	FORCE-X	FORCE-Y	FORCE-Z	MOM-X	MOM-Y	MOM-Z	STAND SPACE	UNITED
1211	1211	0.03	147.14	-134.67	0.00	0.00	0.00	0.00	0.00
1012	1012	0.04	225.51	-81.45	0.00	0.00	0.00	0.00	0.00
1112	1112	0.03	197.22	-105.16	0.00	0.00	0.00	0.00	0.00
1212	1212	0.03	83.66	-106.42	0.00	0.00	0.00	0.00	0.00
1013	1013	0.03	450.34	51.72	0.00	0.00	0.00	0.00	0.00
1113	1113	0.02	459.52	50.21	0.00	0.00	0.00	0.00	0.00

1801	1001	0.05	851.83	-30.04	0.00	0.00	0.00
1011	-0.24	566.88	-14.82	0.00	0.00	0.00	0.00
1111	-0.31	266.58	-134.95	0.00	0.00	0.00	0.00
1211	-0.31	151.32	-136.38	0.00	0.00	0.00	0.00
1012	-0.68	510.47	-117.32	0.00	0.00	0.00	0.00
1112	-0.82	203.11	-106.82	0.00	0.00	0.00	0.00
1212	-0.82	87.87	-108.05	0.00	0.00	0.00	0.00
1013	0.05	735.00	16.89	0.00	0.00	0.00	0.00
1113	0.03	465.07	49.77	0.00	0.00	0.00	0.00
1213	0.02	349.83	48.54	0.00	0.00	0.00	0.00
1014	-0.82	568.46	22.65	0.00	0.00	0.00	0.00
1114	-0.98	270.76	56.48	0.00	0.00	0.00	0.00
1214	-0.99	155.53	55.25	0.00	0.00	0.00	0.00
1021	-98.08	366.07	0.51	0.00	0.00	0.00	0.00
1022	-0.17	439.70	-36.16	0.00	0.00	0.00	0.00
STAAD SPACE							

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SUPPORT REACTIONS -UNIT KN METE STRUCTURE TYPE = SPACE							
JOINT	LOAD	FORCE-X	FORCE-Y	FORCE-Z	MOM-X	MOM-Y	MOM-Z
1121	98.13	605.66	-14.60	0.00	0.00	0.00	0.00
1221	-98.10	532.03	22.07	0.00	0.00	0.00	0.00
1321	-0.19	139.49	10.33	0.00	0.00	0.00	0.00
1012	-0.19	213.11	-26.34	0.00	0.00	0.00	0.00
1112	98.12	379.07	-4.79	0.00	0.00	0.00	0.00
1212	-0.71	505.45	31.88	0.00	0.00	0.00	0.00
1013	-0.37	432.40	-16.21	0.00	0.00	0.00	0.00
1113	-0.41	481.44	-48.73	0.00	0.00	0.00	0.00
1213	-0.41	418.50	-45.14	0.00	0.00	0.00	0.00
1014	-0.72	386.77	-87.79	0.00	0.00	0.00	0.00
1114	-0.82	365.71	-93.12	0.00	0.00	0.00	0.00
1214	-0.82	242.70	-91.52	0.00	0.00	0.00	0.00
1015	-0.01	518.03	-65.54	0.00	0.00	0.00	0.00
1115	-0.01	393.52	-65.56	0.00	0.00	0.00	0.00
1215	-1.19	329.92	-87.66	0.00	0.00	0.00	0.00
1016	-1.37	300.38	-92.97	0.00	0.00	0.00	0.00
1116	-60.41	308.93	-91.38	0.00	0.00	0.00	0.00
1216	-0.01	430.20	-38.25	0.00	0.00	0.00	0.00
1017	60.38	498.54	-0.05	0.00	0.00	0.00	0.00
1117	-60.04	377.57	22.04	0.00	0.00	0.00	0.00
1217	-60.00	383.32	-11.94	0.00	0.00	0.00	0.00
1018	60.35	378.86	-31.42	0.00	0.00	0.00	0.00
1118	-0.03	253.69	26.56	0.00	0.00	0.00	0.00
1218	-0.02	485.70	12.13	0.00	0.00	0.00	0.00
1019	1.29	294.31	-111.52	0.00	0.00	0.00	0.00
1119	1.53	272.83	-137.07	0.00	0.00	0.00	0.00
1219	1.53	155.95	-138.26	0.00	0.00	0.00	0.00
1020	1.45	239.68	-87.69	0.00	0.00	0.00	0.00
1120	1.72	209.10	-109.26	0.00	0.00	0.00	0.00
1220	1.72	92.21	-110.45	0.00	0.00	0.00	0.00
1021	0.00	464.08	48.32	0.00	0.00	0.00	0.00
1121	0.03	470.90	49.41	0.00	0.00	0.00	0.00
1221	0.02	384.01	48.21	0.00	0.00	0.00	0.00
1022	0.45	296.93	53.61	0.00	0.00	0.00	0.00
1122	0.35	273.89	53.58	0.00	0.00	0.00	0.00
1222	0.41	352.00	54.38	0.00	0.00	0.00	0.00
1023	-9.41	322.86	-2.15	0.00	0.00	0.00	0.00
1123	96.45	375.07	-2.96	0.00	0.00	0.00	0.00
1223	-0.30	418.80	36.69	0.00	0.00	0.00	0.00
1024	-96.45	263.77	1.76	0.00	0.00	0.00	0.00
1124	0.31	270.04	-27.94	0.00	0.00	0.00	0.00
1224	96.47	262.22	3.60	0.00	0.00	0.00	0.00
1025	-0.29	305.95	33.31	0.00	0.00	0.00	0.00
STAAD SPACE							

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SUPPORT REACTIONS -UNIT KN METE STRUCTURE TYPE = SPACE							
JOINT	LOAD	FORCE-X	FORCE-Y	FORCE-Z	MOM-X	MOM-Y	MOM-Z
1212	0.58	244.25	-91.39	0.00	0.00	0.00	0.00
1013	-0.02	518.89	-69.06	0.00	0.00	0.00	0.00
1113	-0.01	388.95	-67.72	0.00	0.00	0.00	0.00
1213	0.00	332.04	-87.63	0.00	0.00	0.00	0.00
1014	1.31	302.41	-90.92	0.00	0.00	0.00	0.00
1114	1.32	177.93	-90.92	0.00	0.00	0.00	0.00
1214	-1.65	495.62	-0.04	0.00	0.00	0.00	0.00
1021	0.58	439.73	-39.11	0.00	0.00	0.00	0.00
1121	1.63	311.94	-16.72	0.00	0.00	0.00	0.00
1221	-0.61	377.83	22.34	0.00	0.00	0.00	0.00
1022	-1.65	371.92	4.64	0.00	0.00	0.00	0.00
1122	0.59	306.02	-34.42	0.00	0.00	0.00	0.00
1222	1.64	188.23	-12.04	0.00	0.00	0.00	0.00
1023	-0.60	254.13	27.03	0.00	0.00	0.00	0.00
1123	0.04	473.77	18.09	0.00	0.00	0.00	0.00
1223	0.05	282.00	-103.20	0.00	0.00	0.00	0.00
1024	0.03	282.63	-130.96	0.00	0.00	0.00	0.00
1124	0.06	279.60	-137.66	0.00	0.00	0.00	0.00
1224	0.06	279.60	-137.66	0.00	0.00	0.00	0.00
1025	0.03	159.02	-103.62	0.00	0.00	0.00	0.00
1125	0.04	85.49	-104.35	0.00	0.00	0.00	0.00
1225	0.04	451.57	54.57	0.00	0.00	0.00	0.00
1026	0.02	459.96	53.10	0.00	0.00	0.00	0.00
1126	0.02	346.43	51.38	0.00	0.00	0.00	0.00
1226	0.04	284.73	59.11	0.00	0.00	0.00	0.00
1027	0.02	265.32	58.40	0.00	0.00	0.00	0.00
1127	0.02	151.79	56.68	0.00	0.00	0.00	0.00
1227	0.02	151.79	56.68	0.00	0.00	0.00	0.00
STAAD SPACE							

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SUPPORT REACTIONS -UNIT KN METE STRUCTURE TYPE = SPACE							
JOINT	LOAD	FORCE-X	FORCE-Y	FORCE-Z	MOM-X	MOM-Y	MOM-Z
1212	0.58	244.25	-91.39	0.00	0.00	0.00	0.00
1013	-0.02	518.89	-69.06	0.00	0.00	0.00	0.00
1113	-0.01	388.95	-67.72	0.00	0.00	0.00	0.00
1213	0.00	332.04	-87.63	0.00	0.00	0.00	0.00
1014	1.31	302.41	-90.92	0.00	0.00	0.00	0.00
1114	1.32	177.93	-90.92	0.00	0.00	0.00	0.00
1214	-1.65	495.62	-0.04	0.00	0.00	0.00	0.00
1021	0.58	439.73	-39.11	0.00	0.00	0.00	0.00
1121	1.63	311.94	-16.72	0.00	0.00	0.00	0.00
1221	-0.61	377.83	22.34	0.00	0.00	0.00	0.00
1022	-1.65	371.92	4.64	0.00	0.00	0.00	0.00
1122	0.59	306.02	-34.42	0.00	0.00	0.00	0.00
1222	1.64	188.23	-12.04	0.00	0.00	0.00	0.00
1023	-0.60	254.13	27.03	0.00	0.00	0.00	0.00
1123	0.04	473.77	18.09	0.00	0.00	0.00	0.00
1223	0.05	282.00	-103.20	0.00	0.00	0.00	0.00
1024	0.03	282.63	-130.96	0.00	0.00	0.00	0.00
1124	0.06	279.60	-137.66	0.00	0.00	0.00	0.00
1224	0.06	279.60	-137.66	0.00	0.00	0.00	0.00
1025	0.03	159.02	-103.62	0.00	0.00	0.00	0.00
1125	0.04	85.49	-104.35	0.00	0.00	0.00	0.00
1225	0.04	451.57	54.57	0.00	0.00	0.00	0.00
1026	0.02	459.96	53.10	0.00	0.00	0.00	0.00
1126	0.02	346.43	51.38	0.00	0.00	0.00	0.00
1226	0.04	284.73	59.11	0.00	0.00	0.00	0.00
1027	0.02	265.32	58.40	0.00	0.00	0.00	0.00
1127	0.02	151.79	56.68	0.00	0.00	0.00	0.00
1227	0.02	151.79	56.68	0.00	0.00	0.00	0.00
STAAD SPACE							

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1111	0.00	-68.45	-17.59	0.00	0.00
1211	0.00	-106.04	-17.37	0.00	0.00
1012	0.00	-7.33	-16.81	0.00	0.00
1112	0.00	-58.05	-19.35	0.00	0.00
1212	0.00	-95.63	-19.14	0.00	0.00
1013	-0.01	230.05	-2.79	0.00	0.00
1113	-0.01	218.66	-3.00	0.00	0.00
1213	-0.01	181.07	-2.78	0.00	0.00
1014	0.00	131.39	-3.93	0.00	0.00
1114	0.00	163.85	-4.11	0.00	0.00
1214	0.00	153.86	-0.77	0.00	0.00
1021	-0.01	16.71	-11.09	0.00	0.00
1121	0.00	109.19	-0.73	0.00	0.00
1221	-0.01	244.34	9.59	0.00	0.00
1022	-0.01	105.91	-0.51	0.00	0.00
1122	0.00	-29.24	-10.83	0.00	0.00
1222	0.00	63.24	-0.47	0.00	0.00
1321	0.00	198.39	9.85	0.00	0.00
1322	0.00			0.00	0.00

***** END OF LATEST ANALYSIS RESULT *****

MAIN FRAME

Untitled									
1.14	MIN	-50.37	1212	-485.15	1013	UNTITLED			
1.70	MAX	63.45	1122	560.25	1211				
2.27	MIN	-50.44	1212	-509.15	1013				
2.84	MAX	63.37	1122	573.84	1212				
	MIN	-50.51	1212	-537.13	1122				
	MAX	63.28	1122	602.58	1212				
	MIN	-50.58	1212	-573.11	1122				
	MAX	63.20	1122	631.31	1212				
	MIN	-50.65	1212	-609.03	1122				

MAX/MIN FORCE VALUES FOR MEMB 1904, AMONGST ALL SECT LOCATIONS									
FY/	DIST	LD	LD	LD	LD	LD	LD	LD	LD
FZ	DIST	LD	LD	LD	LD	LD	LD	LD	LD
MAX.	63.61	0.00	1122	631.31	2.84	1212			
MIN.	-50.65	2.84	1212	-609.03	2.84	1122			
	-8.68	2.84	1021	-12.31	2.84	1021			

1905									
0.00	MAX	98.94	1011	633.75	1212				
0.24	MIN	-84.51	1214	-610.47	1122				
0.48	MAX	98.86	1011	621.15	1212				
0.72	MIN	-84.57	1214	-608.57	1122				
0.97	MAX	98.83	1011	627.05	1212				
1.21	MIN	-84.60	1214	-627.85	1122				
		98.79	1011	583.37	1212				
		-64.63	1214	-633.62	1122				
		98.76	1011	570.79	1212				
		-64.66	1214	-639.39	1122				

MAX/MIN FORCE VALUES FOR MEMB 1905, AMONGST ALL SECT LOCATIONS									
FY/	DIST	LD	LD	LD	LD	LD	LD	LD	LD
FZ	DIST	LD	LD	LD	LD	LD	LD	LD	LD
MAX.	98.94	0.00	1011	633.75	0.00	1212			
MIN.	-5.72	0.00	1211	-15.94	0.00	1221			
	-64.66	1.21	1214	-639.39	1.21	1122			
	-5.71	1.21	1221	-15.97	0.00	1121			

1906									
0.00	MAX	68.02	1114	639.39	1122				
0.21	MIN	-80.69	1211	-570.79	1212				
0.42	MAX	67.99	1114	638.85	1122				
0.63	MIN	-80.71	1211	-560.19	1212				
0.84	MAX	67.95	1114	638.32	1122				
1.05	MIN	-80.73	1211	-549.57	1212				
		67.92	1114	637.79	1122				
		-80.75	1211	-538.96	1212				
		67.99	1114	636.27	1122				
		-80.78	1211	-538.76	1122				
		67.88	1114	635.77	1122				
		-80.80	1211	-517.72	1212				

MAX/MIN FORCE VALUES FOR MEMB 1906, AMONGST ALL SECT LOCATIONS									
FY/	DIST	LD	LD	LD	LD	LD	LD	LD	LD
FZ	DIST	LD	LD	LD	LD	LD	LD	LD	LD
MAX.	68.02	0.00	1114	639.39	0.00	1122			
MIN.	-80.80	0.00	1221	-570.79	0.00	1212			
	-5.72	1.05	1211	-9.09	0.00	1221			

1907									
0.00	MAX	111.43	1013	636.76	1122				
0.21	MIN	-51.34	1211	-517.72	1212				
0.42	MAX	111.30	1013	635.56	1122				
0.63	MIN	-51.45	1211	-508.92	1212				
0.84	MAX	111.16	1013	614.39	1122				
1.05	MIN	-51.56	1211	-500.11	1212				
		111.03	1013	603.26	1122				
		-51.67	1211	-491.27	1212				
		110.30	1013	592.14	1122				

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Untitled									
0.00	MAX	98.21	1113	0.00	1322				
0.64	MIN	-121.29	1211	0.00	1322				
1.29	MAX	98.10	1113	78.08	1211				
1.93	MIN	-121.37	1211	-78.17	1211				
2.57	MAX	98.00	1113	156.26	1211				
3.22	MIN	-121.44	1211	-126.26	1113				
		97.89	1113	234.38	1211				
		-121.52	1211	-189.29	1113				
		97.79	1113	312.60	1211				
		-121.60	1211	-252.26	1113				
		97.68	1113	390.88	1211				
		-121.67	1211	-315.13	1113				

MAX/MIN FORCE VALUES FOR MEMB 1901, AMONGST ALL SECT LOCATIONS									
FY/	DIST	LD	LD	LD	LD	LD	LD	LD	LD
FZ	DIST	LD	LD	LD	LD	LD	LD	LD	LD
MAX.	98.21	0.00	1113	390.88	3.22	1211			
MIN.	-121.67	3.22	1211	-315.13	3.22	1113			
	-2.64	3.22	1021	-8.50	3.22	1021			

1902									
0.00	MAX	71.59	1122	390.44	1211				
0.43	MIN	-68.08	1212	-314.99	1211				
0.86	MAX	71.53	1122	415.88	1211				
1.30	MIN	-68.13	1212	-340.16	1113				
1.73	MAX	71.46	1122	441.29	1211				
2.16	MIN	-68.19	1212	-365.30	1113				
		71.40	1122	466.76	1211				
		-68.24	1212	-390.91	1013				
		71.34	1122	492.24	1211				
		-68.29	1212	-418.07	1013				
		71.27	1122	517.75	1211				
		-68.34	1212	-445.19	1013				

MAX/MIN FORCE VALUES FOR MEMB 1902, AMONGST ALL SECT LOCATIONS									
FY/	DIST	LD	LD	LD	LD	LD	LD	LD	LD
FZ	DIST	LD	LD	LD	LD	LD	LD	LD	LD
MAX.	71.59	0.00	1122	517.75	2.16	1211			
MIN.	-13.41	0.00	1321	-16.47	0.00	1021			
	-68.34	2.16	1212	-445.19	2.16	1013			
	-13.56	2.16	1021	-16.30	0.00	1321			

1903									
0.00	MAX	71.74	1122	517.75	1211				
0.05	MIN	-68.30	1212	-445.19	1013				
0.10	MAX	71.74	1122	520.75	1211				
0.15	MIN	-68.31	1212	-448.42	1013				
0.20	MAX	71.73	1122	523.75	1211				
0.25	MIN	-68.32	1212	-451.64	1013				
		71.72	1122	526.75	1211				
		-68.37	1212	-454.96	1013				
		71.71	1122	529.76	1211				
		-68.33	1212	-458.08	1013				
		71.71	1122	532.76	1211				
		-68.33	1212	-461.30	1013				

MAX/MIN FORCE VALUES FOR MEMB 1903, AMONGST ALL SECT LOCATIONS									
FY/	DIST	LD	LD	LD	LD	LD	LD	LD	LD
FZ	DIST	LD	LD	LD	LD	LD	LD	LD	LD
MAX.	71.74	0.00	1122	532.76	0.25	1211			
MIN.	-68.33	0.25	1212	-461.30	0.25	1013			
	-13.56	0.25	1021	-16.26	0.25	1021			

1904									
0.00	MAX	63.61	1122	533.18	1211				
0.57	MIN	-50.30	1212	-461.30	1013				
		63.53	1122	546.69	1211				

Page 1

Untitled										
MIN	-51.78	1211	-482.41	1212	-5.72	1121	-2.22	1321		
MAX	110.78	1013	581.06	1112	5.71	1321	3.43	1021		
MIN	-51.88	1211	-473.53	1212	-5.72	1121	-3.43	1321		
STAAD SPACE										
							PAGE NO.	125		

MAX/MIN FORCE VALUES FOR MEMB												1907. AMONGST ALL SECT LOCATIONS																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																											
	FY/	FZ	LD	DIST	LD	MZ/	DIST	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD	LD

MAX/MIN FORCE VALUES FOR MEMB										
FY/	FZ	LD	DIST	LD	MZ/	DIST	LD	LD	LD	
MAX	0.00	MAX	175.76	1013	581.06	1122	5.71	1221	3.85	1021
MIN	0.21	MIN	-27.49	1212	-473.53	1212	-5.72	1121	-3.85	1321
STAAD SPACE										
MEMB										
1	0.42	MAX	27.68	1212	367.71	1122	5.72	1221	5.06	1321
		MIN	-17.34	1013	-501.11	1212	-5.71	1121	-5.06	1021
2	0.42	MAX	27.34	1013	501.11	1122	5.71	1221	6.26	1321
		MIN	-22.84	1212	-461.87	1212	-5.72	1121	-6.26	1021
3	0.63	MAX	175.13	1013	531.22	1122	5.71	1221	7.47	1021
		MIN	-28.01	1212	-455.98	1212	-5.72	1121	-7.47	1321
4	0.84	MAX	174.93	1013	501.37	1122	5.71	1221	8.67	1221
		MIN	-28.18	1212	-450.06	1212	-5.72	1121	-8.67	1121
5	1.05	MAX	174.72	1013	481.55	1122	5.71	1221	9.87	1221
		MIN	-28.36	1212	-444.10	1212	-5.72	1121	-9.88	1121

MAX/MIN FORCE VALUES FOR MEMB												
	FY/	FZ	LD	DIST	LD	MZ/	DIST	LD	LD	LD	LD	LD
MAX.	175.76	0.00	1013	0.00	1122	581.06	0.00	1122	150.64	C	0.00	1011
	5.71	0.00	1221	0.00	1212	9.87	1.05	1212				
MIN.	-28.36	1.05	1212	-473.53	0.00	-9.88	1.05	1212	48.65	T	1.05	1214
	-5.72	1.05	1121									

MAX/MIN FORCE VALUES FOR MEMB										
FY/	FZ	LD	DIST	LD	MZ/	DIST	LD	LD	LD	
MAX	0.00	MAX	212.74	1013	481.55	1122	22.78	1321	9.25	1221
MIN	0.21	MIN	-27.49	1212	-473.53	1212	-22.79	1021	-9.25	1321
STAAD SPACE										

MAX/MIN FORCE VALUES FOR MEMB										1909, AMONGST ALL SECT LOCATIONS			
	FY/	FZ	LD	DIST	LD	MZ/	DIST	LD	LD	FX	DIST	LD	
MAX.	212.74	0.00	1013	0.00	1122	481.55	0.00	1122					
	22.78	0.00	1321	14.76	1.05	1312							
MIN.	-10.63	1.05	1212	-44.10	0.00	1212				120.26	C	0.00 1011	
	-22.79	1.05	1021	-14.78	1.05	1021				62.21	T	1.05 1214	

MAX/MIN FORCE VALUES FOR MEMB										
FY/	FZ	LD	DIST	LD	MZ/	DIST	LD	LD	LD	
MAX	0.00	MAX	117.23	1122	345.10	1122	6.08	1021	15.16	1321
MIN	0.29	MIN	-77.44	1212	-433.44	1212	-6.06	1321	-15.18	1021
STAAD SPACE										
MAX	0.57	MAX	116.98	1112	331.71	1112	6.08	1021	13.44	1321
MIN	0.86	MIN	-77.64	1212	-411.31	1212	-6.06	1321	-13.45	1021
STAAD SPACE										
MAX	1.14	MAX	116.75	1112	288.39	1112	6.08	1021	11.71	1321
MIN	1.43	MIN	-78.01	1212	-368.06	1212	-6.06	1321	-11.71	1021
STAAD SPACE										
MAX	1.43	MAX	117.89	1112	370.46	1112	6.08	1021	9.98	1321
MIN	1.43	MIN	-78.05	1212	-370.46	1212	-6.06	1321	-9.98	1021
STAAD SPACE										
MAX	1.43	MAX	116.94	1112	222.46	1112	6.08	1021	8.25	1321
MIN	1.43	MIN	-78.76	1212	-333.69	1112	-6.06	1321	-8.25	1021
STAAD SPACE										
MAX	1.43	MAX	115.99	1112	196.97	1322	6.08	1021	6.53	1121
MIN	1.43	MIN	-78.47	1212	-336.83	1112	-6.06	1321	-6.53	1121

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Untitled												
MAX/MIN FORCE VALUES FOR MEMB												
	FY/	FZ	LD	DIST	LD	MZ/	DIST	LD	AMONGST ALL SECT LOCATIONS			
									FX	DIST	LD	
MAX.	117.23		0.00	1122		345.10		0.00	1122			
	6.08		0.00	1031		15.16		0.00	1311			
	-78.47		1.43	1212		-433.44		0.00	1212			
MIN.	-6.06		1.43	1321		-15.13		0.00	1021			
								</				

MAX/MIN FORCE VALUES FOR MEMB										
FY/	FZ	LD	DIST	LD	MZ/	DIST	LD	LD	LD	
MAX	0.00	MAX	115.98	1122	196.97	1322	6.08	1021	6.53	1121
MIN	0.30	MIN	-78.45	1212	-336.83	1112	-6.06	1321	-6.51	1221
STAAD SPACE										

MAX/MIN FORCE VALUES FOR MEMB												
	FY/	FZ	LD	DIST	LD	MZ/	DIST	LD	LD	LD	FX	DIST
MAX.	115.98	0.00	1122	0.00	1322	196.97	0.00	1322	41.73	C	0.00	1011
	6.08	0.00	1021	6.53	0.00	1121						
MIN.	-79.54	1.50	1212	-336.83	0.00	1112						
	-6.06	1.50	1321	-6.51	0.00	1221			90.23	T	1.50	1214

MAX/MIN FORCE VALUES FOR MEMB										
FY/	FZ	LD	DIST	LD	MZ/	DIST	LD	LD	LD	
MAX	0.00	MAX	71.28	1122	63.48	1322	0.41	1321	2.62	1021
MIN	0.68	MIN	-57.79	1212	-315.16	1022	-0.43	1021	-2.58	1321
STAAD SPACE										
MAX	0.68	MAX	70.69	1212	21.77	1322	-0.43	1321	2.33	1021
MIN	1.36	MIN	-78.28	1212	-318.01	1013	-0.43	1021	-2.60	1321
STAAD SPACE										
MAX	1.36	MAX	58.77	1212	34.69	1313	-0.43	1021	1.75	1021
MIN	2.04	MIN	-58.77	1122	-60.74	1322	-0.43	1321	-1.75	1021
STAAD SPACE										
MAX	2.04	MAX	69.51	1122	60.74	1322	0.43	1021	1.74	1321
MIN	2.72	MIN	-68.92	1212	-43.44	1212	-0.43	1321	-1.46	1221
STAAD SPACE										
MAX	3.41	MAX	59.76	1212	-396.85	1013	-0.43	1021	-1.46	1121
MIN		MIN	-68.33	1122	-2.56	1212	-0.43	1021	-1.17	1221
STAAD SPACE										
MAX		MAX	68.33	1212	-2.56	1212	-0.43	1021	-1.18	1121
MIN		MIN	-60.25	1212	-422.33	1013	-0.43	1021	-1.18	1121

MAX/MIN FORCE VALUES FOR MEMB												1912, AMONGST ALL SECT LOCATIONS																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																										
	FY/	FZ	LD	DIST	LD	MZ/	DIST	LD	LD	LD																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																												

1913	0.00	MAX	31.03	1322	-2.56	1212	2.06	1321	1.18	1221
	0.68	MIN	-30.59	1022	-473.53	1013	-2.06	1021	-1.18	1321
		MAX	-72.01	1022	-399.53	1013	-2.06	1021	-0.72	1321
	1.36	MIN	-72.01	1322	74.52	1212	-2.06	1321	1.62	1221
		MAX	-70.62	1022	-356.34	1013	-2.06	1021	-1.62	1321
	2.04	MIN	-70.62	1322	132.56	1212	-2.06	1321	1.62	1221
		MAX	-73.21	1022	-322.75	1013	-2.06	1021	-3.02	1321
	2.72	MIN	-73.21	1322	322.75	1212	-2.06	1321	3.02	1221
		MAX	-73.86	1022	-328.75	1013	-2.06	1021	-4.45	1321
	3.41	MIN	-73.86	1322	192.65	1212	-2.06	1321	5.83	1221
		MAX	-74.39	1022	-237.24	1122	-2.06	1021	-5.86	1321
STAAD SPACE							--	PAGE NO.	128	

MAX/MIN FORCE VALUES FOR MEMB									
FY/	FZ	LD	DIST	LD	MZ/	DIST	LD	LD	LD
MAX	0.00	MAX	31.03	1322	-2.56	1212	2.06	1321	1.18
MIN	0.68	MIN	-30.59	1022	-473.53	1013	-2.06	1021	-1.18
STAAD SPACE									

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	MIN	-85.11	1013	-488.26	1112	unfilled	-8.33	1121	-2.93	1321
1.73	MAX	5.92	1322	5.69	1322		8.29	1221	6.52	1021
	MIN	-85.17	1013	-454.67	1112		-8.33	1121	-6.52	1321
2.16	MAX	5.87	1322	1.14	1322		8.29	1221	10.09	1221
	MIN	-85.24	1013	-421.05	1112		-8.33	1121	-10.11	1121
	STAAD SPACE						--	PAGE NO.		133

	MAX/MIN FORCE VALUES FOR MEMB						1973, AMONGST ALL SECT LOCATIONS					
	FZ/ FY	DIST LD	DIST LD	MZ/ MY	DIST LD	DIST LD	FX	DIST LD	DIST LD	FX	DIST LD	
MAX.	6.11	0.00	1322	14.08	0.00	1322						
	8.29	0.00	1221	10.09	2.16	1221				437.74 C	2.16 1001	
MIN.	-85.24	2.16	1013	-589.47	0.00	1012						
	-8.33	2.16	1121	-10.11	2.16	1121				104.18 C	0.00 1214	

Year	MAX	0.00	MAX	0.55	1322	1.22	1322	1.64	1021	5.19	1321
1924	MIN	-130.93	MAX	-422.14	1112	-1.61	1321	-1.61	1321	-4.25	1021
	0.64	MIN	0.48	1322	0.89	1322	1.64	1021	5.19	1321	
	MIN	-131.04	MAX	-376.85	1322	-1.61	1321	-1.61	1321	-4.21	1021
	1.29	MIN	0.43	1322	-233.89	1112	1.64	1021	5.19	1321	
	MIN	-133.44	MAX	-233.89	1112	-1.64	1021	-1.64	1021	-2.08	1321
	1.93	MIN	0.54	1322	0.36	1322	1.64	1021	5.19	1321	
	MIN	-131.55	MAX	-169.06	1112	-1.61	1321	-1.61	1321	-1.04	1021
	2.57	MIN	0.78	1322	0.16	1322	1.64	1021	5.19	1321	
	MIN	-131.36	MAX	-84.56	1112	-1.61	1321	-1.61	1321	-1.05	1021
	3.22	MIN	0.21	1322	0.00	1322	1.64	1021	5.19	1321	
	MIN	-131.46	MAX	0.00	1322	-1.61	1321	-1.61	1321	0.00	1322

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MIN -75.76 1011 -610.81 1012 -7.02 1321 -9.79 1321
STAAD SPACE -- PAGE NO. 132

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	MAX/MIN FORCE VALUES FOR MEMB						1921, AMONGST ALL SECT LOCATIONS					
	FZ/ FY	DIST LD	DIST LD	DIST LD	DIST LD	DIST LD	MZ/ MY	DIST LD	DIST LD	DIST LD	FX	DIST LD
MAX.	34.43	0.00	0.00	1214	0.00	1322	58.76	0.00	0.00	1322		
	7.19	0.00	0.00	1021	10.16	0.00	1321					
	-75.76	2.84	1011		-770.70	0.00	1012				312.68 c	2.84 1013
MIN.	-7.02	2.84	1321		-10.40	0.00	1021				59.08 c	0.00 1214

Year	0.00	0.05	0.10	0.15	0.20	0.25
1922	MAX	MAX	MAX	MAX	MAX	MAX
	10.30	10.30	10.30	10.30	10.30	10.30
	1372	1372	1372	1372	1372	1372
	15.70	15.70	15.70	15.70	15.70	15.70
	1322	1322	1322	1322	1322	1322
	-611	-611	-611	-611	-611	-611
	98	98	98	98	98	98
	1912	1912	1912	1912	1912	1912
	-607	-607	-607	-607	-607	-607
	24	24	24	24	24	24
	1012	1012	1012	1012	1012	1012
	15.65	15.65	15.65	15.65	15.65	15.65
	1322	1322	1322	1322	1322	1322
	-602	-602	-602	-602	-602	-602
	80	80	80	80	80	80
	1012	1012	1012	1012	1012	1012
	15.12	15.12	15.12	15.12	15.12	15.12
	1322	1322	1322	1322	1322	1322
	-598	-598	-598	-598	-598	-598
	36	36	36	36	36	36
	1012	1012	1012	1012	1012	1012
	14.60	14.60	14.60	14.60	14.60	14.60
	1322	1322	1322	1322	1322	1322
	-593	-593	-593	-593	-593	-593
	92	92	92	92	92	92
	1012	1012	1012	1012	1012	1012
	14.08	14.08	14.08	14.08	14.08	14.08
	1322	1322	1322	1322	1322	1322
	-589	-589	-589	-589	-589	-589
	47	47	47	47	47	47
	1012	1012	1012	1012	1012	1012
	5.77	5.77	5.77	5.77	5.77	5.77
	1021	1021	1021	1021	1021	1021
	-4.45	-4.45	-4.45	-4.45	-4.45	-4.45
	1321	1321	1321	1321	1321	1321
	8.98	8.98	8.98	8.98	8.98	8.98
	1321	1321	1321	1321	1321	1321
	-9.25	-9.25	-9.25	-9.25	-9.25	-9.25
	1021	1021	1021	1021	1021	1021
	-4.45	-4.45	-4.45	-4.45	-4.45	-4.45
	1321	1321	1321	1321	1321	1321
	8.08	8.08	8.08	8.08	8.08	8.08
	1321	1321	1321	1321	1321	1321
	-8.07	-8.07	-8.07	-8.07	-8.07	-8.07
	1021	1021	1021	1021	1021	1021
	5.77	5.77	5.77	5.77	5.77	5.77
	1021	1021	1021	1021	1021	1021
	-7.80	-7.80	-7.80	-7.80	-7.80	-7.80
	1221	1221	1221	1221	1221	1221
	7.87	7.87	7.87	7.87	7.87	7.87
	1221	1221	1221	1221	1221	1221
	-4.45	-4.45	-4.45	-4.45	-4.45	-4.45
	1321	1321	1321	1321	1321	1321

MAX/MIN FORCE VALUES FOR MEMB				1922, AMONGST ALL SECT LOCATIONS							
	FY/	LD		FY/	LD	MY	DIST	LD	FX	DIST	LD
MAX.	10.30	0.00	1322	16.70	0.00	1322					
	5.77	0.00	1021	8.98	0.00	1321					
MIN.	-91.35	0.25	1011	-61.11	0.68	1001			385.22	c	0.25 1001
	-4.45	0.25	1321	-9.25	0.00	1021			116.17	c	0.00 1214

1923	0.00	MAX	6.11	1322	14.08	1322	8.29	1221	7.87	1121
		MIN	-84.92	1012	-589.47	1012	-8.33	1121	-7.80	1221
	0.43	MAX	6.06	1322	11.45	1322	8.29	1221	7.47	1121
		MIN	-84.98	1012	-555.34	1112	-8.33	1121	-0.62	1221
	0.86	MAX	6.02	1322	8.84	1322	8.29	1221	0.77	1121
		MIN	-83.05	1012	-521.82	1112	-8.33	1121	-0.63	1221
	1.30	MAX	5.97	1322	6.26	1322	8.29	1221	2.94	1021

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1930, AMONGST ALL SECT LOCATIONS									
	MAX/MIN FORCE VALUES FOR MEMB						LOCATIONS		
	FY/ FZ	DIST	LD	MZ/ MY	DIST	LD	FX	DIST	LD
MAX.	51.49	0.00	1122	13.03	0.32	1212			
	2.86	0.00	1221	0.93	0.32	1221	19.56	C	0.00 1211
MIN.	-40.40	0.32	1212	-16.61	0.32	1122			
	-4.22	0.32	1121	-1.36	0.32	1121	34.46	T	0.32 1114

1931									
0.00	MAX	28.27	1122	13.03	1212	0.70	1121	0.93	1221
0.48	MIN	-31.22	1212	-16.61	1122	-0.48	1221	-1.36	1121
0.96	MAX	28.04	1122	28.06	1212	0.70	1121	0.70	1221
1.44	MIN	-31.42	1212	-30.12	1122	-0.48	1221	-1.03	1121
1.92	MAX	27.80	1122	43.19	1212	0.70	1121	0.47	1221
2.40	MIN	-31.61	1212	-43.52	1122	-0.48	1221	-0.69	1121
	MAX	27.57	1122	58.41	1212	0.70	1121	0.24	1221
	MIN	-31.80	1212	-56.81	1122	-0.48	1221	-0.35	1121
	MAX	27.34	1122	73.72	1212	0.70	1121	0.01	1221
	MIN	-32.00	1212	-69.99	1122	-0.48	1221	-0.02	1013
	MAX	27.10	1122	89.13	1212	0.70	1121	0.32	1121
	MIN	-32.19	1212	-83.06	1122	-0.48	1221	-0.22	1221

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1931, AMONGST ALL SECT LOCATIONS									
MAX/MIN FORCE VALUES FOR MEMB									
FY/	DIST	LD		MZ/	DIST	LD	FX	DIST	LD
FZ	DIST	LD		MY	DIST	LD			
MAX.	28.27	0.00	1122	89.13	2.40	1212			
	0.70	0.00	1221	0.93	0.00	1221	19.56	C	0.00 1211
MIN.	-32.19	2.40	1212	-83.06	2.40	1122			
	-0.48	2.40	1121	-1.36	0.00	1121	34.34	T	2.40 1114

1932									
0.00	MAX	2.34	1322	89.13	1212	0.09	1221	0.32	1121
0.47	MIN	-50.47	1212	-83.06	1122	-0.13	1121	-0.22	1221
0.94	MAX	2.37	1011	98.52	1122	0.09	1221	0.26	1221
1.42	MIN	-51.20	1011	-73.88	1122	-0.13	1121	-0.19	1121
1.89	MAX	1.83	1322	134.66	1122	0.09	1221	0.13	1121
2.36	MIN	-51.43	1011	-73.32	1322	-0.13	1121	-0.09	1221
	MAX	1.65	1322	152.62	1122	0.09	1221	0.06	1121
	MIN	-51.66	1011	-74.14	1322	-0.13	1121	-0.04	1221
	MAX	1.48	1322	170.71	1122	0.09	1221	0.00	1322
	MIN	-51.89	1011	-74.88	1322	-0.13	1121	0.00	1322

1932, AMONGST ALL SECT LOCATIONS									
MAX/MIN FORCE VALUES FOR MEMB									
	FY/	DIST	LD	MZ/	DIST	LD	FX	DIST	LD
	FZ	DIST	LD	MY	DIST	LD			
MAX.	2.34	0.00	1322	170.71	2.36	1112			
	0.09	0.00	1221	0.32	0.00	1121	17.75	C	0.00 1211
MIN.	-51.89	2.36	1011	-83.06	0.00	1122			
	-0.13	2.36	1121	-0.22	0.00	1221	34.57	T	2.36 1114

1933									
0.00	MAX	33.46	1214	74.88	1322	0.00	1322	0.00	1322
1.08	MIN	-25.80	1022	-170.71	1112	0.00	1322	0.00	1322
2.15	MAX	18.54	1214	59.78	1322	0.00	1322	0.00	1322
	MIN	-27.10	1111	-159.18	1112	0.00	1322	0.00	1322
	MAX	13.96	1322	44.73	1322	0.00	1322	0.00	1322

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4001	0.00	MAX	0.29	1322	0.00	1322	0.28	1322	0.00	1322
		MIN	-2.96	1112	-1.43	1212	-1.43	1212	0.00	1322
	0.80	MAX	0.71	1212	4.32	1013	0.28	1322	0.22	1322
		MIN	-10.48	1013	0.86	1212	-1.43	1212	0.14	1212
	1.60	MAX	4.29	1212	16.79	1013	0.28	1322	0.45	1322
		MIN	-20.67	1013	-1.14	1212	-1.43	1212	-2.29	1212
	2.40	MAX	37.87	1212	57.42	1013	0.28	1322	0.67	1322
		MIN	-11.43	1212	66.21	1013	-1.43	1212	-3.43	1322
	3.20	MAX	11.43	1212	66.21	1013	0.28	1322	0.67	1322
		MIN	-41.04	1013	-13.75	1212	-1.43	1212	-4.58	1322
	4.00	MAX	15.02	1212	103.15	1013	0.28	1322	1.12	1322
		MIN	-51.22	1013	-24.34	1212	-1.43	1212	-5.72	1212

MAX/MIN FORCE VALUES FOR MEMB 4001, AMONGST ALL SECT LOCATIONS

	FY/ FZ	DIST	LD	MZ/ MY	DIST	LD	FX	DIST	LD
MAX.	15.02	4.00	1212	103.15	4.00	1013	3.21	C	4.00 1212
	0.28	0.00	1122	1.12	4.00	1122			
MIN.	-51.22	4.00	1013	-24.34	4.00	1212	2.33	T	4.00 1013
	-1.43	4.00	1212	-5.72	4.00	1212			

4002	0.00	MAX	59.67	1013	103.35	1013	1.23	1012	1.66	1321
		MIN	-25.54	1212	-25.05	1212	-0.20	1021	-3.50	1021
	0.48	MAX	53.53	1013	76.10	1013	1.23	1012	1.56	1321
		MIN	-23.42	1212	-13.26	1212	-0.20	1021	-2.98	1021
	0.96	MAX	47.40	1013	51.81	1013	1.23	1012	1.47	1321
		MIN	-21.29	1212	-2.50	1212	-0.20	1021	-2.46	1021
	1.44	MAX	41.26	1013	30.47	1013	1.23	1012	1.37	1321
		MIN	-19.17	1212	5.05	1321	-0.20	1021	-1.94	1021
	1.93	MAX	35.12	1013	17.89	1012	1.23	1012	1.28	1321
		MIN	-17.05	1212	-1.89	1321	-0.20	1021	-1.42	1021
	2.41	MAX	28.98	1013	23.65	1212	1.23	1012	1.29	1112
		MIN	-14.92	1212	-9.04	1122	-0.20	1321	-0.96	1221

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MAX/MIN FORCE VALUES FOR MEMB 4002, AMONGST ALL SECT LOCATIONS

	FY/ FZ	DIST	LD	MZ/ MY	DIST	LD	FX	DIST	LD
MAX.	59.67	0.00	1013	103.35	0.00	1013	127.78	C	0.00 1012
	1.23	0.00	1012	1.66	0.00	1321			
MIN.	-25.54	0.00	1212	-25.05	0.00	1212	18.10	T	2.41 1322
	-0.20	2.41	1321	-3.50	0.00	1021			

4003	0.00	MAX	35.21	1013	23.65	1212	0.17	1221	1.27	1112
		MIN	-8.16	1212	-5.04	1122	-0.23	1112	-0.97	1221
	1.12	MAX	29.98	1013	34.02	1212	0.17	1221	0.97	1112
		MIN	-6.51	1013	-30.84	1312	-0.13	1312	-0.76	1111
	2.24	MAX	1.11	1322	-30.13	1013	-0.21	1112	-0.58	1221
		MIN	6.70	1212	26.11	1212	0.17	1221	0.51	1112
	3.36	MAX	7.76	1013	-49.46	1013	-0.23	1112	-0.39	1221
		MIN	-11.66	1212	15.83	1212	0.17	1221	0.25	1112
	4.48	MAX	22.08	1013	-32.75	1013	-0.23	1112	-0.19	1221
		MIN	16.61	1212	0.00	1322	0.17	1221	0.00	1322
	5.60	MAX	-36.40	1013	0.00	1322	-0.23	1112	0.00	1322

MAX/MIN FORCE VALUES FOR MEMB 4003, AMONGST ALL SECT LOCATIONS

	FY/	DIST	LD	MZ/	DIST	LD	FX	DIST	LD
MAX.	35.21	0.00	1013	30.84	2.24	1212	127.38	C	0.00 1012
	0.17	0.00	1221	1.27	0.00	1112			
MTN.	-36.40	5.60	1013	-50.13	2.24	1013	19.18	T	5.60 1322
	-0.23	5.60	1112	-0.97	0.00	1221			

4004	0.00	MAX	51.15	1013	0.00	1322	0.00	1322	0.00	1322
		MIN	-17.70	1212	0.00	1322	0.00	1322	0.00	1322
	1.60	MAX	30.69	1013	22.66	1212	0.00	1322	0.00	1322
		MIN	-10.62	1212	-65.49	1013	0.00	1322	0.00	1322

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3.20	MAX	10.23	1013	33.99	1212	0.00	1322	0.00	1322
	MIN	-3.54	1212	-98.24	1013	0.00	1322	0.00	1322
4.80	MAX	3.54	1212	33.99	1212	0.00	1322	0.00	1322
	MIN	-10.23	1013	-98.24	1013	0.00	1322	0.00	1322
6.40	MAX	10.23	1013	22.66	1212	0.00	1322	0.00	1322
	MIN	-30.69	1013	-65.49	1013	0.00	1322	0.00	1322
8.00	MAX	17.70	1212	0.00	1322	0.00	1322	0.00	1322
	MIN	-51.15	1013	0.00	1322	0.00	1322	0.00	1322

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MAX/MIN FORCE VALUES FOR MEMB 4004, AMONGST ALL SECT LOCATIONS

	FY/	DIST	LD	MZ/	DIST	LD	FX	DIST	LD
MAX.	51.15	0.00	1013	33.99	4.80	1212			
	0.00	0.00	1001	0.00	0.00	1001	71.40	C	0.00 1121
MIN.	-51.15	8.00	1013	-98.24	4.80	1013			
	0.00	8.00	1322	0.00	8.00	1322	38.48	T	8.00 1221

4005	0.00	MAX	51.15	1013	0.00	1322	0.00	1322	0.00	1322
		MIN	-17.70	1212	0.00	1322	0.00	1322	0.00	1322
	1.60	MAX	10.69	1013	22.66	1212	0.00	1322	0.00	1322
		MIN	-10.23	1013	-98.24	1013	0.00	1322	0.00	1322
	3.20	MAX	3.54	1212	33.99	1212	0.00	1322	0.00	1322
		MIN	-10.23	1013	-98.24	1013	0.00	1322	0.00	1322
	4.80	MAX	10.23	1013	22.66	1212	0.00	1322	0.00	1322
		MIN	-30.69	1013	-65.48	1013	0.00	1322	0.00	1322
	6.40	MAX	17.70	1212	0.00	1322	0.00	1322	0.00	1322
		MIN	-51.15	1013	0.00	1322	0.00	1322	0.00	1322

MAX/MIN FORCE VALUES FOR MEMB 4005, AMONGST ALL SECT LOCATIONS

	FY/ FZ	DIST	LD	MZ/ MY	DIST	LD	FX	DIST	LD
MAX.	51.15	0.00	1013	33.99	4.80	1212	44.35	C	0.00 1013
	0.00	0.00	1001	0.00	0.00	1001			
MIN.	-51.15	8.00	1013	-98.22	4.80	1013	17.68	T	0.00 1212
	0.00	8.00	1322	0.00	8.00	1322			

4006	0.00	MAX	51.15	1013	0.00	1322	0.00	1322	0.00	1322
		MIN	-17.70	1212	0.00	1322	0.00	1322	0.00	1322
	1.60	MAX	10.69	1013	22.66	1212	0.00	1322	0.00	1322
		MIN	-10.23	1013	-98.24	1013	0.00	1322	0.00	1322
	3.20	MAX	3.54	1212	33.99	1212	0.00	1322	0.00	1322
		MIN	-10.23	1013	-98.24	1013	0.00	1322	0.00	1322
	4.80	MAX	10.23	1013	22.66	1212	0.00	1322	0.00	1322
		MIN	-30.69	1013	-65.48	1013	0.00	1322	0.00	1322
	6.40	MAX	17.70	1212	0.00	1322	0.00	1322	0.00	1322
		MIN	-51.15	1013	0.00	1322	0.00	1322	0.00	1322

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MAX/MIN FORCE VALUES FOR MEMB 4006, AMONGST ALL SECT LOCATIONS

	FY/	DIST	LD	MZ/	DIST	LD	FX	DIST	LD
MAX.	51.15	0.00	1013	33.98	3.20	1212			
	0.00	0.00	1001	0.00	0.00	1001	43.87	C	0.00 1013
MIN.	-51.15	8.00	1013	-98.21	3.20	1013			
	0.00	8.00	1322	0.00	8.00	1322	17.65	T	0.00 1212

4007	0.00	MAX	51.15	1013	0.00	1322	0.00	1322	0.00	1322
		MIN	-17.70	1212	0.00	1322	0.00	1322	0.00	1322
	1.60	MAX	10.69	1013	22.66	1212	0.00	1322	0.00	1322
		MIN	-10.23	1013	-98.21	1013	0.00	1322	0.00	1322
	3.20	MAX	3.54	1212	33.98	1212	0.00	1322	0.00	1322
		MIN	-10.23	1013	-98.21	1013	0.00	1322	0.00	1322
	4.80	MAX	10.23	1013	22.66	1212	0.00	1322	0.00	1322
		MIN	-30.69	1013	-65.48	1013	0.00	1322	0.00	1322

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MAX/MIN FORCE VALUES FOR MEMB										4007, AMONGST ALL SECT LOCATIONS									
FY/ FZ		DIST LD		MZ/ MY		DIST LD		FX		DIST LD		FX		DIST LD		FX		DIST LD	
MAX.	6.40	0.00	10.62	1212	22.65	1212	0.00	0.00	1322	0.00	1322	0.00	1322	0.00	1322	0.00	1322	0.00	1322
MIN.	8.00	0.00	17.70	1212	-65.47	1013	0.00	0.00	1322	0.00	1322	0.00	1322	0.00	1322	0.00	1322	0.00	1322
MAX.	51.15	0.00	0.00	1013	33.98	4.80	1212	0.00	0.00	1001	54.94	C	0.00	1021	0.00	1021	0.00	1021	0.00
MIN.	-51.15	8.00	10.62	1212	-98.21	4.80	1013	0.00	0.00	1322	22.58	T	8.00	1321	0.00	1321	0.00	1321	0.00
STAAD SPACE																			

MAX/MIN FORCE VALUES FOR MEMB										4010, AMONGST ALL SECT LOCATIONS									
FY/ FZ		DIST LD		MZ/ MY		DIST LD		FX		DIST LD		FX		DIST LD		FX		DIST LD	
MAX.	51.15	0.00	0.00	1013	33.98	4.80	1212	0.00	0.00	1001	93.33	C	0.00	1112	0.00	1112	0.00	1112	0.00
MIN.	-51.15	8.00	10.62	1212	-98.21	4.80	1013	0.00	0.00	1322	43.84	T	0.00	1322	0.00	1322	0.00	1322	0.00
STAAD SPACE																			

MAX/MIN FORCE VALUES FOR MEMB										4008, AMONGST ALL SECT LOCATIONS									
FY/ FZ		DIST LD		MZ/ MY		DIST LD		FX		DIST LD		FX		DIST LD		FX		DIST LD	
MAX.	51.15	0.00	0.00	1013	33.98	4.80	1212	0.00	0.00	1001	83.22	C	0.00	1021	0.00	1021	0.00	1021	0.00
MIN.	-51.15	8.00	10.62	1212	-98.21	4.80	1013	0.00	0.00	1322	50.97	T	8.00	1321	0.00	1321	0.00	1321	0.00
STAAD SPACE																			

MAX/MIN FORCE VALUES FOR MEMB										4011, AMONGST ALL SECT LOCATIONS									
FY/ FZ		DIST LD		MZ/ MY		DIST LD		FX		DIST LD		FX		DIST LD		FX		DIST LD	
MAX.	51.15	0.00	0.00	1013	33.98	3.20	1212	0.00	0.00	1001	53.24	C	8.00	1121	0.00	1121	0.00	1121	0.00
MIN.	-51.15	8.00	10.62	1212	-98.21	3.20	1013	0.00	0.00	1322	21.02	T	0.00	1221	0.00	1221	0.00	1221	0.00
STAAD SPACE																			

MAX/MIN FORCE VALUES FOR MEMB										4009, AMONGST ALL SECT LOCATIONS									
FY/ FZ		DIST LD		MZ/ MY		DIST LD		FX		DIST LD		FX		DIST LD		FX		DIST LD	
MAX.	51.15	0.00	0.00	1013	33.98	4.80	1212	0.00	0.00	1001	111.68	C	8.00	1021	0.00	1021	0.00	1021	0.00
MIN.	-51.15	8.00	10.62	1212	-98.21	4.80	1013	0.00	0.00	1322	79.44	T	0.00	1321	0.00	1321	0.00	1321	0.00
STAAD SPACE																			

MAX/MIN FORCE VALUES FOR MEMB										4012, AMONGST ALL SECT LOCATIONS									
FY/ FZ		DIST LD		MZ/ MY		DIST LD		FX		DIST LD		FX		DIST LD		FX		DIST LD	
MAX.	51.15	0.00	0.00	1013	33.98	4.80	1212	0.00	0.00	1001	42.96	C	8.00	1013	0.00	1013	0.00	1013	0.00
MIN.	-51.15	8.00	10.62	1212	-98.21	4.80	1013	0.00	0.00	1322	20.91	T	8.00	1212	0.00	1212	0.00	1212	0.00
STAAD SPACE																			

MAX/MIN FORCE VALUES FOR MEMB										4013, AMONGST ALL SECT LOCATIONS									
FY/ FZ		DIST LD		MZ/ MY		DIST LD		FX		DIST LD		FX		DIST LD		FX		DIST LD	
MAX.	51.15	0.00	0.00	1013	33.98	4.80	1212	0.00	0.00	1001	0.00	1322	0.00	1322	0.00	1322	0.00	1322	0.00
MIN.	-51.15	8.00	10.62	1212	-98.21	4.80	1013	0.00	0.00	1322	0.00	1322	0.00	1322	0.00	1322	0.00	1322	0.00
STAAD SPACE																			

MAX/MIN FORCE VALUES FOR MEMB										4013, AMONGST ALL SECT LOCATIONS									
FY/ FZ		DIST LD		MZ/ MY		DIST LD		FX		DIST LD		FX		DIST LD		FX		DIST LD	
MAX.	51.15	0.00	0.00	1013	33.98	4.80	1212	0.00	0.00	1001	0.00	1322	0.00	1322	0.00	1322	0.00	1322	0.00
MIN.	-51.15	8.00	10.62	1212	-98.21	4.80	1013	0.00	0.00	1322	0.00	1322	0.00	1322	0.00	1322	0.00	1322	0.00
STAAD SPACE																			

-1.00 4.00 1321 -5.72 0.00 1212 3.04 T 0.00 1121
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MAX.	51.15	0.00 1013	33.99	4.80 1212	44.68 C	8.00 1021
MIN.	-51.15	8.00 1013	-98.24	4.80 1011	20.98 T	8.00 1212
	0.00	8.00 1322	0.00	8.00 1322		

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4014	0.00	MAX	36.45 1013	0.00 1322	0.35 1021	0.00 1322
	1.12	MIN	-16.57 1212	0.00 1322	-0.30 1321	0.00 1322
	2.24	MAX	22.13 1013	15.79 1212	0.35 1021	0.39 1021
	3.36	MIN	-11.82 1212	-32.81 1013	-0.30 1321	-0.34 1321
	4.48	MAX	7.81 1013	26.02 1212	0.35 1021	0.78 1021
	5.60	MIN	-9.69 1212	-49.77 1013	-0.30 1321	-0.67 1321
			-6.57 1013	20.71 1212	0.35 1021	1.17 1021
			-3.25 1212	-29.84 1013	-0.30 1321	-1.01 1321
			-20.83 1013	-34.98 1212	0.35 1021	1.36 1021
			-35.16 1013	23.42 1212	-0.30 1321	-1.95 1321
				-9.35 1121	-0.30 1321	-1.68 1321
						--- PAGE NO. 202

STAAD SPACE

MAX/MIN FORCE VALUES FOR MEMB 4014, AMONGST ALL SECT LOCATIONS						
	FY/	DIST	LD	MZ/	DIST	LD
	FZ	DIST	LD	MY	DIST	LD
MAX.	36.45	0.00 1013	30.71	3.36 1212		
MIN.	-35.16	8.00 1013	-50.00	5.60 1013	123.79 C	5.60 1012
	-0.30	5.60 1321	-1.68	5.60 1321	21.61 T	0.00 1321

4015	0.00	MAX	14.84 1212	23.42 1212	0.15 1322	1.95 1021
	0.48	MIN	-29.10 1013	-9.35 1121	-1.19 1012	-1.68 1321
	0.96	MAX	16.96 1212	17.64 1112	0.15 1322	1.61 1221
	1.44	MIN	-35.23 1013	-1.67 1321	-1.19 1012	-1.75 1121
	1.93	MAX	19.08 1212	30.30 1013	0.15 1322	1.31 1221
	2.41	MIN	-41.37 1013	5.33 1214	-1.19 1012	-1.88 1121
			21.21 1212	51.69 1013	0.15 1322	1.01 1221
			-47.51 1013	-2.61 1212	-1.19 1012	-2.00 1121
			23.33 1212	76.04 1013	0.15 1322	0.72 1221
			-13.33 1212	-13.33 1212	-1.19 1012	-2.12 1121
			-55.79 1013	-25.07 1212	0.15 1322	0.42 1221
					-1.19 1012	-2.25 1121

MAX/MIN FORCE VALUES FOR MEMB 4015, AMONGST ALL SECT LOCATIONS						
	FY/	DIST	LD	MZ/	DIST	LD
	FZ	DIST	LD	MY	DIST	LD
MAX.	25.46	2.41 1212	103.35	2.41 1013		
MIN.	-59.79	0.00 1322	1.95	0.00 1021	124.19 C	2.41 1012
	-1.19	2.41 1013	-25.07	2.41 1212	20.53 T	0.00 1321
			-2.25	2.41 1121		

4016	0.00	MAX	51.22 1013	103.15 1013	1.43 1212	4.02 1321
	0.80	MIN	-15.72 1212	-4.36 1212	-1.00 1321	-5.72 2212
	1.60	MAX	41.04 1013	19.76 1113	1.43 1212	3.22 1321
	2.40	MIN	-11.41 1212	-37.49 1013	-1.00 1321	-4.57 1212
	3.20	MAX	30.85 1013	37.49 1013	1.43 1212	2.41 1321
	4.00	MIN	-7.87 1212	-6.02 1212	-1.00 1321	-1.63 1321
			20.67 1013	16.79 1013	1.43 1212	-2.70 1213
			-4.30 1212	-1.15 1212	-1.00 1321	0.80 1321
			10.48 1013	4.32 1013	1.43 1212	-1.14 1212
			-0.72 1212	0.14 1321	-1.00 1321	0.00 1322
			2.96 1112	0.00 1322	1.43 1212	0.00 1322
			-1.61 1321	0.00 1322	-1.00 1321	0.00 1322
						--- PAGE NO. 203

STAAD SPACE

MAX/MIN FORCE VALUES FOR MEMB 4016, AMONGST ALL SECT LOCATIONS						
	FY/	DIST	LD	MZ/	DIST	LD
	FZ	DIST	LD	MY	DIST	LD
MAX.	51.22	0.00 1013	103.15	0.00 1013		
MIN.	-15.02	0.00 1212	-24.36	0.00 1212	3.20 C	0.00 1212

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MAX/MIN FORCE VALUES FOR MEMB									
6007, AMONGST ALL SECT LOCATIONS									
FY/	FZ	DIST	LD	MZ/	MY	DIST	LD	FX	DIST
MAX.	18.27	0.00	1013	9.43	3.20	1212			
MIN.	-18.27	8.00	1001	-35.07	3.20	1013		16.67 C	8.00 1013
	0.00	8.00	1322	0.00	8.00	1322		1.74 C	0.00 1322

MAX/MIN FORCE VALUES FOR MEMB									
6008, AMONGST ALL SECT LOCATIONS									
FY/	FZ	DIST	LD	MZ/	MY	DIST	LD	FX	DIST
MAX.	0.00	MAX	18.27 1013	0.00	1322	0.00	1322	0.00	1322
MIN.	1.60	MIN	-4.91 1212	0.00	1322	0.00	1322	0.00	1322
	3.20	MAX	10.96 1013	6.29 1212	0.00	1322	0.00	1322	0.00
	3.20	MIN	-2.95 1212	-23.38 1013	0.00	1322	0.00	1322	0.00
	4.80	MAX	0.98 1212	-35.07 1013	0.00	1322	0.00	1322	0.00
	6.40	MAX	0.98 1212	-35.07 1013	0.00	1322	0.00	1322	0.00
	8.00	MIN	-10.96 1013	-23.38 1013	0.00	1322	0.00	1322	0.00
	8.00	MIN	-18.27 1013	0.00	1322	0.00	1322	0.00	1322

STAAD SPACE

MAX/MIN FORCE VALUES FOR MEMB									
6009, AMONGST ALL SECT LOCATIONS									
FY/	FZ	DIST	LD	MZ/	MY	DIST	LD	FX	DIST
MAX.	18.27	0.00	1013	9.43	4.80	1212			
MIN.	-18.27	8.00	1001	-35.07	4.80	1013		19.29 C	8.00 1021
	0.00	8.00	1322	0.00	8.00	1322		3.98 T	0.00 1321

MAX/MIN FORCE VALUES FOR MEMB									
6010, AMONGST ALL SECT LOCATIONS									
FY/	FZ	DIST	LD	MZ/	MY	DIST	LD	FX	DIST
MAX.	0.00	MAX	18.27 1013	0.00	1322	0.00	1322	0.00	1322
MIN.	1.60	MIN	-4.91 1212	0.00	1322	0.00	1322	0.00	1322
	3.20	MAX	10.96 1013	6.29 1212	0.00	1322	0.00	1322	0.00
	3.20	MIN	-2.95 1212	-23.38 1013	0.00	1322	0.00	1322	0.00
	4.80	MAX	0.98 1212	-35.07 1013	0.00	1322	0.00	1322	0.00
	6.40	MAX	0.98 1212	-35.07 1013	0.00	1322	0.00	1322	0.00
	8.00	MIN	-10.96 1013	-23.38 1013	0.00	1322	0.00	1322	0.00
	8.00	MIN	-18.27 1013	0.00	1322	0.00	1322	0.00	1322

STAAD SPACE

MAX/MIN FORCE VALUES FOR MEMB									
6011, AMONGST ALL SECT LOCATIONS									
FY/	FZ	DIST	LD	MZ/	MY	DIST	LD	FX	DIST
MAX.	18.27	0.00	1013	9.43	3.20	1212			
MIN.	-18.27	8.00	1001	-35.07	3.20	1013		16.67 C	8.00 1013
	0.00	8.00	1322	0.00	8.00	1322		1.74 C	0.00 1322

MAX/MIN FORCE VALUES FOR MEMB									
6012, AMONGST ALL SECT LOCATIONS									
FY/	FZ	DIST	LD	MZ/	MY	DIST	LD	FX	DIST
MAX.	0.00	MAX	18.27 1013	0.00	1322	0.00	1322	0.00	1322
MIN.	1.12	MIN	-4.91 1212	0.00	1322	0.00	1322	0.00	1322
	2.24	MAX	10.96 1013	6.29 1212	0.00	1322	0.00	1322	0.00
	3.36	MAX	0.98 1212	-35.07 1013	0.00	1322	0.00	1322	0.00
	4.48	MAX	0.98 1212	-35.07 1013	0.00	1322	0.00	1322	0.00
	5.60	MIN	-10.96 1013	-23.38 1013	0.00	1322	0.00	1322	0.00
	5.60	MIN	-18.27 1013	0.00	1322	0.00	1322	0.00	1322

STAAD SPACE

MAX/MIN FORCE VALUES FOR MEMB									
6010, AMONGST ALL SECT LOCATIONS									
FY/	FZ	DIST	LD	MZ/	MY	DIST	LD	FX	DIST
MAX.	18.27	0.00	1013	9.43	3.20	1212			
MIN.	-18.27	8.00	1001	-35.07	3.20	1013		26.66 C	8.00 1013
	0.00	8.00	1322	0.00	8.00	1322		4.60 C	0.00 1211

MAX/MIN FORCE VALUES FOR MEMB									
6011, AMONGST ALL SECT LOCATIONS									
FY/	FZ	DIST	LD	MZ/	MY	DIST	LD	FX	DIST
MAX.	0.00	MAX	18.27 1013	0.00	1322	0.00	1322	0.00	1322
MIN.	1.60	MIN	-4.91 1212	0.00	1322	0.00	1322	0.00	1322
	3.20	MAX	10.96 1013	6.29 1212	0.00	1322	0.00	1322	0.00
	3.20	MIN	-2.95 1212	-23.38 1013	0.00	1322	0.00	1322	0.00
	4.80	MAX	0.98 1212	-35.07 1013	0.00	1322	0.00	1322	0.00
	6.40	MAX	0.98 1212	-35.07 1013	0.00	1322	0.00	1322	0.00
	8.00	MIN	-10.96 1013	-23.38 1013	0.00	1322	0.00	1322	0.00
	8.00	MIN	-18.27 1013	0.00	1322	0.00	1322	0.00	1322

MAX/MIN FORCE VALUES FOR MEMB									
6011, AMONGST ALL SECT LOCATIONS									
FY/	FZ	DIST	LD	MZ/	MY	DIST	LD	FX	DIST
MAX.	18.27	0.00	1013	9.44	3.20	1212			
MIN.	-18.27	8.00	1001	-35.08	3.20	1013		35.22 C	8.00 1013
	0.00	8.00	1322	0.00	8.00	1322		6.98 C	0.00 1211

MAX/MIN FORCE VALUES FOR MEMB									
6012, AMONGST ALL SECT LOCATIONS									
FY/	FZ	DIST	LD	MZ/	MY	DIST	LD	FX	DIST
MAX.	0.00	MAX	18.27 1013	0.00	1322	0.00	1322	0.00	1322
MIN.	1.12	MIN	-4.91 1212	0.00	1322	0.00	1322	0.00	1322
	2.24	MAX	10.96 1013	6.29 1212	0.00	1322	0.00	1322	0.00
	3.36	MAX	0.98 1212	-35.07 1013	0.00	1322	0.00	1322	0.00
	4.48	MAX	0.98 1212	-35.07 1013	0.00	1322	0.00	1322	0.00
	5.60	MIN	-10.96 1013	-23.38 1013	0.00	1322	0.00	1322	0.00
	5.60	MIN	-18.27 1013	0.00	1322	0.00	1322	0.00	1322

STAAD SPACE

MAX/MIN FORCE VALUES FOR MEMB									
6012, AMONGST ALL SECT LOCATIONS									
FY/	FZ	DIST	LD	MZ/	MY	DIST	LD	FX	DIST
MAX.	12.79	0.00	1013	4.62	3.36	1212			
MIN.	-12.79	5.60	1001	-17.19	3.36	1013		47.00 C	5.60 1013
	0.00	5.60	1322	0.00	5.60	1322		9.58 C	0.00 1322

STAAD SPACE

PERIMETER COLUMNS

1.22	MAX	0.01 1013	0.02 1212	10.59 1112	3.57 1114
	MIN	0.00 1212	-0.10 1013	-3.39 1322	-1.32 1211
1.83	MAX	0.01 1013	0.02 1212	10.59 1112	6.74 1112
	MIN	0.00 1212	-0.11 1013	-3.39 1322	-0.10 1322
2.44	MAX	0.01 1013	0.02 1212	10.59 1112	13.20 1112
	MIN	0.00 1212	-0.12 1013	-3.39 1322	-2.17 1322
3.05	MAX	0.01 1013	0.03 1212	10.59 1112	19.66 1112
	MIN	0.00 1212	-0.12 1013	-3.39 1322	-4.24 1322

STAAD SPACE

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MAX/MIN FORCE VALUES FOR MEMB									
FY/	FZ/	DIST	LD	MZ/	DIST	LD	FX	DIST	LD
MAX.	0.01	0.00 1013	0.03	3.05 1212	19.66	3.05 1112	226.57 C	0.00 1013	
MIN.	10.59	0.00 1112	-0.12	3.05 1013	-12.68	0.00 1111	84.27 T	3.05 1211	

233, AMONGST ALL SECT LOCATIONS

230	0.00	MAX	0.01 1013	0.00 1322	9.85 1322	0.00 1322			
	0.64	MIN	0.00 1212	-19.35 1112	-19.35 1112	6.30 1322			
	1.28	MAX	0.01 1013	0.00 1212	9.85 1322	12.39 1112			
	1.92	MIN	0.00 1212	-19.35 1112	-19.35 1112	12.39 1112			
	2.56	MAX	0.01 1013	0.00 1212	9.85 1322	18.90 1322			
	3.20	MIN	0.00 1212	-19.35 1112	-19.35 1112	25.21 1322			

MAX/MIN FORCE VALUES FOR MEMB									
FY/	FZ/	DIST	LD	MZ/	DIST	LD	FX	DIST	LD
MAX.	0.01	0.00 1013	0.01	3.20 1212	31.51	3.20 1322	244.34 C	0.00 1122	
MIN.	9.85	0.00 1322	-0.03	3.20 1013	-61.93	3.20 1112	107.96 T	3.20 1211	

230, AMONGST ALL SECT LOCATIONS

231	0.00	MAX	0.01 1013	0.01 1212	8.81 1112	31.51 1322			
	0.88	MIN	0.00 1212	-0.03 1013	-4.54 1322	-61.93 1112			
	1.76	MAX	0.01 1013	0.01 1212	8.81 1112	27.51 1322			
	2.64	MIN	0.00 1212	-0.04 1013	-4.54 1322	-54.19 1112			
	3.52	MAX	0.01 1013	0.01 1212	8.81 1112	23.52 1322			
	4.40	MIN	0.00 1212	-0.05 1013	-4.54 1322	-46.44 1112			

231, AMONGST ALL SECT LOCATIONS

232	0.00	MAX	0.01 1013	0.02 1212	8.81 1112	11.54 1322			
	0.24	MIN	0.00 1212	-0.08 1013	-4.54 1322	-3.19 1112			
	0.48	MAX	0.01 1013	0.02 1212	8.81 1112	10.45 1322			
	0.72	MIN	0.00 1212	-0.09 1013	-4.54 1322	-21.08 1112			
	0.96	MAX	0.01 1013	0.02 1212	8.81 1112	9.36 1322			
	1.20	MIN	0.00 1212	-0.09 1013	-4.54 1322	-18.97 1112			

232, AMONGST ALL SECT LOCATIONS

233	0.00	MAX	0.01 1013	0.02 1212	8.81 1112	10.59 1112			
	0.61	MIN	0.00 1212	-0.09 1013	-4.54 1322	-3.39 1322			
	1.20	MAX	0.01 1013	0.02 1212	8.81 1112	10.59 1112			
	1.92	MIN	0.00 1212	-0.10 1013	-4.54 1322	-6.84 1112			

MAX/MIN FORCE VALUES FOR MEMB									
FY/	FZ/	DIST	LD	MZ/	DIST	LD	FX	DIST	LD
MAX.	0.01	0.00 1013	0.02	1.20 1212	228.44 C	0.00 1013			
MIN.	8.81	0.00 1112	-0.09	1.20 1013	81.71 T	1.20 1211			

233, AMONGST ALL SECT LOCATIONS

233	0.00	MAX	0.01 1013	0.02 1212	10.59 1112	6.10 1322			
	0.61	MIN	0.00 1212	-0.09 1013	-3.39 1322	-12.68 1111			
	1.20	MAX	0.01 1013	0.02 1212	10.59 1112	4.03 1322			
	1.92	MIN	0.00 1212	-0.10 1013	-3.39 1322	-6.84 1111			

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2 FOUNDATION DESIGN

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Rev A3

Date: 12 June 2007

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Calculation/Sketch

Meinhardt Infrastructure Pte Ltd
Consulting Engineers, Planners & Managers



Site	Main Depot Rashidiya - Building 3	Date	May '07	Job No.	068010
	Footing Summary	Designed by	Rudy	Sheet No.	

Allowable soil Bearing Pressure = 150 kN/m² (Main Building)
= 300 kN/m² (Lathan Pit Area)
= 20 kN/m³
= 24 kN/m³

Soil Density
Concrete Density

GL	Footing Type	Footing Length (m)	Footing Width (m)	Footing Depth (m)	STAAD Column ref	STAAD Column Reaction (ULS) (kN)*	Max Vertical Crane Load Combination (kN)	Footing Weight (kN)	Soil Depth (m)	Soil Weight (kN)	Total Load (kN)	Bearing Pressure (kN/m ²)	Remark
1/A	F21	5.00	2.00	0.750	1101 + 3011	668	237	180	1.20	240	1134	113	Ok
1/A'	F4	1.60	1.60	0.600	3013	172		37	1.20	61	221	86	Ok
1/B	F21	5.00	2.00	0.750	1125 + 3015	673	237	180	1.20	240	1137	114	Ok
2/A	F25	3.30	1.80	0.750	1301	436	237	107	1.20	143	798	134	Ok
2/B	F25	3.30	1.80	0.750	1325	458	237	107	1.20	143	814	137	Ok
3/A	F25	3.30	1.80	0.750	1401	461	237	107	1.20	143	816	137	Ok
3/B	F25	3.30	1.80	0.750	1425	422	237	107	1.20	143	788	133	Ok
4/A	F25	3.30	1.80	0.750	1501	460	237	107	1.20	143	815	137	Ok
4/B	F25	3.30	1.80	0.750	1525	524	237	107	1.20	143	861	145	Ok
5/A	F25	3.30	1.80	0.750	1601	473	237	107	1.20	143	824	139	Ok
5/B	F25	3.30	1.80	0.750	1625	525	237	107	1.20	143	862	145	Ok
6/A	F25	3.30	1.80	0.750	1701	476	237	107	1.20	143	826	139	Ok
6/B	F25	3.30	1.80	0.750	1725	526	237	107	1.20	143	862	145	Ok
7/A	F25	3.30	1.80	0.750	1801	852		107	1.20	143	858	144	Ok
7/B	F25	3.30	1.80	0.750	1825	532	237	107	1.20	143	867	146	Ok
8/A	F25	3.30	1.80	0.750	1901	490	237	107	1.20	143	836	141	Ok
8/B	F25	3.30	1.80	0.750	1925	539	237	107	1.20	143	871	147	Ok
9/A	F25	3.30	1.80	0.750	2001	487	237	107	1.20	143	834	140	Ok
9/B	F25	3.30	1.80	0.750	2025	533	237	107	1.20	143	867	146	Ok
10/A	F25	3.30	1.80	0.750	2101	460	237	107	1.20	143	815	137	Ok
10/B	F25	3.30	1.80	0.750	2125	525	237	107	1.20	143	861	145	Ok
11/A	F25	3.30	1.80	0.750	2201	474	237	107	1.20	143	825	139	Ok
11/B	F25	3.30	1.80	0.750	2225	527	237	107	1.20	143	862	145	Ok
12/A	F25	3.30	1.80	0.750	2301	434	237	107	1.20	143	796	134	Ok

Calculation/Sketch

Meinhardt Infrastructure Pte Ltd
Consulting Engineers, Planners & Managers



Site	Main Depot Rashidiya - Building 3	Date	May '07	Job No.	068010
	Footings Summary	Designed by	Rudy	Sheet No.	

Allowable soil Bearing Pressure = 150 kN/m² (Main Building)
= 300 kN/m² (Lathan Pit Area)
= 20 kN/m³
= 24 kN/m³

Soil Density
Concrete Density

GL	Footings Type	Footings Length (m)	Footings Width (m)	Footings Depth (m)	STAAD Column ref	STAAD Column Reaction (ULS) (kN)*	Max Vertical Crane Load Combination (kN)	Footings Weight (kN)	Soil Depth (m)	Soil Weight (kN)	Total Load (kN)	Bearing Pressure (kN/m ²)	Remark
12/B	F25	3.30	1.80	0.750	2325	451	237	107	1.20	143	808	136	Ok
13/A	F21	5.00	2.00	0.750	2501 + 3111	664	237	180	1.20	240	1131	113	Ok
13/A'	F4	1.60	1.60	0.600	3113	171		37	1.20	61	221	86	Ok
13/B	F21	5.00	2.00	0.750	2525 + 3115	704	237	180	1.20	240	1159	116	Ok
1.1/C	F4	1.60	1.60	0.600	1234	103		37	1.20	61	172	67	Ok
2/C	F4	1.60	1.60	0.600	1334	123		37	1.20	61	186	73	Ok
3/C	F4	1.60	1.60	0.600	1434	108		37	1.20	61	176	69	Ok
4/C	F4	1.60	1.60	0.600	1534	109		37	1.20	61	176	69	Ok
5/C	F4	1.60	1.60	0.600	1634	109		37	1.20	61	176	69	Ok
6/C	F4	1.60	1.60	0.600	1734	109		37	1.20	61	176	69	Ok
7/C	F4	1.60	1.60	0.600	1834	126		37	1.20	61	188	73	Ok
8/C	F4	1.60	1.60	0.600	1934	126		37	1.20	61	189	74	Ok
9/C	F4	1.60	1.60	0.600	2034	110		37	1.20	61	177	69	Ok
10/C	F4	1.60	1.60	0.600	2134	109		37	1.20	61	176	69	Ok
11/C	F4	1.60	1.60	0.600	2234	108		37	1.20	61	176	69	Ok
12/C	F4	1.60	1.60	0.600	2334	117		37	1.20	61	182	71	Ok
12.1/C	F4	1.60	1.60	0.600	2434	84		37	1.20	61	158	62	Ok

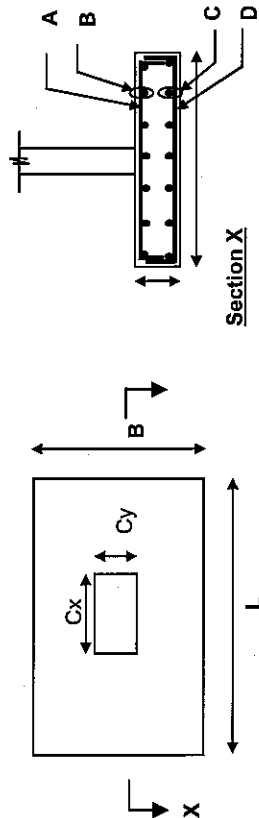
Calculation/Sketch

Meinhardt Infrastructure Pte Ltd
Consulting Engineers, Planners & Managers



Site	Main Depot Rashidiya - Building 3	Date	May '07	Job No.	068010
	Footing Detail	Designed by	Rudy	Sheet No.	

Summary



Footing Type	L	B	H	Soil Bearing pressure (kPa)	A	Spacing (mm)	B	Spacing (mm)	C	Spacing (mm)	D	Spacing (mm)
F4	1600	1600	600	150	12T12	135	12T16	135	12T16	135	12T12	135
F21	2000	5000	750	150	14T16	145	25T16	204	25T20	203	14T25	144
F25	1800	3300	750	150	12T16	153	22T16	152	22T20	151	12T20	153

Calculation/Sketch

Meinhardt Infrastructure Pte Ltd
Consulting Engineers, Planners & Managers



Site	Main Depot Rashidiya - Building 20	Date	Apr '07	Job No.	068010.01
	Footing F3				
	For soil bearing pressure = 150kPa	Designed by	Rudy	Sheet No.	

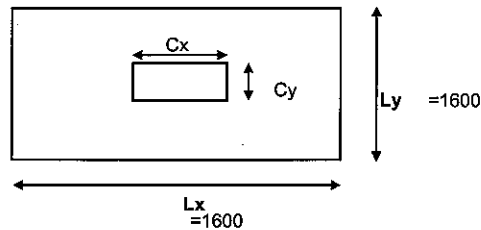
REINFORCED CONCRETE FOOTING DESIGN TO BS8110

1. Material Properties

Concrete grade	f_{cu}	=	40	N/mm ²
Steel grade	f_y	=	460	N/mm ²
Concrete density	γ_c	=	24	KN/m ³
Steel density	γ_s	=	78	KN/m ³

2. Design Data

Depth of Footing	h	=	600	mm
Width of Footing	L_y	=	1600	mm
Length of Footing	L_x	=	1600	mm
Width of Column	C_y	=	450	mm
Length of Column	C_x	=	450	mm
Concrete Ten. Cover	c	=	50	mm
Concrete Comp. Cover	c	=	50	mm
Tension bar dia. (x-dir)	D_x	=	12	mm
Tension bar dia. (y-dir)	D_y	=	16	mm
Comp. bar diameter (x-dir)	D_c	=	12	mm
Comp. bar diameter (y-dir)	D_c	=	16	mm
Effect.depth (x-dir)	d_x	=	544	mm
Effect.depth (y-dir)	d_y	=	530	mm
Area of footing	A	=	2.56	m ²



3. Loadings

Safe bearing pressure	p_b	=	150	KN/m
Load factor	f	=	1.5	
Total axial load (SLS)	N	=	384.00	kN
Eccentricity	e	=	0	mm

4. Analysis

Net pressure (SLS)	p_{max}	=	135.60	KN/m			
Net pressure (ULS)	p_{max}	=	203.40	KN/m			
Shear force (x-dir)	V_x	=	187.13	kN			
Shear force (y-dir)	V_y	=	187.13	kN			
Moment (x-dir)	M_x	=	53.80	kNm			
Moment (y-dir)	M_y	=	53.80	kNm			
Steel required (x-dir)	A_{sreq}	=	1248	mm ²	Nos	ϕ	Spacing
Steel provided (x-dir)	A_{sprov}	=	1357	mm ²	O.K.	12	135 x-dir
Steel required (y-dir)	A_{sreq}	=	1248	mm ²	Nos	ϕ	
Steel provided (y-dir)	A_{sprov}	=	2413	mm ²	O.K.	12	16 135 y-dir
Top bars required (x-dir)	A'_{sreq}	=	1248	mm ²	Nos	ϕ	
Top bars provided (x-dir)	A'_{sprov}	=	1357	mm ²	O.K.	12	12 135 x-dir
Top bars required (y-dir)	A'_{sreq}	=	1248	mm ²	Nos	ϕ	
Top bars provided (y-dir)	A'_{sprov}	=	2413	mm ²	O.K.	12	16 135 y-dir

Calculation/Sketch

Meinhardt Infrastructure Pte Ltd
Consulting Engineers, Planners & Managers



Site	Main Depot Rashidiya - Building 20	Date	Apr '07	Job No.	068010.01
	Footing F3				
	For soil bearing pressure = 150kPa	Designed by	Rudy	Sheet No.	

5. Shear Check

Checking for punching shear:

Shear force at face $V_f = 530.44$ kN
Shear stress at face $v_f = 0.54$ N/mm² **O.K.**

Distance from Support (mm)	av (mm)	V (kN)	v (N/mm ²)	v_c (N/mm ²)	Enhanced v_c (N/mm ²)	Remarks
d	544	43.78	0.01	0.40	0.80	No Shear Links
2d	1088	0.00	0.00	0.40	0.40	No Shear Links

Checking for shear (x-dir):

Shear force at face (x-dir) $V_c = 207.00$ kN
Shear stress at face (x-dir) $v_c = 0.24$ N/mm² **O.K.**

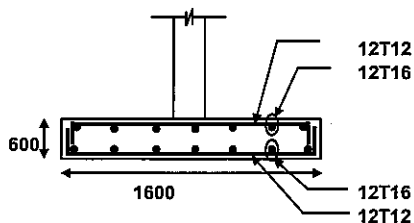
Distance from Support (mm)	av (mm)	V (kN)	v (N/mm ²)	v_c (N/mm ²)	Enhanced v_c (N/mm ²)	Remarks
d	544	11.16	0.01	0.40	0.80	No Shear Links
2d	1088	0.00	0.00	0.40	0.40	No Shear Links

Checking for shear (y-dir):

Shear force at face (y-dir) $V_c = 207.00$ kN
Shear stress at face (y-dir) $v_c = 0.24$ N/mm² **O.K.**

Distance from Support (mm)	av (mm)	V (kN)	v (N/mm ²)	v_c (N/mm ²)	Enhanced v_c (N/mm ²)	Remarks
d	530	16.20	0.02	0.49	0.97	No Shear Links
2d	1060	0.00	0.00	0.49	0.49	No Shear Links

6. Detailing

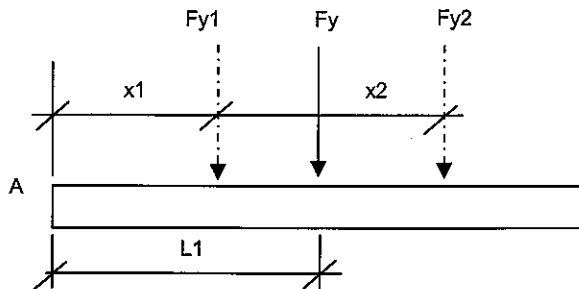


Calculation/Sketch

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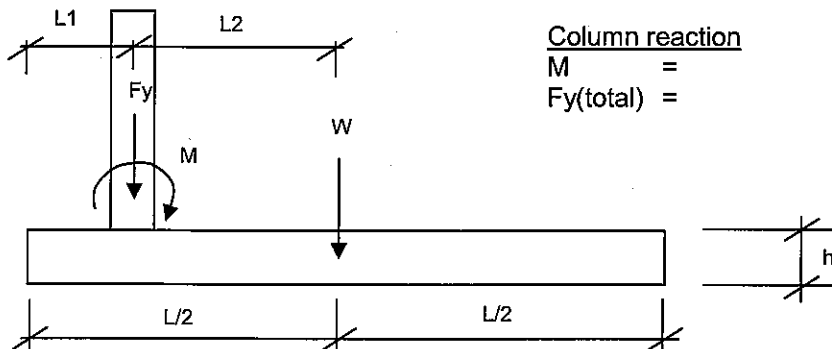


Site	Dubai Depot Diesel Engine Workshop (Rashidiya) F21	Date	June '07	Job No.	068010
		Designed by Rudy		Sheet No.	



$$\begin{aligned}
 Fy1 &= 696.1 \text{ kN} & (\text{ULS}) \\
 Fy2 &= 244.3 \text{ kN} & (\text{ULS}) \\
 x1 &= 1.85 \text{ m} \\
 x2 &= 4.35 \text{ m} \\
 Fy &= Fy1 + Fy2 \\
 &= 940.4 \text{ kN} \\
 L1 &= (Fy1 \cdot x1 + Fy2 \cdot x2) / (Fy1 + Fy2) \\
 &= 2.50 \text{ m}
 \end{aligned}$$

Footing Design Check



Column reaction

$$\begin{aligned}
 M &= 0 \text{ kNm} & (\text{ULS}) \\
 Fy(\text{total}) &= 940.4 \text{ kN} & (\text{ULS})
 \end{aligned}$$

Assume footing base dimension

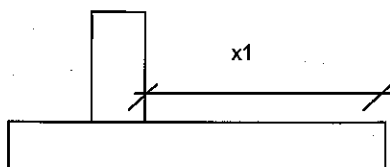
$$\begin{aligned}
 L &= 5.00 \text{ m} & \text{Depth of soil above the footing, } H_s &= 1.2 \text{ m} \\
 B &= 2.00 \text{ m} & \text{Density of the backfill soil, } \gamma_{\text{soil}} &= 20.0 \text{ kN/m}^3 \\
 h &= 0.75 \text{ m} & W(\text{soil}) &= 240.0 \text{ kN} \\
 L1 &= 2.50 \text{ m} \\
 L2 &= 0.00 \text{ m} & \text{Safe bearing capacity} &= 150.0 \text{ kN/m}^2 \\
 W(\text{selfwt}) &= 180.0 \text{ kN}
 \end{aligned}$$

Equivalent bending moment at the center of the footing, M_c

$$\begin{aligned}
 &= (M - Fy \cdot L2) / \gamma_L & \gamma_L &= 1.4 \\
 &= 0.3 \text{ kNm}
 \end{aligned}$$

Total vertical load at the center of the footing, F_t

$$\begin{aligned}
 &= Fy / \gamma_L + W(\text{soil}) + W(\text{selfwt}) \\
 &= 1091.7 \text{ kN}
 \end{aligned}$$

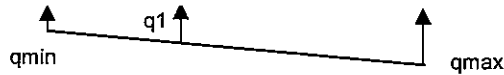


Calculation/Sketch

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Site	Dubai Depot	Date	June '07	Job No.	068010
	Diesel Engine Workshop (Rashidiya)				
	F21	Designed by Rudy		Sheet No.	



Eccentricity due to the resultant force, e

$$= \frac{M_c}{F_t} = 0.000 \text{ m} \geq \frac{L}{6} = 0.833 \text{ m}$$

The whole base is under compression

$$\begin{aligned} \text{Effective stress length, } L' &= 5 \text{ m} \\ \text{Base pressure, } q_{\max} &= 109.27 \text{ kN/m}^2 \quad \text{Ok} \\ q_{\min} &= 109.08 \text{ kN/m}^2 \end{aligned}$$

Design of footing reinforcement

$$\begin{aligned} \text{Distance between the footing edge to the column edge, } x_1 &= 2.500464 \text{ m} \\ \text{Stress at the face of the column, } q_1 &= 109.17 \text{ kN/m}^2 \\ \text{cover to the main reinforcement} &= 83 \text{ mm} \\ \text{effective depth of the footing, } d &= 667 \text{ mm} \\ d' &= 83 \text{ mm} \\ f_{cu} &= 40 \text{ N/mm}^2 \\ f_y &= 460 \text{ N/mm}^2 \end{aligned}$$

Design bending reinforcement (along the length)

$$\begin{aligned} \text{Maximum ULS moment at the face of column, } M_1 &= \\ = B[0.5q_1(x_1^2/3) + 0.5q_{\max}(2x_1^2/3)]\gamma_L &= 956.2 \text{ kNm} \end{aligned}$$

$$\begin{aligned} K' &= 0.156 \\ K = M_1 / f_{cu} B d^2 &= 0.0269 \\ z &= 385.0 \text{ mm} \\ x &= 626.6 \text{ mm} \end{aligned}$$

$$\begin{aligned} A_s' \text{ req} &= 0.0 \text{ mm}^2 \\ A_s \text{ req} &= 6205.7 \text{ mm}^2 \\ A_s \text{ req (min)} &= 975.0 \text{ mm}^2 / \text{m} \end{aligned}$$

$$\begin{aligned} \text{Provide } 14 \text{ T25} \\ A_s \text{ provided} &= 6872.2 \text{ mm}^2 \text{ (spacing} = 141.1 \text{ mm)} \quad \text{Ok} \\ &= 3479.48 \text{ mm}^2 / \text{m} \end{aligned}$$

Design shear reinforcement

$$\begin{aligned} \text{Maximum ULS shear at the face of column, } V_1 &= \\ = B \cdot 0.5[q_1 + q_{\max}]x_1\gamma_L &= 764.7 \text{ kN} \end{aligned}$$

$$\begin{aligned} V_1 / b d &= 0.57 \text{ N/mm}^2 & 100A_s / B d &= 0.51516 \\ & & 400 / d &= 0.5997 \end{aligned}$$

$$\begin{aligned} v_c &= 0.54 \text{ N/mm}^2 \\ v_c + 0.4 &= 0.94 \text{ N/mm}^2 \\ 0.8f_{cu}^{0.5} &= 5.0 \text{ N/mm}^2 \leq 5 \text{ N/mm}^2 \end{aligned}$$

Calculation/Sketch

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Site	Dubai Depot Diesel Engine Workshop (Rashidiya) F21	Date	June '07	Job No.	068010
		Designed by Rudy		Sheet No.	

Asv/sv req = 1.999 mm² / mm
Provide T @300 c/c (2 legs)
Asv/sv prov = 0 mm² / mm

Design bending reinforcement (along the width)

$q_{ave} = q_1 + q_{max}$
 $= 109.2 \text{ N/mm}^2$

Maximum ULS moment at the face of column, M2
 $= q_{ave} * (0.5*B)^2 / 2$
 $= 54.6 \text{ kNm/m length}$

$K = M2 / f_{cub} d^2 = 0.0033$ $d = 642 \text{ mm}$
 $z = 442.6 \text{ mm}$ $b = 1000 \text{ mm}$
 $x = 571.1 \text{ mm}$

As req = 308.3 mm² / m
As req (min) = 834.6 mm² / m

Provide 10 T16

As provided = 2010.6 mm² (spacing = 203.8 mm) Ok
= 986.67 mm² / m

Calculation/Sketch

Meinhardt Infrastructure Pte Ltd
Consulting Engineers, Planners & Managers



Site	Dubai LRT	Date	6/11/2007	Job No.	068010.01
	Footing F25				
	For soil bearing pressure = 150kPa	Designed by	Sebastian	Sheet No.	

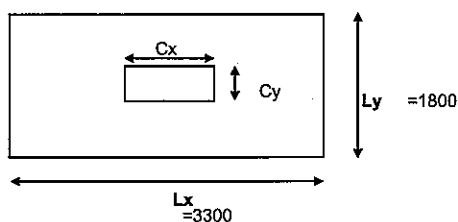
REINFORCED CONCRETE FOOTING DESIGN TO BS8110

1. Material Properties

Concrete grade	f_{cu}	=	40	N/mm ²
Steel grade	f_y	=	460	N/mm ²
Concrete density	γ_c	=	24	KN/m ³
Steel density	γ_s	=	78	KN/m ³

2. Design Data

Depth of Footing	h	=	750	mm
Width of Footing	L_y	=	1800	mm
Length of Footing	L_x	=	3300	mm
Width of Column	C_y	=	450	mm
Length of Column	C_x	=	450	mm
Concrete Ten. Cover	c	=	50	mm
Concrete Comp. Cover	c	=	50	mm
Tension bar dia. (x-dir)	D_x	=	20	mm
Tension bar dia. (y-dir)	D_y	=	20	mm
Comp. bar diameter (x-dir)	D_c	=	16	mm
Comp. bar diameter (y-dir)	D_c	=	16	mm
Effect.depth (x-dir)	d_x	=	690	mm
Effect.depth (y-dir)	d_y	=	670	mm
Area of footing	A	=	5.94	m ²



3. Loadings

Safe bearing pressure	p_b	=	150	KN/m
Load factor	f	=	1.5	
Total axial load (SLS)	N	=	891.00	kN
Eccentricity	e	=	0	mm

4. Analysis

Net pressure (SLS)	p_{max}	=	132.00	KN/m			
Net pressure (ULS)	p_{max}	=	198	KN/m			
Shear force (x-dir)	V_x	=	507.87	kN			
Shear force (y-dir)	V_y	=	441.05	kN			
Moment (x-dir)	M_x	=	361.86	kNm			
Moment (y-dir)	M_y	=	148.85	kNm			
Steel required (x-dir)	A_{sreq}	=	1755	mm ²			
Steel provided (x-dir)	A_{sprov}	=	3770	mm ²	O.K.	Nos	ϕ Spacing
Steel required (y-dir)	A_{sreq}	=	3218	mm ²		12	20 153 x-dir
Steel provided (y-dir)	A_{sprov}	=	6912	mm ²	O.K.	Nos	ϕ Spacing
Top bars required (x-dir)	A'_{sreq}	=	1755	mm ²		22	20 151 y-dir
Top bars provided (x-dir)	A'_{sprov}	=	2413	mm ²	O.K.	Nos	ϕ Spacing
Top bars required (y-dir)	A'_{sreq}	=	3218	mm ²		12	16 153 x-dir
Top bars provided (y-dir)	A'_{sprov}	=	4423	mm ²	O.K.	Nos	ϕ Spacing
						22	16 152 y-dir

Calculation/Sketch

Meinhardt Infrastructure Pte Ltd
Consulting Engineers, Planners & Managers



Site	Dubai LRT	Date	6/11/2007	Job No.	068010.01
	Footing F25				
	For soil bearing pressure = 150kPa	Designed by	Sebastian	Sheet No.	

5. Shear Check

Checking for punching shear:

Shear force at face $V_f = 1290.94$ kN
Shear stress at face $v_f = 1.04$ N/mm² **O.K.**

Distance from Support (mm)	av (mm)	V (kN)	v (N/mm ²)	v_c (N/mm ²)	Enhanced v_c (N/mm ²)	Remarks
d	690	583.00	0.12	0.50	0.99	No Shear Links
2d	1380	0.00	0.00	0.50	0.50	No Shear Links

Checking for shear (x-dir):

Shear force at face (x-dir) $V_c = 577.13$ kN
Shear stress at face (x-dir) $v_c = 0.46$ N/mm² **O.K.**

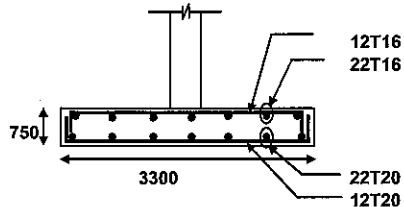
Distance from Support (mm)	av (mm)	V (kN)	v (N/mm ²)	v_c (N/mm ²)	Enhanced v_c (N/mm ²)	Remarks
d	690	297.68	0.24	0.50	0.99	No Shear Links
2d	1380	18.23	0.01	0.50	0.50	No Shear Links

Checking for shear (y-dir):

Shear force at face (y-dir) $V_c = 501.19$ kN
Shear stress at face (y-dir) $v_c = 0.23$ N/mm² **O.K.**

Distance from Support (mm)	av (mm)	V (kN)	v (N/mm ²)	v_c (N/mm ²)	Enhanced v_c (N/mm ²)	Remarks
d	670	3.71	0.00	0.50	1.00	No Shear Links
2d	1340	0.00	0.00	0.50	0.50	No Shear Links

6. Detailing



3 BEAM DESIGN

Document No.: **DM001-E-MAG-CDP-DR-DCC-371943**

Rev A3

Date: 12 June 2007

Page

Calculation/Sketch

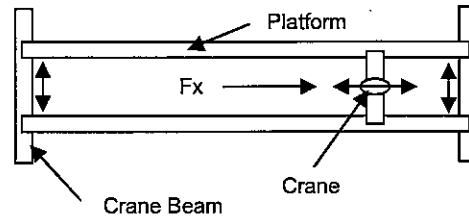
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Site	Dubai Depot - RAB6	Date	June '07	Job No.	068010
	Crane Beam Loading				
	(10 T / 15m Bridge)	Designed by Rudy		Sheet No.	

Loading

Crane Capacity	=	100 kN
Crane Self-weight	=	74.5 kN
Platform Load	=	0.45*15
	=	6.75 kN
Beam Self-weight, w	=	1.16 kN/m
Axial Load, Fx	=	(32+2.65)/2
	=	17.3 kN
Impact Load factor	=	1.25



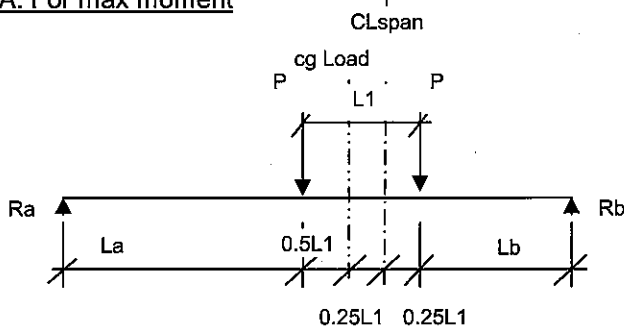
Vertical Crane Wheel Load (assuming the crane is at one end)

$$P_v' = 100/2 + 74.5/4 + 9/4 = 70.3 \text{ kN}$$

Vertical Crane Wheel Load Including Impact, P_v = 87.9 kN

Horizontal surge load Including Impact, P_h = 21.7 kN

A. For max moment



L_s	=	8 m
L_1	=	2.9 m
L_a	=	$0.5L_s - 0.75L_1$
	=	1.825 m
L_b	=	$0.5L_s - 0.25L_1$
	=	3.275 m

1. Vertical direction

(a) due to Crane Vertical Load with impact

$$\begin{aligned} R_a' &= P_v' * (L_s - L_a) / L_s + P_v' * L_b / L_s = 103.8 \text{ kN} \\ R_b' &= 2 * P_v' - R_a' = 72.0 \text{ kN} \\ M_v' &= R_b' * L_b = 235.7 \text{ kNm} \end{aligned}$$

(b) due to Crane Vertical Load without impact

$$\begin{aligned} R_a &= P_v * (L_s - L_a) / L_s + P_v * L_b / L_s = 83.1 \text{ kN} \\ R_b &= 2 * P_v - R_a = 57.6 \text{ kN} \\ M_v &= R_b * L_b = 188.5 \text{ kNm} \end{aligned}$$

(c) due to selfweight of the beam

$$\begin{aligned} R_{wa} &= R_{wb} = 0.5 * w * L_s = 4.6 \text{ kN} \\ M_w \text{ (at location of max vertical moment)} &= 0.5 * w * L_s^2 * \{ [(L_a + L_1) / L_s] - [(L_a + L_1) / L_s]^2 \} \\ &= 9.0 \text{ kNm} \end{aligned}$$

2. Horizontal direction due to crane surge load (including impact factor)

$$\begin{aligned} R_{ha} &= 21.1 \text{ kN} \\ R_{hb} &= 22.2 \text{ kN} \\ M_{h'} &= 72.6 \text{ kNm} \end{aligned}$$

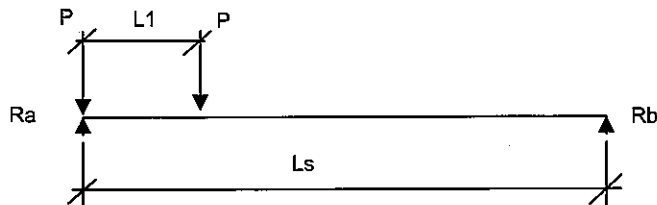
Calculation/Sketch

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Site	Dubai Depot - RAB6	Date	June '07	Job No.	068010
	Crane Beam Loading				
	(10 T / 15m Bridge)	Designed by Rudy		Sheet No.	

B. For max shear



1. Vertical direction

(a) due to Crane Vertical Load with impact

$$\begin{aligned} R_a' &= P_v' * [1 + (L_s - L_1) / L_s] &= & 143.9 \text{ kN} \\ R_b' &= 2 * P_v' - R_a &= & 31.9 \text{ kN} \\ M_v' &= R_b' * (L_s - L_1) &= & 162.5 \text{ kNm} \end{aligned}$$

(b) due to Crane Vertical Load without impact

$$\begin{aligned} R_a &= P_v' * [1 + (L_s - L_1) / L_s] &= & 115.1 \text{ kN} \\ R_b &= 2 * P_v' - R_a &= & 25.5 \text{ kN} \\ M_v &= R_b * (L_s - L_1) &= & 130.0 \text{ kNm} \end{aligned}$$

(c) due to selfweight of the beam

$$\begin{aligned} R_{wa} = R_{wb} &= 0.5 * w L_s &= & 4.6 \text{ kN} \\ M_w \text{ (at location of max vertical moment)} &= 0.5 w L_s^2 * \{ [L_1 / L_s] - [L_1 / L_s]^2 \} &= & 8.6 \text{ kNm} \end{aligned}$$

2. Horizontal direction due to crane surge load (including impact factor)

$$\begin{aligned} R_{ha} &= 35.5 \text{ kN} \\ R_{hb} &= 7.9 \text{ kN} \\ M_h &= 40.0 \text{ kNm} \end{aligned}$$

C. Load Combination

Case 1 : Vertical load with impact only

$$\begin{aligned} M_v(\text{design}) &= 1.4 * M_w + 1.6 * M_v' &= & 389.7 \text{ kNm} \\ V_v(\text{design}) &= 1.4 * R_{wa} + 1.6 * R_a' &= & 236.8 \text{ kN} \end{aligned}$$

Case 2 : Vertical load + Horizontal load without impact

$$\begin{aligned} M_v(\text{design}) &= 1.4 * M_w + 1.6 * M_v &= & 314.2 \text{ kNm} \\ M_h(\text{design}) &= 1.6 * M_h &= & 116.1 \text{ kNm} \end{aligned}$$

Case 3 : Vertical load + Horizontal load with impact

$$\begin{aligned} M_v(\text{design}) &= 1.4 * (M_w + M_v') &= & 342.5 \text{ kNm} \\ M_h(\text{design}) &= 1.4 * M_h' &= & 101.6 \text{ kNm} \end{aligned}$$

Calculation/Sketch

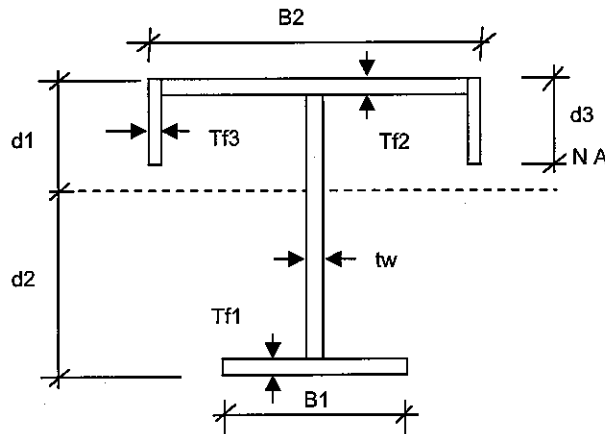
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Site	Dubai Depot - RAB6	Date	May '07	Job No.	068010
	Crane Beam Section	Designed by	Rudy	Sheet No.	

Calculation of section capacity

B1	=	250 mm
B2	=	380 mm
D	=	650 mm
d3	=	100 mm
Tf1	=	12 mm
Tf2	=	9.5 mm
Tf3	=	17.5 mm
tw	=	8 mm
Af1	=	3000.0 mm ²
Af2	=	3277.5 mm ²
Af3 (top)	=	3419.6 mm ²
Af3 (bot)	=	80.4 mm ²



To find the equal area axis

Assume d1	=	97.7 mm
Aweb (top)	=	705.6 mm ²
Aweb (bot)	=	4322.4 mm ²

Plastic modulus

Sxx	=	3309492 mm ⁴
Syy	=	1764114 mm ⁴

Total area above the neutral axis
Atop = 7402.8 mm²

Total area below the neutral axis
Abot = 7402.8 mm²

Atop - Abot = 0 mm²
d2 = 552.3 mm

Atotal = 14805.5 mm²

Assumption:

- Moment in the vertical direction is resisted by equal flange I section of B1 flange width
- Horizontal moment is resisted by the Tf3 flanges

A) For vertical load.

Check equivalent of the top flange width

$$B' = (B2 - Tf3) * Tf2 / Tf1$$

$$= 273 \text{ mm} > B1, \text{ use } B1 \text{ width}$$

Refer to the subsequent pages for section capacity check

Calculation/Sketch

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Site	Dubai Depot - RAB6 Crane Beam Section	Date	May '07	Job No.	068010
		Designed by	Rudy	Sheet No.	

B) For horizontal load,

$$d3/Tf3 = 5.71 < 9E$$

Section is plastic

$$py = 325 \text{ N/mm}^2$$

$$\begin{aligned} Ag &= 0.9d3 \cdot Tf3 \\ &= 1575 \text{ mm}^2 \end{aligned}$$

$$\begin{aligned} Ph &= pyAg \\ &= 511.9 \text{ kN} \end{aligned}$$

Force due to horizontal moment

$$\begin{aligned} Fh &= Mh / (B2 - Tf3) \\ &= 320.4 \text{ kN} && \text{(without impact factor)} && \text{Ok} \\ &= 280.3 \text{ kN} && \text{(with impact factor)} && \text{Ok} \end{aligned}$$

Project: Dubai LRT
Subject: Crane Beam Vertical capacity
Discreption

Job No.: 068010.021

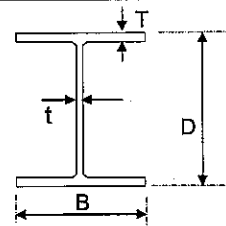
Designed by: Pudy

Sheet No.

Design of Steel Beam with no or little axial force — Predominantly Flexural Member**1. Design Forces**

ULS Bending Moment $M_{max} = 389.7$ kN-m
ULS Shear Force $V_{max} = 236.8$ kN
Spacing = 1.0 m

Axial Force $P_{max} = 0$ kN
Load factor, $\gamma_f = 1.5$
Ultimate strut load = 0 kN

**2. Individual Member Properties (For Equal Flange Built Up I Section Only)**

Section size	650 x 250 x 30 kg/m	E =	2.05E+05	N/mm ²
Beam depth, D	650 mm	Depth between fillets, d =	626	mm
Beam width, B	250 mm	* Plastic modulus, $S_x =$	2.70E+06	mm ³
Flange thickness, T	12 mm	* Plastic modulus, $S_y =$	3.64E+04	mm ³
Web thickness, t	8 mm	[*calculated based on contribution from flanges & web]		
Sectional area, A_g	1.10E+04 mm ²	Radius of gyration, $r_x =$	265.20	mm
Moment of inertia, I_x	7.74E+08 mm ⁴	Radius of gyration, $r_y =$	53.30	mm
Moment of inertia, I_y	3.13E+07 mm ⁴	$A_{eff} =$	11008	mm ²
Section Modulus, Z_x	2.38E+06 mm ³	$Z_{x,eff} =$	1.88E+06	mm ³
Section Modulus, Z_y	2.50E+05 mm ³	$Z_{y,eff} =$	2.50E+05	mm ³

3. Material Properties

Steel grade = 50
Design strength, $p_y = 355$ N/mm²
Elastic modulus = 205 kN/mm²

4. Section Classification

Epsilon = $[275/p_y]^{0.5} = 0.880$
Outstand element of compression flange, b/T
Plastic - 7.9
Compact - 8.8
Semi-compact - 13.2
Actual $b/T = 10.083$
Section classified as **Semi-compact**
Web subject to compression throughout, d/t
Plastic - 70.4 $r_1 = 0.00$
Compact - 88.0
Semi-compact - 105.62 $r_2 = 0.00$
Actual $d/t = 78.25$
Flange is semi-compact. Web is compact.

5. Design Capacity Check**5.1 Local/Section Capacity Check**

Shear capacity of beam, $P_v = 0.6p_y A_v = 1108$ kN > Design Value, OK
Since d/t is more than 63 $\epsilon \rightarrow$ Shear buckling resistance check is required Clause 4.2.3
Moment Capacity of beam, $P_y Z_x$ Clause 4.2.5
 M_{cx} beam section = 845.6 kN-m > Design Value, OK

5.2 Lateral Torsional Buckling Check

$(A_{eff}/A_g)^{0.5} = 1.00$
Effective length, $L_{xx} = 5.1$ m
Slenderness Ratio: $\lambda = 19.2$
Design $\lambda_{max} = 95.7$
Effective length, $L_{yy} = 5.10$ m
Slenderness Ratio: $\lambda_y = 95.68$
Clause 4.3

Equivalent Slenderness Ratio, $\lambda_{LT} = u \nu \lambda \sqrt{\beta_w} = 78.0$
 $\beta_w = 0.88$
Buckling parameter, $u = 0.9$
Slenderness factor, $\nu = 0.96$ } For Equal Flanges
Torsional index, $x = 54.2$
 $\lambda/x = 1.8$
Slenderness factor $\nu = 1 / ((1 + 0.05(\lambda/x)^2)^{0.25})$

Bending strength, $P_b = \frac{P_E P_y}{\phi_{LT} \sqrt{(\phi_{LT}^2 - P_E P_y)}} = 195.6$ N/mm²
Euler strength P_E
 $P_E = (\pi^2 E / \lambda_{LT}^2) = 332.2$
 $\phi_{LT} = \frac{P_y + (\eta_{LT} + 1) P_E}{2} = 399$
Limiting Slenderness Ratio $\lambda_{L0} = 30.20$
 $\alpha_{LT} = 7.00$
 $\eta_{LT} = \alpha_{LT} (\lambda_{LT} - \lambda_{L0}) / 1000 = 0.33$
It shall be ≥ 0.0
 $\alpha_{LT} = 0.33$

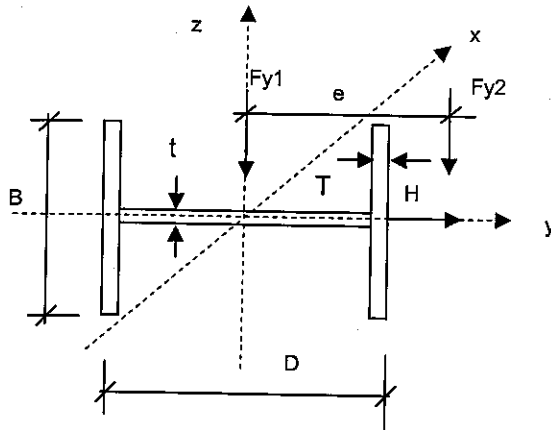
Buckling Resis. Momt. $M_b = 466.1$ kNm > Design Value, OK
Ratio of ULS BM / Capacity = 0.84

Use Simpli Supported Beam to calculate UDL & Deflection for the applied bending moment

Equivalent SLS UDL (SS) = 79.9 kN/m
Span/deflection Ratio = 1150.0 < 250 OK
Deflection at Mid-Span (SS) $\delta = \frac{5 w l^4}{384 E I_{xx}} = 4.4$ mm



Site	Dubai Depot Diesel Engine Workshop (Rashidiya - Bldg 6) Side Beam Design	Date	June '07	Job No.	068010
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Member Properties

Beam depth, D	=	300 mm
Beam width, B	=	250 mm
Flange thickness, T	=	15 mm
Web thickness, t	=	10 mm
Depth btw fillets, d	=	270 mm
Sectional area, Ag	=	10200 mm ²
Moment of inertia, Iz	=	1.69E+08 mm ⁴
Moment of inertia, Iy	=	3.91E+07 mm ⁴
Section Modulus, Zz	=	1.13E+06 mm ³
Section Modulus, Zy	=	3.13E+05 mm ³

* Plastic modulus, Sz	=	1.26E+06 mm ³
* Plastic modulus, Sy	=	3.50E+04 mm ³
[*calculated based on contribution from flanges & web]		
Radius of gyration, rz	=	128.66 mm
Radius of gyration, ry	=	61.90 mm
A,eff	=	10200 mm ²
Zz,eff	=	1.02E+06 mm ³
Zy,eff	=	3.13E+05 mm ³

E	=	205000 N/mm ²
v	=	0.3
G	=	$E / 2 * (1 + v)$
	=	78846 N/mm ²
Steel grade	=	50
Strength, py	=	355 N/mm ²

Section Classification

Epsilon = $[275/py]^{0.5}$	=	0.880
Outstand element of compression flange, b/T		
Plastic	=	7.9
Compact	=	8.8
Semi-compact	=	13.2
Actual b/T	=	8

Flange is compact.Section classified as **Semi-compact**

Web subject to compression throughout, d/t		
Axial Force P _{max}	=	0.00 kN
r ₁	=	0.00
Plastic	=	70.4
Compact	=	88.0
r ₂	=	0.00
Semi-compact	=	105.62
Actual d/t	=	27.00
Web is plastic.		

For symmetrical I & H section

Torsional constant, J	=	$(1/3) * [2BT^3 + (D-2T) * t^3]$	=	7.65E+05
Warping constant, H	=	$0.25 * Iy * (D-T)^2$	=	7.937E+11
Warping static moment, Sws	=	$(D-T) * B^2 * T / 16$	=	1.67E+07
Normalised warping function, Wns	=	$(D-T)B/4$	=	1.78E+04

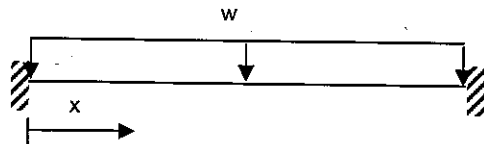
Calculation/Sketch

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Site	Dubai Depot Diesel Engine Workshop (Rashidiya - Bldg 6) Side Beam Design	Date	June '07	Job No.	068010
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Loading



$$\begin{aligned} \text{Span Length, } L_x &= 8.0 \text{ m} \\ M_x &= 0.5wL_x^2 \left[\left(\frac{x}{L_x} \right) - \left(\frac{x}{L_x} \right)^2 - \frac{1}{6} \right] \quad (\text{fixed end}) \\ M_x &= 0.5wL_x^2 \left[\left(\frac{x}{L_x} \right) - \left(\frac{x}{L_x} \right)^2 \right] \quad (\text{simply supported}) \\ e, \text{ eccentricity} &= 350 \text{ mm} \end{aligned}$$

$$F_{y1} - \text{Self wt} = 0.80 \text{ kN/m}$$

Fy2 - Purlin + Cladding + Insulation

$$\begin{aligned} \text{Height} &= 4.4 \text{ m} \\ \text{Weight} &= 0.5 \text{ kN/m}^2 \\ \text{Load} &= 2.2 \text{ kN/m} \end{aligned}$$

H - Wind Load

$$\begin{aligned} \text{Pressure} &= 1.24 \text{ kN/m}^2 \\ \text{Load} &= 5.4 \text{ kN/m} \end{aligned}$$

For bending along minor axis ($w = 1.4DL$) - simply supported

$$\begin{aligned} \text{At } x = 0.5L & \quad M_y(\text{max}) = \frac{wLx^2}{8} = 33.5 \text{ kNm} \\ \text{At } x = 0.25L \text{ and } x = 0.75L & \quad M_y = \frac{3wL^2}{32} = 25.1 \text{ kNm} \\ \text{At } x = 0 \text{ and } x = L & \quad M_y = 0.0 \text{ kNm} \end{aligned}$$

$$\begin{aligned} m_{LT} &= 0.2 + (0.15M_2 + 0.5M_3 + 0.15M_4) / M_{\text{max}} = 0.93 \\ m_{LT} * M_{\text{ymax}} &= 31.0 \text{ kNm} \end{aligned}$$

For Bending along major axis ($w = H$) - fixed support

$$\begin{aligned} \text{At } x = 0 \text{ and } x = L & \quad M_z(\text{max}) = \frac{wLx^2}{12} = 29.0 \text{ kNm} \\ \text{At } x = 0.25L \text{ and } x = 0.75L & \quad M_z = \frac{wLx^2}{96} = \\ \text{At } x = 0.5L & \quad M_z = \frac{wLx^2}{24} = \end{aligned}$$

$$\begin{aligned} m_{LT} &= 0.2 + (0.15M_2 + 0.5M_3 + 0.15M_4) / M_{\text{max}} = 0.70 \\ m_{LT} * M_{z\text{max}} &= 20.3 \text{ kNm} \end{aligned}$$

$$\text{Torsional moment (} w = 1.4F_{y2} \text{)} \quad T_q(\text{max}) = w * e * L_x = 8.6 \text{ kNm}$$

$$\text{Shear in z-direction (} w = 1.4DL \text{)} \quad F_{vx} = \frac{wLx}{2} = 16.7 \text{ kN}$$

$$\text{Shear in y-direction (} w = H \text{)} \quad F_{vy} = \frac{wLx}{2} = 21.7 \text{ kN}$$

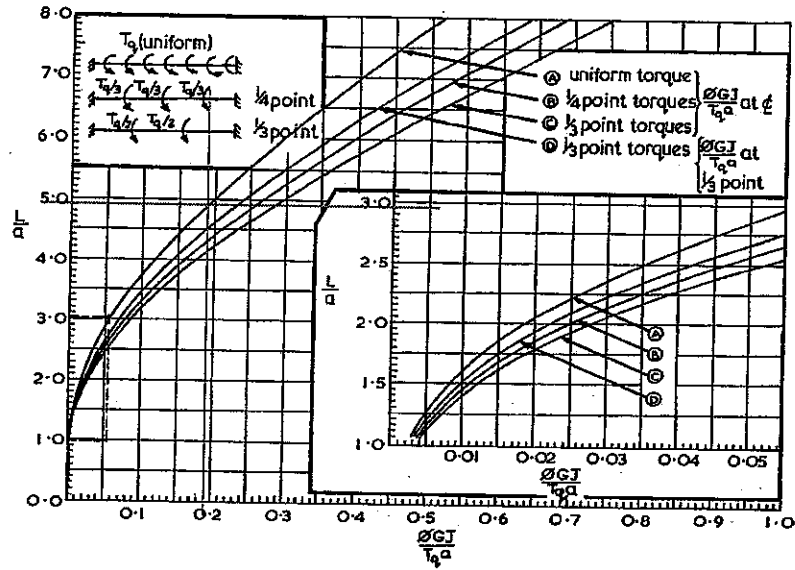
$$\text{Resultant shear} \quad F_r = (F_{vx}^2 + F_{vy}^2)^{0.5} = 27.4 \text{ kN}$$

Site Dubai Depot
Diesel Engine Workshop (Rashidiya - Bldg 6)
Side Beam Design

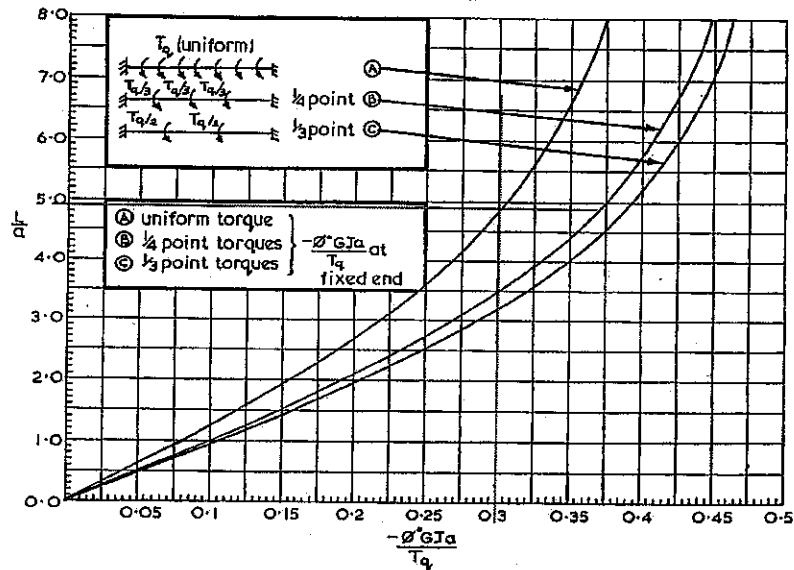
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GRAPH 7



GRAPH 8



Calculation/Sketch

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Site	Dubai Depot Diesel Engine Workshop (Rashidiya - Bldg 6) Side Beam Design	Date	June '07	Job No.	068010
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Shear Capacity Check

Shear capacity of beam, $P_v = 0.6p_y A_v = 639 \text{ kN} > \text{Design Value, OK}$

Since d/t is not more than 63

Shear buckling resistance check is not required

Clause 4.2.3

Local capacity check

Torsional bending constant, $a = (EH/GJ)^{0.5} = 1642.3884 \text{ mm}$

With uniform torque and $L/a = 4.9$

From the Graph 7

$\phi = 0.19 T_q a / GJ$
 $= 0.044 \text{ rad}$

$M_z t = \phi M_z$
 $= 1.3 \text{ kNm}$

$\sigma_{byt} = M_y t / Z_y$
 $= 4.11 \text{ N/mm}^2$

$\sigma_{by} = M_y / Z_y$
 $= 107.12 \text{ N/mm}^2$

From the Graph 8

$\phi'' = 0.3 T_q / GJ a$
 $= 2.60E-08$

$\sigma_w = E W_{ns} \phi''$
 $= 94.9 \text{ N/mm}^2$

$\sigma_{bz} = M_z / Z_z$
 $= 25.73 \text{ N/mm}^2$

$\sigma_{bz} + \sigma_{by} + \sigma_{byt} + \sigma_w = 231.9 \text{ N/mm}^2$ Ok, less than p_y

Overall buckling check

$(A_{eff}/A_g)^{0.5} = 1.00$

Effective length, $L_{zz} = 10.2 \text{ m}$ Effective length, $L_{yy} = 10.2 \text{ m}$

Slenderness Ratio: $= 79.3$ Slenderness Ratio: $= 164.78$

Design Lamda, $\lambda = 164.8$

$\beta_w = 0.89$

Torsional index, $x = 20.0$

$\lambda/x = 8.2$

For equal flanges

Buckling parameter, $u = 0.9$
Slenderness factor, $v = 1 / [(1 + 0.05 (\lambda/x)^2)^{0.25}] = 0.69$

Equivalent Slenderness Ratio $\lambda_{LT} = u v \lambda (\beta_w)^{0.5} = 96.6$

Limiting Slenderness Ratio $\lambda_{L0} = 0.4 (\pi^2 E / p_y)^{0.5} = 17$

$\alpha_{LT} = 7$ (Appendix B - Cl 2.2)

$\eta_{LT} = \alpha_{LT} (\lambda_{LT} - \lambda_{L0}) / 1000 = 0.557$

Euler strength, $p_e = (\pi^2 E / \lambda_{LT}^2) = 216.7 \text{ N/mm}^2$

$\phi_{LT} = 0.5 [p_y + (\eta_{LT} + 1) p_e] = 346.2 \text{ N/mm}^2$

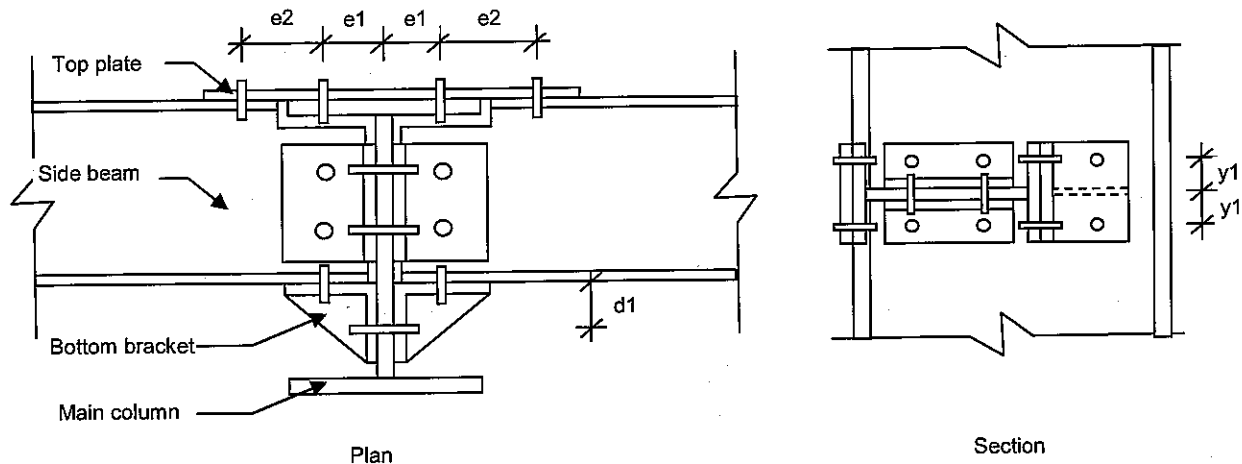
Bending strength, $p_{bz} = p_e p_y / [\phi_{LT} + (\phi_{LT}^2 - p_e p_y)^{0.5}] = 139.0 \text{ N/mm}^2$

Buckling Resis. Momt. $M_{bz} = 156.5 \text{ kNm}$

$m_{LT} M_z / M_{bz} + m_{LT} M_y / p_y Z_y + (\sigma_{byt} + \sigma_w) / p_y * [1 + 0.5 (m_{LT} M_y / M_{bz})]$
 $= 0.44$ Ok, less than 1

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Side Beam End Connection Design



From the side beam design

Vertical bending moment	$M_y =$	0.0 kNm
Horizontal bending moment	$M_z =$	29.0 kNm
Torsional bending moment	$T_q =$	8.6 kNm
Vertical shear	$F_y =$	21.7 kN
Horizontal shear	$F_z =$	16.7 kN

Assumption

- 1) Vertical bending moment to be resisted by top and bottom bolt connected to main column
- 2) Horizontal and torsional bending moment to be resisted by bolt connected to side beam's flange
- 3) Vertical shear to be resisted by bolt on the side beam's flange and main column's web
- 4) Horizontal shear to be resisted by bolt on side beam's and main column's web

Side beam dimension

Depth	D1	=	300 mm
Width	B1	=	250 mm
Web thickness	t1	=	10 mm
Flange thickness	T1	=	20 mm

Main column dimension

Depth	D2	=	440 mm
Width	B2	=	250 mm
Web thickness	t2	=	10 mm
Flange thickness	T2	=	15 mm

e1	=	75 mm
e2	=	100 mm
d1	=	50 mm
y1	=	75 mm

Calculation/Sketch

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Site	Dubai Depot Diesel Engine Workshop (Rashidiya - Bldg 6) Side Beam Design	Date	June '07	Job No.	068010
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1) Weld connecting side beam web to main column's web

a) Shear due to vertical load
 $V_y = F_y / \text{no of bolt} = 5.4 \text{ kN}$ no of bolt = 4

b) Shear due to horizontal load
 $V_z = F_z / \text{no of bolt} = 4.2 \text{ kN}$ no of bolt = 4

Effective weld length = 250 mm x 2 = 500 mm
Weld size, s = 6 mm (up to 15mm thick)
Weld strength, pw = 1.05 kN/mm (based on longitudinal capacity)
Weld length required = $(V_y + V_z) / pw = 9 \text{ mm}$
= 5 mm per face Ok

2) Bolts at the side beam's flange

a) Shear due to horizontal bending moment
 $V_{hs} = M_z / (D1 * \text{no of row of bolt}) = 48.3 \text{ kN}$ no of row of bolt = 2

b) Shear due to torsional bending moment
 $V_t = T_q / (D1 * \text{no of bolt}) = 14.3 \text{ kN}$ no of bolt = 2

c) Shear due to vertical load
 $V_y = F_y / \text{no of bolt} = 10.9 \text{ kN}$ no of bolt = 2

d) Resultant shear
 $V_r = [V_{hs}^2 + (V_t + V_y)^2]^{0.5} = 54.4 \text{ kN}$

Bolt diameter, $\phi = 20 \text{ mm}$
Shear capacity, $V_b = 91.9 \text{ kN}$ Ok

3) Bolt at the side beam's web

a) Shear due to horizontal load
 $V_z = F_z / \text{no of bolt} = 8.4 \text{ kN}$ no of bolt = 2

Bolt diameter, $\phi = 20 \text{ mm}$
Shear capacity, $V_b = 91.9 \text{ kN}$ Ok

4) Bolts at the bracket connected to main column's web

a) Tension due to horizontal bending moment
 $T = M_z / [(D1 + d1) * \text{no of row of bolt}] = 41.4 \text{ kN}$ no of row of bolt = 2

Bolt diameter, $\phi = 20 \text{ mm}$
Bolt tensile capacity, $T_b = 110 \text{ kN}$ Ok

4 CONNECTION DESIGN

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Page

Calculation/Sketch

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Site	Dubai Depot Diesel Engine Workshop (Rashidiya)	Date	June '07	Job No.	068010
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Column/Beam Splice Connection Design

From the STAAD Pro analysis

$F_{xc} = 222 \text{ kN (compression)}$
 $F_{xt} = 1 \text{ kN (tension)}$
 $F_y = 81 \text{ kN}$
 $M_z = 639 \text{ kNm}$

Total Tension

$T = (n \cdot T_b / d) \cdot [(d-a) + (d-y_1-a) + (d-y_2-a) + \dots]$
 where n = no of row of bolts and T_b is bolt tension capacity

Total Compression

$= 0.5a \cdot p_c \cdot b$
 where b = stiff bearing length
 $= \text{web thickness} + 2 \cdot \text{plate thickness}$

Moment Capacity

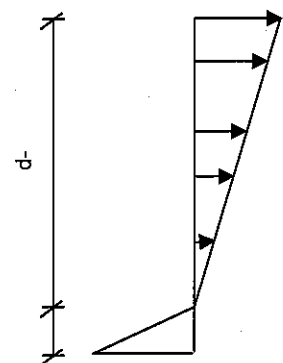
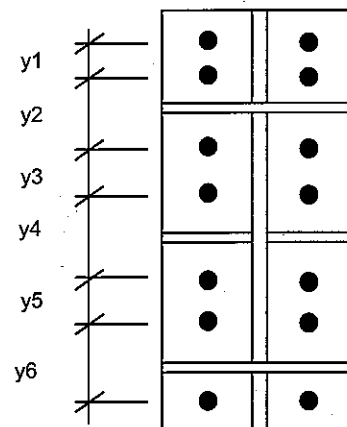
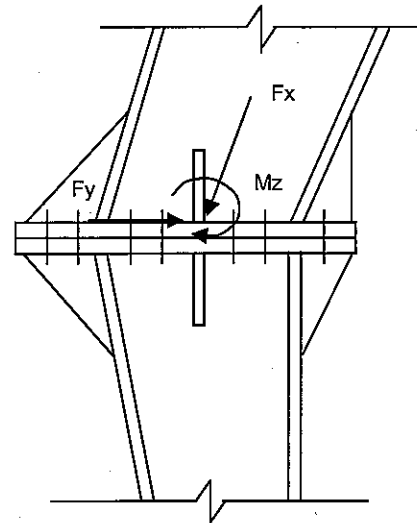
$M_c = [n T_b / (d-a)] \cdot [(d-a)^2 + (d-a-y_1)^2 + (d-a-y_2)^2 + \dots]$

For Gr 8.8 bolt

Bolt size	=	27 mm		
Tb	=	207 kN		
Vb	=	172 kN		
n	=	2		
n total	=	16		
web, tw	=	8 mm		
plate, tp	=	36 mm		
b	=	80 mm		
y1	95	Trial a	=	110 mm
y2	125	T	=	1427.2 kN
y3	120	C	=	1427.2 kN
y4	230	T-C	=	0.00 kN
y5	120	Mc	=	815.4 kNm
y6	170			
Depth, d	<u>860</u>			

Tension force on the bolt, T_1 = $(M_z / M_c) \cdot T_b$
 = 162 kN

$[T_1 + (F_{xt} / n \text{ total})] / T_b + F_y / (n \text{ total} \cdot V_b) = 0.81$
 $< 1.4 \text{ Ok}$



$p_c = 325 \text{ N/mm}^2$

Calculation/Sketch

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Site	Dubai Depot Diesel Engine Workshop (Rashidiya)	Date	June '07	Job No.	068010
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Check the effective weld length on the stiffener plate to resist bolt's tension

Weld size, s = 6 mm (up to 15mm thick)
Weld strength, pw = 1.05 kN/mm (based on longitudinal capacity)
Weld length required = $(T1 + 0.5 \cdot F_{xt}) / pw$
= 155 mm
= 78 mm per face

Calculation/Sketch

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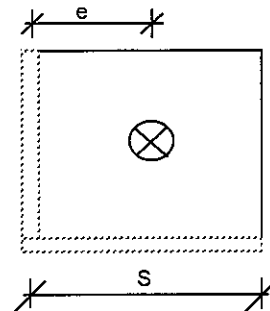


Site Dubai Depot Date June '07 Job No. 068010
Diesel Engine Workshop (Rashidiya)

Designed by Rudy Sheet No.

From the plate analysis (refer attached) using
Prying Force, $T_o = 10$ kN

Plate width, S_o (mm)	Ave. support bending moment, M_o (kNm)
75	0.020
100	0.027
125	0.035
150	0.040



Actual - Prying force, $T = 162$ kN
Side, $S = 75$ mm
Bending Moment, $M = (T/T_o) * M_o = 0.327$ kNm

Bolt Diameter, $\phi = 27$ mm
Bolt distance, $e = 75$ mm
Thickness of plate, $t_p = 36$ mm
 p_y allow $= 345$ N/mm²

Effective width of the plate resisting the bending moment, $b_p = \min(2e + \phi, S) = 75$ mm

Section of modulus of the plate, $Z_p = b_p * t_p^2 / 6 = 16200$ mm²

$M / Z_p = 20$ N/mm² Ok

Calculation/Sketch

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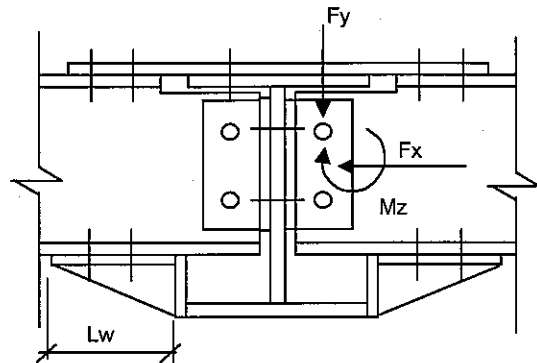
Site	Dubai Depot	Date	Jun '07	Job No.	068010
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Secondary Beam Connection Design

From the analysis

$F_y = 51 \text{ kN}$
 $F_x = 112 \text{ kN}$
 $M_z = 0 \text{ kNm}$

Eccentricity of load, $e = 60 \text{ mm}$



- Assume vertical and horizontal shear is resisted by bolt on the web

Check shear on the web

- for the secondary beam web

No of bolt provided	=	3
Shear per bolt	=	40.9 kN
Bolt diameter, ϕ	=	20 mm
Shear capacity, V_b	=	91.9 kN

- for the main beam web

No of bolt provided	=	3
Distance between bolt	=	150 mm
Tensile load per bolt	=	57.0 kN
Bolt diameter, ϕ	=	20 mm
Tensile capacity, V_b	=	110 kN

Calculation/Sketch

Meinhardt Infrastructure Pte Ltd
Consulting Engineers, Planners & Managers



Site	Dubai Depot Diesel Engine Workshop (Rashidiya)	Date	Jun '07	Job No.	068010
		Designed by	Rudy	Sheet No.	

For the cantilever end

- Assume shear due to bending is resisted by top and bottom bolt
 - Assume vertical and horizontal shear is resisted by bolt on the web
- Eccentricity of load, $e = 60 \text{ mm}$

$M_{\max} = 103 \text{ kNm}$
 $V_{\max} = 60 \text{ kN}$

Check shear on the web

- for the secondary beam web

No of bolt provided $= 2$
Shear per bolt $= 29.8 \text{ kN}$
Bolt diameter, $\phi = 20 \text{ mm}$
Shear capacity, $V_b = 91.9 \text{ kN}$

- for the main beam web

No of bolt provided $= 2$
Distance between bolt $= 100 \text{ mm}$
Tensile load per bolt $= 35.8 \text{ kN}$
Bolt diameter, $\phi = 20 \text{ mm}$
Tensile capacity, $V_b = 110 \text{ kN}$

Check bolts on the top flange

Beam depth - flange thickness, $D_f = 292 \text{ mm}$
 $M_{\max}$ bending moment (incl. moment due to ecc), $M_e = 106.9 \text{ kNm}$

Shear on the flange, $V_f = \frac{M_e}{D_f} = 366.2 \text{ kN}$

No of bolt provided $= 6$
Shear per bolt $= 61.0 \text{ kN}$
Bolt diameter, $\phi = 20 \text{ mm}$
Shear capacity, $V_b = 91.9 \text{ kN}$

Calculation/Sketch

Meinhardt Infrastructure Pte Ltd
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Site Dubai Depot
Diesel Engine Workshop (Rashidiya)

Date Jun '07

Job No. 068010

Designed by Rudy

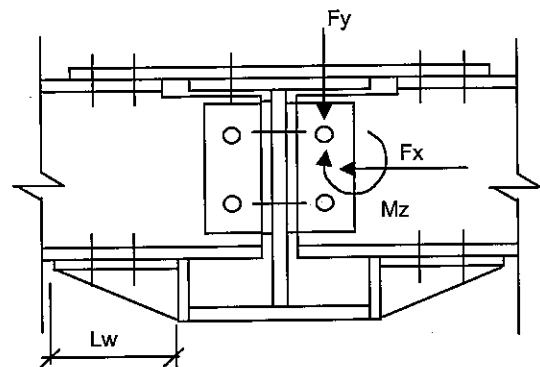
Sheet No.

Office Frame to Main Frame Connection Design

From the analysis

$F_y = 51 \text{ kN}$
 $F_x = 34 \text{ kN}$
 $M_z = 0 \text{ kNm}$

Eccentricity of load, $e = 60 \text{ mm}$



- Assume vertical and horizontal shear is resisted by bolt on the web

Check shear on the web

- for the secondary beam web

No of bolt provided	=	2
Shear per bolt	=	31.0 kN
Bolt diameter, ϕ	=	16 mm
Shear capacity, V_b	=	58.9 kN

- for the main beam web

No of bolt provided	=	2
Distance between bolt	=	150 mm
Tensile load per bolt	=	37.2 kN
Bolt diameter, ϕ	=	16 mm
Tensile capacity, V_b	=	70.7 kN

Calculation/Sketch

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For the continuous end

- Assume shear due to bending is resisted by top and bottom bolt
 - Assume vertical and horizontal shear is resisted by bolt on the web
- Eccentricity of load, e = 60 mm

M_{max} = 171 kNm
 V_{max} = 52 kN

Check shear on the web

- for the secondary beam web

No of bolt provided = 2
Shear per bolt = 25.9 kN
Bolt diameter, ϕ = 16 mm
Shear capacity, V_b = 58.9 kN

- for the main beam web

No of bolt provided = 2
Distance between bolt = 100 mm
Tensile load per bolt = 31.1 kN
Bolt diameter, ϕ = 16 mm
Tensile capacity, V_b = 70.7 kN

Check bolts on the top flange

Beam depth - flange thickness, D_f = 292 mm
 M_{max} bending moment (incl. moment due to ecc), M_e = 173.8 kNm

Shear on the flange, V_f = M_e / D_f
= 595.3 kN

No of bolt provided = 6
Shear per bolt = 99.2 kN
Bolt diameter, ϕ = 24 mm
Shear capacity, V_b = 132 kN

Calculation/Sketch

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Site	Dubai Depot Diesel Engine Workshop (Rashidiya)	Date	Jun '07	Job No.	068010
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Column to Main Beam Connection Design (Office Area)

From the STAAD Pro analysis

Fxc =	126	kN (compression)
Fxt =	0	kN (tension)
Fy =	472	kN (include shear due to beam moment)
Mz =	171	kNm

Total Tension

$$T = (n \cdot T_b / d) [(d-a) + (d-y_1-a) + (d-y_2-a) + \dots]$$

where n = no of row of bolts and T_b is bolt tension capacity

Total Compression

$$= 0.5a \cdot p_c \cdot b$$

where b = stiff bearing length

= web thickness + 2 * plate thickness

Moment Capacity

$$M_c = [n T_b / (d-a)] * [(d-a)^2 + (d-a-y_1)^2 + (d-a-y_2)^2 + \dots]$$

For Gr 8.8 bolt

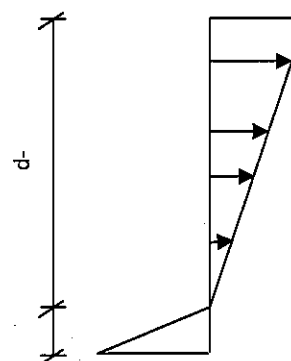
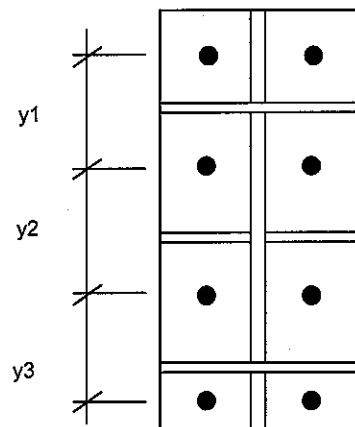
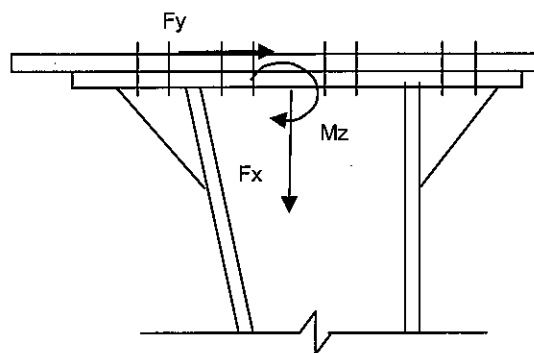
Bolt size	=	30 mm
T _b	=	252 kN
V _b	=	210 kN
n	=	2
n total	=	8

web, t _w	=	8 mm
plate, t _p	=	15 mm
b	=	38 mm

y1	150	Trial a	=	130 mm
y2	150	T	=	803.1 kN
y3	150	C	=	803.1 kN
y4	0	T-C	=	0.00 kN
y5	0	M _c	=	207.4 kNm
y6	0			
Depth, d	450			

$$\text{Tension force on the bolt, } T_1 = (M_z / M_c) \cdot T_b = 207 \text{ kN}$$

$$[T_1 + (F_{xt} / n_{\text{total}})] / T_b + F_y / (n_{\text{total}} \cdot V_b) = 1.10 < 1.4 \text{ Ok}$$



$$p_c = 325 \text{ N/mm}^2$$

Calculation/Sketch

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Site	Dubai Depot Diesel Engine Workshop (Rashidiya)	Date	Jun '07	Job No.	068010
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Check the effective weld length on the stiffener plate to resist bolt's tension

Weld size, s = 6 mm (up to 15mm thick)
Weld strength, pw = 1.05 kN/mm (based on longitudinal capacity)
Weld length required = $\max(T1 + 0.5 \cdot F_{xt}, F_y) / pw$
= 449 mm
= 225 mm per face

Calculation/Sketch

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Site Dubai Depot
Diesel Engine Workshop (Rashidiya)

Date Jun '07

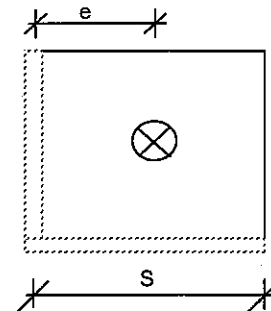
Job No. 068010

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From the plate analysis (refer attached) using
Prying Force, $T_o = 10$ kN

Plate width, S_o (mm)	Ave. support bending moment, M_o (kNm)
75 312	0.020
100 63.4	0.027
125 6.61	0.035
150 41.6	0.040



Actual - Prying force, $T = 207$ kN
Side, $S = 125$ mm
Bending Moment, $M = (T/T_o) * M_o = 0.732$ kNm

Bolt Diameter, $\phi = 20$ mm
Bolt distance, $e = 50$ mm
Thickness of plate, $t_p = 15$ mm
 p_y allow $= 355$ N/mm²

Effective width of the plate resisting the bending moment, $b_p = \min(2e + \phi, S) = 120$ mm

Section of modulus of the plate, $Z_p = b_p * t_p^2 / 6 = 4500$ mm²

$M / Z_p = 163$ N/mm² Ok

Calculation/Sketch

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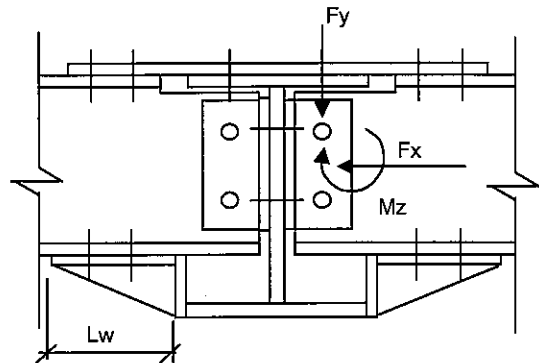
Site	Dubai Depot Diesel Engine Workshop (Rashidiya)	Date	Jun '07	Job No.	068010
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Secondary Beam Connection Design (Office Area)

From the analysis

$F_y = 18 \text{ kN}$
 $F_x = 35 \text{ kN}$
 $M_z = 0 \text{ kNm}$

Eccentricity of load, $e = 60 \text{ mm}$



- Assume vertical and horizontal shear is resisted by bolt on the web

Check shear on the web

- for the secondary beam web

No of bolt provided	=	2
Shear per bolt	=	19.8 kN
Bolt diameter, ϕ	=	16 mm
Shear capacity, V_b	=	58.9 kN

- for the main beam web

No of bolt provided	=	2
Distance between bolt	=	150 mm
Tensile load per bolt	=	24.7 kN
Bolt diameter, ϕ	=	16 mm
Tensile capacity, V_b	=	70.7 kN